THE
POPULAR SCIENCE
MONTHLY

EDITED BY

J. McKEEN CATTELL

VOL. LVII

MAY TO OCTOBER, 1900

NEW YORK AND LONDON
McCLURE, PHILLIPS AND COMPANY
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SEPTEMBER, 1900.

THE MODERN OCCULT.

By Professor JOSEPH JASTROW,
UNIVERSITY OF WISCONSIN.

IF that imaginary individual so convenient for literary illustration, a visitor from Mars, were to alight upon our planet at the present time, and if his intellectual interests induced him to take a survey of mundane views of what is "in heaven above, or on the earth beneath or in the waters under the earth," of terrestrial opinions in regard to the great problems of mind and matter, of government and society, of life and death—our Martian observer might conceivably report that a limited portion of mankind were guided by views that were the outcome of accumulated toil, and generations of studious devotion, representing a slow and tortuous, but progressive growth through error and superstition, and at the cost of persecution and bloodshed; that they maintained institutions of learning where the fruits of such thought could be imparted and the seeds cultivated to bear still more richly, but that outside of this respectable yet influential minority there were endless upholders of utterly unlike notions and of widely diverging beliefs, clamoring like the builders of the tower of Babel in diverse tongues.

It is well at least occasionally to remember that our conceptions of science and of truth, of the nature of logic and of evidence, are not so universally held as we unreflectingly assume or as we hopefully wish. Almost every one of the fundamental and indisputable tenets of science is regarded as hopelessly in error by some ardent would-be reformer. One Hampden declares that the earth is a motionless plane with the North Pole as the center; one Carpenter gives a hundred remarkable reasons why the earth is not round, with a challenge to the scientists of America to disprove them; one Symmes regarded the earth as hollow and habitable within, with openings at the poles which he offered to explore for the consideration of the "patronage of this and the new worlds"; while Symmes, Jr., explains how the interior is lighted, and that it probably forms the home of the lost tribes of Israel; and one Teed announces on equally conclusive evidence that the earth is a “stationary concave cell ... with people, Sun, Moon, Planets and Stars on the inside,” the whole constituting an “alchemico-organic structure, a Gigantic Electro-Magnetic Battery.” If we were to pass from opinions regarding the shape of the earth to the many other and complex problems that appeal to human interests, it would be equally easy to collect ‘ideas’ comparable to these in value, evidence and eccentricity. With the conspicuously pathological outgrowth of brain-functioning—although its representatives in the literature of my topic are neither few nor far between—I shall not specifically deal; and yet the general abuse of logic, the helpless flounderings in the mire of delusive analogy, the baseless assumptions, which characterize insane or ‘crank’ productions, are readily found in modern occult literature.

The occult consists of a mixed aggregate of movements and doctrines, which may be the expressions of kindred interests and dispositions but present no essential community of content. Such members of this cluster of beliefs as in our day and generation have attained a considerable adherence or still retain it from former generations constitute the modern occult. The prominent characteristic of the occult is its marked divergence in trend and belief from the recognized standards and achievements of human thought. This divergence is one of attitude and logic and general perspective. It is a divergence of intellectual temperament that distorts the normal reactions to science and evidence and to the general significance and values of the factors of our complicated natures and our equally complicated environment. At least it is this in extreme and pronounced forms; and shades from it through an irregular variety of tints to a vague and often unconscious susceptibility for the unusual and eccentric, combined with an instability of conviction regarding established beliefs that is more often the expression of the weakness of ignorance than of the courage of independence. Occult doctrines are also likely to involve and to proceed upon mysticism and superstition; and their theme centers about such problems as the nature of mental action, the conception of life and death, the effect of cosmic conditions upon human events and endowment, the delineation of character, the nature and treatment of disease, or indeed about any of the larger or smaller realms of knowledge that combine with a strong human and possibly a practical interest, a considerable complexity of basal principles and general relations.

In surveying the more notable instances of the modern occult, it is well while bearing in mind the particular form of occultism or mysticism, or it may be merely of superstition and error, which one or another of the occult movements exhibits, to emphasize the importance of the intellectual motive or temperament that inclines to the occult. It is important to inquire not only what is believed, but what is the nature of the evidence that induces belief, what attracts and then makes converts, what the influences by which the belief spreads. Two classes of motives or interests are conspicuous; the one prominently intellectual or theoretical, the other moderately or grossly practical. Movements in which the former interest dominates contain elements that command respect even when they do not engage sympathy; they frequently appeal, though it may be unwisely, to worthy impulses and lofty aspirations. Amongst the movements presenting prominent practical aspects are to be found instances of the most irreverent and pernicious, as well as of the most vulgar, ignorant and fraudulent schemes which have been devised to mislead the human mind. Most occult movements, however, are of a mixed character, and in their career the speculative and the practical change in importance at different times or in different lands, or at the hands of variously minded leaders. Few escape and some seem especially designed for the partisanship of that class who are seeking whom they may devour; and stimulated by the greed for gain or the love for notoriety, set their snares for the eternally gullible. Fortunately, it must be added that the interest in the occult is under the sway of the law of fashion, and many a mental garment which is donned in spite of the protest of reason and propriety, is quietly laid aside when the dictum of the hour pronounces it unbecoming.

Historically considered, the occult points back to distant epochs and foreign civilizations; to ages when the facts of nature were but weakly grasped, when belief was largely dominated by the authority of tradition, when even the ablest minds fostered or assented to superstition, when the social conditions of life were inimical to independent thought and the mass of men were cut off from intellectual growth of even the most elementary kind. Pseudo-science flourished in the absence of true knowledge, and imaginative insight and unfounded belief held the office intended for inductive reason. Ignorance inevitably led to error and false views to false practices. In a sympathetic environment of this kind the occultist flourished and displayed the impressive insignia of exclusive wisdom. His attitude was that of one seeking to solve an enigma, to find the key to a strange puzzle; his search was for some mystic charm, some talismanic formula, some magical procedure, which shall dispel the mist that hides the face of nature and expose her secrets to his ecstatic gaze. By one all-encompassing, masterful effort the correct solution was to be discovered or revealed; and at once and for all, ignorance would give place to true knowledge, science and nature become as an open book, doubt and despair be replaced by the serenity of perfect wisdom. As our ordinary senses and faculties are obviously insufficient to accomplish such ends, supernatural powers must be appealed to, a transcendental sphere of spiritual activity must be cultivated capable of perceiving through the hidden symbolism of apparent phenomena, the underlying relations of cosmic structure and final purposes. Long periods of training and devotion, seclusion from the world, contemplation of inner mysteries, lead the initiate through the various stages of adeptship up to the final plane of communion with the infinite and the comprehension of truth in all things. This form of occultism reaches its fullest and purest expression in Oriental wisdom-religions. These vie in interest to the historian with the mythology and philosophy of Greece and Rome; and we of the Occident feel free to profit by their ethical and philosophical content, and to cherish the impulses which gave them life. But when such views are forcibly transplanted to our age and clime, when they are decked in garments so unlike their original vestments, particularly when they are associated with dubious practices and come into violent conflict with the truth that has accumulated since they first had birth, their aspect is profoundly altered and they come within the circle of the modern occult.

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Of this character is Theosophy, an occult movement brought into recent prominence by the works and personality of Mme. Blavatsky. The story of the checkered career of that remarkable woman is fairly accessible. Born in Russia in 1831 as Helen Petrovna, daughter of Colonel Hahn, of the Russian army, she was married at the age of seventeen to an elderly gentleman, M. Blavatsky. She is described in girlhood as a person of passionate temper and wilful and erratic disposition. She separated or escaped from her husband after a few months of married life and entered upon an extended period of travel and adventure, in which ‘psychic’ experiences and the search for unusual persons and beliefs were prominent. She absorbed Hindu wisdom from the adepts of India; she sat at the feet of a thaumaturgist at Cairo; she journeyed to Canada to meet the medicine man of the Red Indians, and to New Orleans to observe the practices of Voodoo among the negroes. It is difficult to know what to believe in the accounts prepared by her enthusiastic followers. Violations of physical laws were constantly occurring in her presence, and “sporadic outbreaks of rappings and feats of impulsive pots, pans, beds and chairs insisted on making themselves notorious.” In 1873 she came to New York and sat in ‘spiritualistic’ circles, assuming an assent to their theories, but claiming to see through and beyond the manifestations the operations of her theosophic guides in astral projection. At one of these séances she met Colonel Olcott and assisted him in the foundation of the Theosophical Society in New York in October, 1875. Mme. Blavatsky directed the thought of this society to the doctrines of Indian occultism, and reported the appearance in New York of a Hindu Mahatma, who left a turban behind him as evidence of his astral visit. Later Mme. Blavatsky and Colonel Olcott (who remained her staunch supporter, but whom she referred to in private as a ‘psychologized baby’) went to India and at Adyar established a shrine from which were mysteriously issued answers to letters placed within its recesses, from which inaccessible facts were revealed and a variety of interesting marvels performed. Discords arose within her household and led to the publication by M. and Mme. Coulomb, her confederates, of letters illuminating the tricks of the trade by which the miracles had been produced. Mme. Blavatsky pronounced the letters to be forgeries, but they were sufficiently momentous to bring Mr. Hodgson to India to investigate for the Society for Psychical Research. He was able to deprive many of the miracles of their mystery, to show how the ‘shrine’ from which the Mahatma’s messages emanated was accessible to Mme. Blavatsky by the aid of sliding panels and secret drawers, to show that these messages were in style, spelling and handwriting the counterpart of Mme. Blavatsky’s, to show that many of the phenomena were the result of planned collusion and that others were created by the limitless credulity and the imaginative exaggeration of the witnesses—‘domestic imbeciles,’ as madame confidentially called them. The report of the society convicted ‘the

Priestess of Isis' of "a long continued combination with other persons to produce by ordinary means a series of apparent marvels for the support of the Theosophic movement"; and concludes with these words: "For our own part, we regard her neither as the mouthpiece of hidden seers nor as a mere vulgar adventuress; we think that she has achieved a title to permanent remembrance as one of the most accomplished, ingenious and interesting impostors in history." Mme. Blavatsky died in 1891, and her ashes were divided between Adyar, London and New York.

The Theosophic movement continues, though with abated vigor, owing partly to the above-mentioned disclosures, but probably more to the increasing propagandism of other cults, to the lack of a leader of Mme. Blavatsky's genius, or to the inevitable ebb and flow of such interests. Mme. Blavatsky continued to expound Theosophy after the exposures, and Mrs. Besant, Mr. Sinnett and others were ready to take up the work at her death. However, miracles are no longer performed, and no immediately practical ends are proclaimed. Individual development and evolution, mystic discourses on adeptship and Karma and Maya and Nirvana, communion with the higher ends of life, the cultivation of an esoteric psychic insight, form the goal of present endeavor. The Mahatmas are giving "intellectual instructions, enormously more interesting than even the exhibition of their abnormal powers." ... The modern Theosophist seeks to appeal to men and women of philosophical inclinations, for whom an element of mysticism has its charm, and who are intellectually at unrest with the conceptions underlying modern science and modern life. Such persons are quite likely to be well-educated, refined and sincere. We may believe them intellectually misguided; we may recognize the fraud to which their leader resorted to glorify her creed, but we must equally recognize the absence of many pernicious tendencies in their teachings which characterize other and more practical occult movements.

Spiritualism, another member of the modern occult family, presents a combination of features rather difficult to portray; but its public career of half a century has probably rendered its tenets and practices fairly familiar. For, like other movements, it presents both doctrines and manifestations, and, like other movements, it achieved its popularity through its manifestations and emphasized the doctrines to maintain the interest and solidarity of its numerous converts. Deliberate fraud has been repeatedly demonstrated in a large number of alleged ‘spiritualistic’ manifestations; in many more the very nature of the phenomena and of the conditions under which they appear is so strongly suggestive of trickery as to render any other hypothesis of their origin improbable and unnecessary. Unconscious deception, exaggerated and distorted reports, defective and misleading observation have been demonstrated to be most potent reagents, whereby alleged miracles are made to throw off their mystifying envelopings and to leave a simple deposit of intelligible and often commonplace fact. That the methods of this or that medium have not been brought within the range of such explanation may be admitted, but the admission carries with it no bias in favor of the spiritualistic hypothesis. It may be urged, however, that where there is much smoke there is apt to be some fire; yet there is little prospect of discovering the nature of the fire until the smoke has been completely cleared away. Perhaps it has been snatched from heaven by a materialized Prometheus; perhaps it may prove to be the trick of a *ridiculus mus* gnawing at a match. However, the main point to be insisted upon with regard to such manifestations is that their interpretation and their explanation demand technical knowledge and training, or at least special adaptability to such pursuits. “The problem cannot be solved and settled by amateurs, nor by ‘common sense’ that

> Delivers brawling judgments all day long,
> On all things unashamed.”

Spiritualism represents a systematization of popular beliefs and superstitions, modified by echoes of religious and philosophical doctrines; and is thus not wholly occult. Its main purpose was to establish the reality of communication with departed spirits; the means which at first spontaneously presented themselves and later were devised for this purpose were in large measure not original. The rappings are in accord with the traditional folk-lore behavior of ghosts, though their transformation into a signal code may have been due to the originality of the Fox children; the planchette has its analogies in Chinese and European modes of divination; clairvoyance was incorporated from the phenomena of artificial somnambulism, as practiced by the successors of Mesmer; the ‘sensitive’ or ‘medium’ suggests the same origin as well as the popular belief in the gift of supernatural powers to favored individuals; others of the phenomena such as ‘levitation’ and ‘cabinet performances’ have counterparts in Oriental magic; ‘slate-writing,’ ‘form materializations,’ ‘spirit-messages’ and ‘spirit photographs’ are, in the main, modern contributions. These various phenomena as ordinarily presented breed the typical atmosphere of the séance chamber, which resists precise analysis, but in which it is easy to detect morbid credulity, blind prepossession and emotional contagion; while the dependence of the phenomena on the character of the medium offers strong temptation alike to shrewdness, eccentricity and dishonesty. On the side of his teachings the spiritualist is likewise not strikingly original. The relations of his beliefs to those that grew about the revelations of Swedenborg, to the speculations of the German ‘pneumatologists’ and to other philosophical doctrines, though perhaps not intimate, are yet traceable and interesting; and in another view the ‘spiritualist’ is as old as man himself and finds his antecedents in the necromancer of Chaldea, or in the Shaman of Siberia, or the Angekok of Greenland, or the spirit-doctor of the Karens. The modern mediums are simply repeating with new costumes and improved scenic effects the mystic drama of primitive man.

Spiritualism thus appeals to a deep-seated craving in human nature, that of assurance of personal immortality and of communion with the departed. Just so long as a portion of mankind will accept material evidence of such a belief, and will even countenance the irreverence, the triviality and the vulgarity surrounding the manifestations, just so long as these persons will misjudge their own powers of detecting how the alleged supernatural appearances were really produced and remain unimpressed by the principles upon which alone a consistent explanation is possible, just so long will spiritualism and kindred delusions flourish.

As to the present-day status of this cult it is not easy to speak positively. Its clientèle has apparently greatly diminished; it still numbers amongst its adherents men and women of culture and education and many more who cannot be said to possess these qualities. There seems to be a considerable class of persons who believe that natural laws are insufficient to account for their personal experiences and those of others, and who temporarily or permanently incline to a spiritualistic hypothesis in preference to any other. Spiritualists of this intellectual temper can, however, form but a small portion of those who are enrolled under its creed. If one may judge by the tone and contents of current spiritualistic literature, the rank and file to which Spiritualism appeals present an unintellectual occult company, credulously accepting what they wish to believe, utterly regardless of the intrinsic significance of evidence or hypothesis, vibrating from one extreme or absurdity to another, and blindly following a blinder or more fanatic leader or a self-interested charlatan. While for the most extravagant and unreasonable expressions of Spiritualism one would probably turn to the literature of a few decades ago, yet the symptoms presented by the Spiritualism of to-day are unmistakably of the same character, and form a complex as characteristic as the symptom-complex of hysteria or epilepsy, and which, *faute de mieux*, may be termed occult. It is a type of occultism of a particularly pernicious character because of its power to lead a parasitic life upon the established growths of religious beliefs and interests, and at the same time to administer to the needs of an unfortunate but widely prevalent passion for special signs and omens and the interpretation of personal experiences. It is a weak though comprehensible nature that becomes bewildered in the presence of a few experiences that seem homeless among the generous provisions of modern science, and runs off panic-stricken to find shelter in a system that satisfies a narrow personal craving at the sacrifice of broadly established principles, nurtured and grown strong in the hardy and beneficent atmosphere of science. It is a weaker and an ignorant nature that is attracted to the cruder forms of such beliefs, be it by the impulsive yielding to emotional susceptibility, by the contagion of an unfortunate mental environment, or by the absence of the steadying power of religious faith or of logical vigor or of confidence in the knowledge of others. Spiritualism finds converts in both camps and assembles them under the flag of the occult.[A]

[A] To prevent misunderstanding it is well to repeat that I am speaking of the general average of thorough-going spiritualists. The fact that a few mediums have engaged the attention of scientifically minded investigators has no bearing on the motives which lead most persons to make a professional call on a medium, or to join a circle. The further fact that these investigators have at times found themselves baffled by the medium's performances, and that a few of them have announced their readiness to accept the spiritualistic hypothesis is of importance in some aspects, but does not determine the general trend of the spiritualistic movement in the direction in which it is considered in the present discussion. It may also prevent misunderstanding of other parts of my presentation to continue this footnote by adding that I desire to distinguish sharply between the occult and what has unwisely been termed Psychical Research—unwisely because such research is either truly psychological and requires no differentiation from other allied and legitimate research, or it is something other than psychological which is inaptly expressed by calling it 'psychical.' I admit and emphasize that the majority of such research is the result of a scientific motive and is far removed from the occult. I therefore shall say nothing of Psychical Research and regret that it is necessary even to deny its possible inclusion in the occult. Such inclusion is, however, suggested by much that is talked of and written under the name of Psychic Research, and there can be no doubt that the interest of many members of Psychic Research Societies and of readers of their publications, is essentially of an occult nature. Whatever in these publications seems to favor mystery and to substantiate supernormal powers is readily absorbed, and its bearings fancifully interpreted and exaggerated; the more critical and successfully explanatory papers meet with a less extended and less sensational reception. Unless most wisely directed, Psychic Research is likely, by not letting the right hand know what the left hand is doing, to foster the undesirable propensities of human nature as rapidly as it antagonizes them. Like indiscriminate alms-giving it has the possibilities of affording relief and of making paupers at the same time. While I regard the acceptance of telepathy as an established phenomenon, as absolutely unwarranted and most unfortunate, and

> while I feel a keen personal regret that men whose ability and opinions I estimate highly have announced their belief in a spiritualistic explanation of their personal experiences with a particular medium, yet my personal regret and my logical disapproval of these conclusions have obviously no bearing upon the general questions under discussion. The scientific investigation of the same phenomena which have formed the subject matter of occult beliefs, is radically different in motive, method and result from the truly occult.

The wane in the popularity of Spiritualism may be due in part to frequent exposures, in part to the passing of the occult interest to pastures new, and in part to other and less accessible causes. Such interest may again become dominant by the success or innovations of some original medium or by the appearance of some unforeseen circumstances; at present there is a disposition to take up 'spiritual healing' and 'spiritual readings of the future' rather than mere assurances from the dead, and thus to emulate the practical success of more recently established rivals. The history of Spiritualism, by its importance and its extravagance of doctrine and practice, forms an essential and an instructive chapter in the history of belief; and there is no difficulty in tracing the imprints of its footsteps on the sands of the occult.

The impress of ancient and mediæval lore upon latter-day occultism is conspicuous in the survivals of Alchemy and Astrology. Phrenology represents a more recent pseudo-science, but one sufficiently obsolete to be considered under the same head, as may also Palmistry, which has relations both to an ancient form of divination and to a more modern development after the manner of Physiognomy. The common characteristic of these is their devotion to a practical end. Alchemy occupies a somewhat distinct position. The original alchemists sought the secret of converting the baser metals into gold, in itself a sufficiently alluring and human occupation. There is no reason why such a problem should assume an occult aspect, except the sufficient one that ordinary procedures have not proved capable to effect the desired end. It is a proverbial fault of ambitious inexperience to attack valiantly large problems with endless confidence and sweeping aspiration. It is well enough in shaping your ideas to hitch your wagon to a star; yet the temporary utility of horses need not be overlooked; but shooting arrows at the stars is apt to prove an idle pastime. If we are willing to forget for the moment that the same development of logic and experiment that makes possible the mental and material equipment of the modern chemist makes impossible his consideration of the alchemist's search, we may note how far the inherent constitution of the elements, to say nothing of their possible transmutation, has eluded his most ultimate analysis. How immeasurably farther it was removed from the grasp of the alchemist can hardly be expressed. But this is a scientific and not an occult view of the matter; it was not by progressive training in marksmanship that the occultist hoped to send his arrows to the stars. His was a mystic search for the magical transmutation, the elixir of life or the philosopher's stone. One might suppose that once the world has agreed that these ends are past finding out, the alchemist, like the maker of stone arrow-heads, would have found his occupation gone and have left no successor. His modern representative, however, is an interesting and by no means extinct species. He seems to flourish in France, but may be found in Germany, in England and in this country. He is rarely a pure alchemist (although so recently as 1854 one of them offered to manufacture gold for the French mint), but represents the pure type of occultist. He calls himself a Rosicrucian; he establishes a university of the higher studies and becomes a Professor of Hermetic Philosophy. His thought is mystic, and symbolism has an endless fascination for him. The mystic significance of numbers, extravagant analogies of correspondence, the traditional hidden meanings of the Kabbalah fairly intoxicate him; and verbose accounts of momentous relations and of unintelligible discoveries run riot in his writings. His science is not a mere Chemistry, but a Hyper-Chemistry; his transmutations are not merely material but spiritual. Like all followers of an esoteric belief, he must stand apart from his fellow-men; he must cultivate the higher 'psychic' powers so that eventually he may be able by the mere action of his will to cause the atoms to group themselves into gold.

The modern alchemist is, however, a general occultist; he may be also an astrologer or a magnetist or a theosophist. But he is foremost an ardent enthusiast for exclusive and unusual lore—not the common and superficial possessions of misguided democratic science. He goes through the forms of study, remains superior to the baser practical ends of life, and finds his reward in the self-satisfaction of exclusive wisdom. In Paris, at least, he forms part of a rather respectable salon, speaking socially, or a ‘company of educated charlatans,’ speaking scientifically. His class does not constitute a large proportion of modern occultists, but they present a prominent form of its intellectual temperament. “There are also people,” says Mr. Lang, “who so dislike our detention in the prison house of old unvarying laws that their bias is in favor of anything which may tend to prove that science in her contemporary mood is not infallible. As the Frenchman did not care what sort of a scheme he invested money in, provided that it annoys the English, so many persons do not care what they invest belief in, provided that it irritates men of science.” Of such is the kingdom of alchemists and their brethren.

Astrology, phrenology, physiognomy and palmistry have in common a search for knowledge whereby to regulate the affairs of life, to foretell the future, to comprehend one’s destiny and capabilities. They aim to secure success or at least to be forearmed against failure by being forewarned. This is a natural, a practical, and in no essential way, an occult desire. It becomes occult, or better, superstitious, when it is satisfied by appeals to relations and influences which do not exist, and by false interpretation of what may be admitted as measurably and vaguely true and about equally important. When not engaged in their usual occupation of building most startling superstructures on the weakest foundations, practical occultists are like Dr. Holmes’ katydid, “saying an undisputed thing in such a solemn way.” They will not hearken to the experience of the ages that success cannot be secured nor character read by discovering their mystic stigmata; they will not learn from physiology and psychology that the mental capabilities, the moral and emotional endowment of an individual are not stamped on his body so that they may be revealed by half an hour’s use of the calipers and tape-measure; they will not listen when science and common sense unite in teaching that the knowledge of mental powers is not such as may be applied by rule of thumb to individual cases, but that like much other valuable knowledge, it proceeds by the exercise of sound judgment, and must as a rule rest content with suggestive generalizations and imperfectly established correlations. An educated man with wholesome interests and a vigorous logical sense can consider a possible science of character and the means of aiding its advance without danger and with some profit. But this meat is sheer poison to those who are usually attracted to such speculations, while it offers to the unscrupulous charlatan a most convenient net to spread for the unwary. In so far as these occult mariners, the astrologists and phrenologists and *id genus omne* are sincere, and in so far represent superstition rather than commercial fraud, they simply ignore through obstinacy or ignorance the light-houses and charts and the other aids to modern navigation, and persist in steering their craft by an occult compass. In some cases they are professedly setting out, not for any harbor marked on terrestrial maps, but their expedition is for the golden fleece or for the apples of the Hesperides; and with loud-voiced advertisements of their skill as pilots, they proceed to form stock companies for the promotion of the enterprise and to sell the shares to credulous speculators.

It would be a profitless task to review the alleged data of astrology or phrenology or palmistry except for the illustrations which they readily yield of the nature of the conceptions and the logic which command a certain popular interest and acceptance. The interest in these notions, is, as Mr. Lang argues about ghosts and rappings and bogles, in how they come to be believed rather than in how much or how little they chance to be true. In examining the professed evidence for the facts and laws and principles (*sit venia verbis*) that pervade astrology or phrenology or palmistry or dream-interpretation, or beliefs of that ilk, we find the flimsiest kind of texture that will hardly bear examination and holds together only so long as it is kept secluded from the light of day. Far-fetched analogy, baseless assertion, the uncritical assimilation of popular superstitions, a great deal of prophecy after the event—it is wonderful how clearly the astrologer finds the indications of Napoleon’s career in his horoscope, or the phrenologist reads them in the Napoleonic cranial protuberances—much fanciful elaboration of detail, ringing the variations on a sufficiently complex and non-demonstrable proposition, cultivating a convenient vagueness of expression together with an apologetic skill in providing for and explaining exceptions, the courage to ignore failure and the shrewdness to profit by coincidences and half-assimilated smatterings of science; and with it all an insensibility to the moral and intellectual demands of the logical decalogue, and you have the skeleton which clothed with one flesh becomes astrology, and with another phrenology and with another palmistry or solar biology or descriptive mentality or what not. Such pseudo-sciences thrive upon that widespread and intense craving for practical guidance of our individual affairs, which is not satisfied with judicious applications of general principles, with due consideration of the probabilities and uncertainties of human life, but demands an impossible and precise revelation. Not all that passes for, and in a way is, knowledge, is or is likely soon to become scientific; and when a peasant parades in an academic gown the result is likely to be a caricature.

To achieve fortune, to judge well and command one's fellow-men, to foretell and control the future, to be wise in worldly lore, are natural objects of human desire; but still another is essential to happiness. Whether we attempt to procure these good fortunes by going early to bed and early to rise, or by more occult procedures, we wish to be healthy as well as wealthy and wise. The maintenance of health and the perpetuity of youth were not absent from the mediæval occultist's search, and formed an essential part of the benefits to be conferred by the elixir of life and the philosopher's stone. A series of superstitions and extravagant systems are conspicuous in the antecedents and the bye-paths of the history of medicine, and are related to it much as astrology is to astronomy or alchemy to chemistry; and because medicine in part remains, and to previous generations was conspicuously an empirical art rather than a science, it offers great opportunity for practical error and misapplied partial knowledge. It is not necessary to go back to early civilizations or to primitive peoples, among whom the medicine-man and the priest were one and alike appealing to occult powers, nor to early theories of disease which beheld in insanity the obsession of demons and resorted to exorcism to cast them out; it is not necessary to consider the various personages who acquired notoriety as healers by laying on of hands or by appeal to faith, or who like Mesmer introduced the system of Animal Magnetism, or like some of his followers, sought directions for healing from the clairvoyant dicta of somnambules; it is not necessary to ransack folk-lore superstitions and popular remedies for the treatment of disease; for the modern forms of 'irregular' healing offer sufficient illustrations of occult methods of escaping the ills that flesh is heir to.

The existence of a special term for a medical impostor is doubtless the result of the prevalence of the class thus named, but quackery and occult medicine though mutually overlapping, can by no means be held accountable for one another's failings. Many forms of quackery proceed on the basis of superstitions or fanciful or exaggerated notions containing occult elements, but for the present purpose it is wise to limit attention to those in which this occult factor is distinctive; for medical quackery in its larger relations is neither modern nor occult. Occult healing takes its distinctive character from the theory underlying the practice rather than from the nature of the practice. It is not so much what is done as why it is done or pretended to be done or not done, that determines its occult character. A factor of prominence in modern occult healing is indeed one that in other forms characterized many of its predecessors and was rarely wholly absent from the connection between the procedure and the result; this is the mental factor, which may be called upon to give character to a theory of disease, or be utilized consciously or unconsciously as a curative principle. It is not implied that ‘mental medicine’ is necessarily and intrinsically occult, but only that the general trend of modern occult notions regarding disease may be best portrayed in certain typical forms of ‘psychic’ healing. The legitimate recognition of the importance of mental conditions in health and disease is one of the results of the union of modern psychology and modern medicine. An exaggerated and extravagant as well as pretentious and illogical over-statement and misstatement of this principle may properly be considered as occult.

Among such systems there is one which by its momentary prominence overshadows all others, and for this reason as well as for its more explicit or rather extended statement of principles, must be accorded special attention. I need hardly say that I refer to that egregious misnomer, Christian Science. This system is said to have been discovered by or revealed to Mrs. Mary Baker Glover Eddy in 1866. Several of its most distinctive positions (without their religious setting) are to be found in the writings and were used in the practice of Mr. or Dr. P. P. Quimby (1802–1866), whom Mrs. Eddy professionally consulted shortly before she began her own propagandum. On its theoretical side the system presents a series of quasi-metaphysical principles, and also a professed interpretation of the Scriptures; on its practical side it offers a means of curing or avoiding disease and includes under disease also what is more generally described as sin and misfortune. With Christian Science as a religious movement I shall not directly deal; I wish, however, to point out that this assumption of a religious aspect finds a parallel in Spiritualism and Theosophy and doubtless forms one of the most potent reasons for the success of these occult movements. It would be a most dangerous principle to admit that the treatment of disease and the right to ignore hygiene can become the perquisite of any religious faith. It would be equally unwarranted to permit the principles which are responsible for such beliefs to take shelter behind the ramparts of religious tolerance; for the essential principles of Christian Science do not constitute a form of Christianity any more than they constitute a science; but in so far as they do not altogether elude description, pertain to the domain over which medicine, physiology and psychology hold sway. As David Harum, in speaking of his church-going habits, characteristically explains, "the one I stay away from when I don't go's the Prespyteriun," so the doctrines which Christian Science 'stays away from' are those over which recognized departments of academic learning have the authority to decide.

Mrs. Eddy's magnum opus serving at once as the text-book of the 'science' and as a revised version of the Scriptures—Science and Health, with Key to the Scriptures—has been circulated to the extent of one hundred and seventy thousand copies. I shall not give an account of this book nor subject its more tangible tenets to a logical review; I must be content to recommend its pages as suggestive reading for the student of the occult and to set forth in the credentials of quotation marks some of the dicta concerning disease. Yet it may be due to the author of this system to begin by citing what are declared to be its fundamental tenets, even if their connection with what is built upon them is far from evident.

> "The fundamental propositions of Christian Science are summarized in the four following, to me *self-evident* propositions. Even if read backward, these propositions will be found to agree in statement and proof:
>
> 1. God is All in all.
>
> 2. God is good. Good is Mind.
>
> 3. God, Spirit, being all, nothing is matter.
>
> 4. Life, God, omnipotent Good, deny death, evil, sin, disease—Disease, sin, evil, death, deny Good, omnipotent God, Life."
>
> "What is termed disease does not exist." "Matter has no being." "All is mind." "Matter is but the subjective state of what is here termed *mortal mind*." "All disease is the result of education, and can carry its ill-effects no farther than mortal mind maps out the way."

“The fear of dissevered bodily members, or a belief in such a possibility, is reflected on the body, in the shape of headache, fractured bones, dislocated joints, and so on, as directly as shame is seen rising to the cheek. This human error about physical wounds and colics is part and parcel of the delusion that matter can feel and see, having sensation and substance.” “Insanity implies belief in a diseased brain, while physical ailments (so-called) arise from belief that some other portions of the body are deranged.... A bunion would produce insanity as perceptible as that produced by congestion of the brain, were it not that mortal mind calls the bunion an unconscious portion of the body. Reverse this belief and the results would be different.” “We weep because others weep, we yawn because they yawn, and we have small-pox because others have it; but mortal mind, not matter, contains and carries the infection.” “A Christian Scientist never gives medicine, never recommends hygiene, never manipulates.” “Anatomy, Physiology, Treatises on Health, sustained by what is termed material law, are the husbandmen of sickness and disease.” “You can even educate a healthy horse so far in physiology that he will take cold without his blanket.” “If exposure to a draught of air while in a state of perspiration is followed by chills, dry cough, influenza, congestive symptoms in the lungs, or hints of inflammatory rheumatism, your Mind-remedy is safe and sure. If you are a Christian Scientist, such symptoms will not follow from the exposure; but if you believe in laws of matter and their fatal effects when transgressed, you are not fit to conduct your own case or to destroy the bad effects of belief. When the fear subsides and the conviction abides that you have broken no law, neither rheumatism, consumption nor any other disease will ever result from exposure to the weather.” “Destroy fear and you end the fever.” “To prevent disease or cure it mentally let spirit destroy the dream of sense. If you wish to heal by argument, find the type of the ailment, get its name and array your mental plea against the physical. Argue with the patient (mentally, not audibly) that he has no disease, and conform the argument to the evidence. Mentally insist that health is the everlasting fact, and sickness the temporal falsity. Then realize the presence of health and the corporeal senses will respond, so be it.” “My publications alone heal

> more sickness than an unconscientious student can begin to reach." "The quotient when numbers have been divided by a fixd rule, are not more unquestionable than the scientific tests I have made of the effects of truth upon the sick." "I am never mistaken in my scientific diagnosis of disease." "Outside of Christian Science all is vague and hypothetical, the opposite of Truth." "Outside Christian Science all is error."

Surely this is a remarkable product of mortal mind! It would perhaps be an interesting *tour de force*, though hardly so entertaining as 'Alice in Wonderland,' to construct a universe on the assertions and hypotheses which Christian Science presents; but it would have less resemblance to the world we know than has Alice's Wonderland. For any person for whom logic and evidence are something more real than ghosts or myths, the feat must always be relegated to the airy realm of the imagination and must not be brought in contact with earthly realities. And yet the extravagance of Mrs. Eddy's book, its superb disdain of vulgar fact, its transcendental self-confidence, its solemn assumption that reiteration and variation of assertion somehow spontaneously generate proof or self-evidence, its shrewd assimilation of a theological flavor, its occasional successes in producing a presentable travesty of scientific truth—all these distinctions may be found in many a dust-covered volume, that represents the intensity of conviction of some equally enthusiastic and equally inspired occultist, but one less successful in securing a chorus to echo his refrain.

I cannot dismiss 'Eddyism' without illustrating the peculiar structures under which, in an effort to be consistent, it is forced to take shelter. Since disease is always of purely mental origin, it follows that disease and its symptoms cannot ensue without the conscious coöperation of the patient; since "Christian Science divests material drugs of their imaginary power," it follows that the labels on the bottles that stand on the druggist's shelves are correspondingly meaningless. And it becomes an interesting problem to inquire how the consensus of mortal mind came about that associates one set of symptoms with prussic acid, and another with alcohol, and another with quinine. Inhaling oxygen or common air would prepare one for the surgeon's knife, and prussic acid or alcohol have no more effect than water, if only a congress of nations would pronounce the former to be anæsthetic and promulgate a decree that the latter shall be harmless. Christian Science does not flinch from this position. “If a dose of poison is swallowed through mistake and the patient dies, even though physician and patient are expecting favorable results, does belief, you ask, cause this death? Even so, and as directly as if the poison had been intentionally taken. In such cases a few persons believe the potion swallowed by the patient to be harmless; but the vast majority of mankind, though they know nothing of this particular case and this special person, believe the arsenic, the strychnine, or whatever the drug used, to be poisonous, for it has been set down as a poison by mortal mind. The consequence is that the result is controlled by the majority of opinions outside, not by the infinitesimal minority of opinions in the sick chamber.” But why should the opinions of οἱ πολλοι {hoi polloi} be of influence in such a case, and the enlightened minorities be sufficient to effect the marvellous cures in all the other cases? Christian Scientists do not take cold in draughts in spite of the contrary opinions or illusions of misguided majorities. The logical Christian Scientist need not eat, “for the truth is food does not affect the life of man,” and should not renounce his faith by adding, “but it would be foolish to venture beyond our present understanding, foolish to stop eating, until we gain more goodness and a clearer comprehension of the living God.” And if he is a mental physician he must be a mental surgeon, too, and not plead that, “Until the advancing age admits the efficacy and supremacy of mind, it is better to leave the adjustment of broken bones and dislocations to the fingers of surgeons.” But it is unprofitable to consider the weakness of any occult system in its encounters with actual science and actual fact. It is simply as a real and prominent menace to rationality that these doctrines naturally attract consideration. As illustrations of present-day occult beliefs we are naturally tempted to inquire what measure of (perverted) truth they may contain; but the more worthy question is, How do such perversions come to find so large a company of ‘supporting listeners’? For to any one who can read and be convinced by the sequence of words of this system, ordinary logic has no power, and to him the world of reality brings no message. No form of the modern occult antagonizes the foundations of science so brusquely as this one. The possibility of science rests on the thorough and absolute distinction between the subjective and the objective. In what measure a man loses the power to draw this distinction clearly and as other men do, in that measure he becomes irrational and insane. The objective exists; and no amount of thinking it away, or thinking it differently, will change it. That is what is understood by ultimate scientific truth; something that will endure unmodified by passing ways of viewing it, open to every one's verification who can come equipped with the proper means to verify—a permanent objective to be ascertained by careful logical inquiry, not to be determined by subjective opinion. Logic is the language of science; Christian Science and what sane men call science can never communicate because they do not speak the same language.

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It would be unfortunate if in emphasizing the popular preëminence of Christian Science, one were to overlook the significance of the many other forms of 'drugless healing' which bid for public favor by appeal to ignorance and to occult and superstitious instincts. Some are allied to Christian Science and like it assimilate their cult to a religious movement; others are unmistakably the attempts of charlatans to lure the credulous by noisy advertisements of newly discovered and scientifically indorsed systems of 'psychic force,' or some personal 'ism.' For many purposes it would be unjust to group together such various systems, which in the nature of things must include sinner and saint, the misguided sincere, the half-believers who think 'there may be something in it,' or 'that it is worth a trial,' along with scheming quacks and adepts in commercial fraud. They illustrate the many and various roads traveled in the search for health by pilgrims who are dissatisfied with the highways over which medical science goes its steady, though it may be, uncertain gait. Among them there is both plausible exaggeration and ignorant perversion and dishonest libel of the relations that bind together body and mind. Among the several schisms from the Mother Church of Christian Science there is one that claims to be the 'rational phase of the mental healing doctrine,' that acknowledges the reality of disease and the incurability of serious organic disorders and resents any connection with the "half-fanatical personality worship" [of Mrs. Eddy] as quite as foreign to its tenets as would be the views of the 'Free Religious Association' to the 'Pope of Rome.' 'Divine Healing' exhibits its success in one notable instance, in the establishment of a school and college, a bank, a land and investment association, a printing and publishing office and sundry Divine Healing Homes; and this prosperity is now to be extended by the foundation of a city or colony of converts who shall be united by the common bond of faith in divine healing as transmitted in the personal power of their leader. The official organ of this movement announces that the personification of their faith “makes her religion a business and conducts herself upon sound business principles.” With emphatic protest on the part of each that he alone holds the key to salvation, and that his system is quite original and unlike any other, comes the procession of Metaphysical Healer and Mind-Curist and Viticulturist and Magnetic Healer and Astrological Health Guide and Phrenopathist and Medical Clairvoyant and Psychic Scientist and Mesmerist and Occultist. Some use or abuse the manipulations of Hypnotism; others claim the power to concentrate the magnetism of the air and to excite the vital fluids by arousing the proper mental vibrations, or by some equally lucid and demonstrable procedure; some advertise magnetic cups and positive and negative powders and absent treatment by outputs of ‘psychic force’ and countless other imposing devices. In truth, they form a motley crew, and with their ‘Colleges of Fine Forces’ and ‘Psychic Research Companies,’ offering diplomas and degrees for a three weeks’ course of study or the reading of a book, represent the slums of the occult. An account of their methods is likely to be of as much interest to the student of fraud as to the student of opinion.

There can be no doubt that many of these systems have been stimulated into life or into renewed vigor by the success of ‘Christian Science’; this is particularly noticeable in the introduction of absent treatment as a plank in their diverse platforms. This ingenious method of restoring the health of their patients and their own exchequers appealed to all the band of healing occultists from Spiritualist to Vibrationist, as easily adaptable to their several systems. In much the same way Mesmer, more than a hundred years ago, administered to the practice which had exhausted the capacity of his personal attention by magnetizing trees and selling magnetized water. The absent treatment represents the occult ‘extension movement’; and unencumbered by the hampering restrictions of physical forces, superior even to wireless telegraphy, carries its influence into the remotest homes. From ocean to ocean and from North to South these absent healers set apart some hour of the day when they mentally convey their healing word to the scattered members of their flock. On the payment of a small fee you are made acquainted with the ‘soul-communion time-table’ for your longitude and may know when to meet the healing vibrations as they pass by. Others disdain any such temporal details and assure a cure merely on payment of the fee; the healer will know sympathetically when and how to transmit the curative impulses. Poverty and bad habits as well as disease readily succumb to the magic of the absent treatment. Here is the hysterical edict of one of them: 'Join the Success Circle.' ... "The Centre of that Circle is my omnipotent WORD. Daily I speak it. Its vibrations radiate more and more powerfully day by day.... As the sun sends out vibrations ... so my WORD radiates Success to 10,000 lives as easily as to one."

It is impossible to appreciate fully the extravagances of these occult healers unless one makes a sufficient sacrifice of time and patience to read over a considerable sample of the periodical publications with which American occultism is abundantly provided. And when one has accomplished this task he is still at sea to account for the readers and believers who support these various systems so undreamt of in our philosophy. It would really seem that there is no combination of ideas too absurd to fail entirely of a following. Carlyle without special provocation concluded that there were about forty million persons in England, mostly fools; what would have been his comment in the face of this vast array of human folly! If it be urged in rejoinder that beneath all this rubbish heap a true jewel lies buried, that the wonderful cures and the practical success of these various systems indicate their dependence upon an essential and valuable factor in the cure of disease and the formation of habits, it is possible with reservation to assent and with emphasis to demur. Such success, in so far as it is rightly reported, exemplifies the truly remarkable function of the mental factor in the control of normal as of disordered physiological functions. This truth has been recognized and utilized in unobtrusive ways for many generations, and within recent years has received substantial elaboration from carefully conducted experiments and observations. Specifically the therapeutic action of suggestion, both in its more usual forms and as hypnotic suggestion, has shown to what unexpected extent such action may proceed in susceptible individuals. The well-informed and capable physician requires no instruction on this point; his medical education furnishes him with the means of determining the symptoms of true organic disorder, of functional derangement and of the modifications of these under the more or less unconscious interference of an unfortunate nervous system. It is quite as human for the physician as for other mortals to err, and there is doubtless as wide a range among them as among other pursuits, of ability, tact and insight. 'But when all is said and done’ the fundamental fact remains that the utilization of the mental factor in the alleviation of disease will be best administered by those who are specifically trained in the knowledge of bodily and mental symptoms of disease. Such application of an established scientific principle may prove to be a jewel of worth in the hands of him who knows how to cut and set it. The difference between truth and error, between science and superstition, between what is beneficent to mankind and what is pernicious, frequently lies in the interpretation and the spirit as much as or more than in the fact. The utilization of mental influences in health and disease becomes the one or the other according to the wisdom and the truth and the insight into the real relations of things that guide its application. As far removed as chemistry from alchemy, as astronomy from astrology, as the doctrine of the localization of function in the brain from phrenology, as ‘animal magnetism’ from hypnotic suggestion, are the crude and perverse notions of Christian Scientist or Metaphysical Healer removed from the rational application of the influence of the mind over the body.

The growth and development of the occult forms an interesting problem in the psychology of belief. The motives that induce the will to believe in the several doctrines that have been passed in review are certainly not more easy to detect and to describe than would be the case in reference to the many other general problems—philosophical, scientific, religious, social, political or educational—on which the right to an opinion seems to be regarded as an inalienable heritage of humanity or at least of democracy. Professor James tells us that often “our faith is faith in some one else’s faith, and in the greatest matters this is most the case.” Certainly the waves of popularity of one cult and another reflect the potent influence of contagion in the formation of opinion and the direction of conduct. When we look upon the popular delusions of the past through the achromatic glasses which historical remoteness from present conditions enables us to adjust to our eyes, we marvel that humanity could have been so grossly misled, that obvious relations and fallacies could have been so stupidly overlooked, that worthless and prejudiced evidence could have been accepted as sound and significant. But the opinions to which we incline are all colored o’er with the deep tinge of emotional reality, which is the living expression of our interest in them or our inclination toward them. What they require is a more vigorous infusion of the pale cast of thought; for the problem of the occult and the temptations to belief which it holds out are such as can be met only by a vigorous and critical application of a scientific logic. As logical acumen predominates over superficial plausibility, as belief comes to be formed and evidence estimated according to its intrinsic value rather than according to its emotional acceptability, the propagandum of the occult will meet with greater resistance and aversion.

The fixation of belief proceeds under the influence of both general and special forces; the formation of a belief is at once a personal and a social reaction—a reaction to the evidence which recorded and personal experience presents and to the beliefs current in our environment, and this reaction is further modified by the temperament of the reagent. And although individual beliefs, however complex, are neither matters of chance nor are their causes altogether past finding out, yet some of their contributing factors are so vague and so inaccessible that they are most profitably considered as particular results of more or less clearly discerned general principles; and in many respects there is more valid interest in the general principles than in the particular results. It is interesting and it may be profitable to investigate why this area is wooded with oak and that with maple, but it is somewhat idle to speculate why this particular tree happens to be a maple rather than an oak, even if it chances to stand on our property, and to have an interest to us beyond all other trees. It is this false concentration of the attention to the personal and individual result that is responsible for much unwarranted belief in the occult. It is likely that no single influence is more potent in this direction than this unfortunate over-interest in one's own personality and the consequent demand for a precise explanation of one's individual experiences. This habit seems to me a positive vice, and I am glad to find support in Professor James: "The chronic belief of mankind that events may happen for the sake of their personal significance is an abomination." Carried over to the field of subjective experiences, this habit sees in coincidences peculiarly significant omens and portents, not definitely and superstitiously, it may be, but sufficiently to obscure the consideration of the experience in any other than a personal light. The victim of this habit will remain logically unfit to survive the struggle against the occult. Only when the general problem is recognized as more significant for the guidance of belief than the attempted explicit personal explanations will these problems stand out in their true relations. It is interesting to note that the partaking of mince-pie at evening may induce bad dreams, but it is hardly profitable to speculate deeply why my dream took the form of a leering demon with the impolite habit of squatting on my chest. The stuff that dreams are made of is not susceptible of that type of analysis. The most generous allowance must be made for coincidences and irrelevancies, and it must be constantly remembered that the obscure phenomena of psychology, and, indeed, the phenomena of more thoroughly established and intrinsically more definite sciences, cannot be expected to pass the test of detailed and concrete combinations of circumstances. In other classes of knowledge the temptation to demand such explicit explanations of observations and experiences is not so strong because of the absence of an equally strong personal interest; but that clearly does not affect the logical status of the problem. The reply to this argument I can readily anticipate; and I confess that my admiration of Hamlet is somewhat dulled by reason of that ill-advised remark to Horatio about there being more things in heaven and earth than are dreamt of in our philosophies. The occultist always seizes upon that citation to refute the scientist. He prints it as his motto on his books and journals, and regards it as a slow poison that will in time effect the destruction of the rabble of scientists and reveal the truth of his own Psycho-Harmonic Science or Heliocentric Astrology. It is one thing to be open-minded and to realize the incompleteness of scientific knowledge and to appreciate how often what was ignored by one generation has become the science of the next; and it is a very different thing to be impressed with coincidences and dreams and premonitions, and to regard them as giving the keynote to the conceptions of nature and reality, and to look upon science as a misdirected effort. Such differences of attitude depend frequently upon a difference of temperament as well as upon intellectual discernment; the man or the woman who flies to the things not dreamt of in our philosophy quite commonly does not understand the things which our philosophy very creditably accounts for. The two types of mind are different, and (I am again citing Professor James) “the scientific-academic mind and the feminine-mystical mind shy from each other’s facts just as they fly from each other’s temper and spirit.”

Certain special influences combine with these fundamental differences of attitude to favor the spread of belief in the occult; and of these the character of the beliefs as of the believers furnish some evidence. At various stages of the discussion I have referred to the deceptive nature of the argument by analogy; to the dominating sympathy with a conclusion and the resulting assimilation and overestimation of apparent evidence in its favor;

to the frequent failure to understand that the formation of valid opinion and the interpretation of evidence in any field of inquiry require somewhat of expert training and special aptitude, obviously so in technical matters, but only moderately less so in matters misleadingly regarded as general; to bias and superstition, to the weakness that bends easily to the influences of contagion, to unfortunate educational limitations and perversions and, not the least, to a defective grounding in the nature of scientific fact and proof. The mystery attaching to the behavior of the magnet led Mesmer to call his curative influence ‘animal magnetism’—a conception that still prevails among latter-day occultists. The principle of sympathetic vibration, in obedience to which a tuning-fork takes up the vibrations of another in unison with it, is violently transferred to imaginary brain vibrations and to still more imaginary telepathic currents. The X-ray and wireless telegraphy are certain to be utilized in corroboration of unproven modes of mental action, and will be regarded as the key to clairvoyance and rapport, just as well-known electrical phenomena have given rise to the notions of positive and negative temperaments and mediumistic polar attraction and repulsion. All this results from the absurd application of analogies; for analogies even when appropriate are little more than suggestive or at least corroborative of relations or conceptions which owe their main support to other and more sturdy evidence. Analogy under careful supervision may make a useful apprentice, but endless havoc results when the servant plays the part of the master.

No better illustrations could be desired of the effects of mental prepossession and the resulting distortion of evidence and of logical insight, than those afforded by Spiritualism and Christian Science. In both these movements the assimilation of a religious trend has been of inestimable importance to their dissemination. Surely it is not merely or mainly the evidences obtainable in the séance chamber, nor the irresistible accumulation of cures by argument and thought-healings, that account for the organized gatherings of Spiritualists and the costly temples and thriving congregations of Christ Scientist. It is the presentation of a practical doctrine of immortality and of the spiritual nature of disease in conjunction with an accepted religious system, that is responsible for these vast results. The ‘Key to the Scriptures’ has immeasurably reinforced the ‘Science and Health,’ and brought believers to a new form of Christianity who never would have been converted to a new system of medicine presented on purely intellectual grounds. Rationality is doubtless a characteristic tendency of humanity, but logicality is an acquired possession and one by no means firmly established in the race at large. So long as we are reproved by the discipline of nature and that rather promptly, we tend to act in accordance with the established relations of things; and that is rationality. But the more remote connections between antecedent and consequent and the development of habits of thought which shall lead to reliable conclusions in complex situations; and again, the ability to distinguish between the plausible and the true, the firmness to support principle in the face of paradox and seeming non-conformity, to think clearly and consistently in the absence of the practical reproof of nature—that is logicality. It is only as the result of a prolonged and conscientious training aided by an extensive experience and a knowledge of the historical experience of the race, that the inherent rational tendencies develop into established logical habits and principles of belief. For many this development remains stunted or arrested; and they continue as children of a larger growth, leaning much on others, rarely venturing abroad alone and wisely confining their excursions to familiar ground. When they unfortunately become possessed with the desire to travel, their lack of appreciation of the sights which their journeys bring before them gives to their reports the same degree of reliability and value as attaches to the much ridiculed comments of the philistine *nouveaux riches*.

For these sufficient reasons it is Utopian to look forward to the day when the occult shall have disappeared, and the lion and the lamb shall feed and grow strong on the same nourishment. Doubtless new forms and phases of the occult will arise to take the place of the old as their popularity declines; and the world will be the more interesting and more characteristically a human dwelling-place for containing all sorts and conditions of minds. None the less it is the plain duty and privilege of each generation to utilize every opportunity to dispel error and superstition, and to oppose the dissemination of irrational beliefs. It is particularly the obligation of the torch-bearers of science to illuminate the path of progress and to transmit the light to their successors with undiminished power and brilliancy; this flame must burn both as a beacon-light to guide the wayfarer along the highways of advance and as a warning against the will-of-the-wisps that shine seductively in the bye-ways. The safest and most efficient antidote to the spread of the pernicious tendencies inherent in the occult lies in the cultivation of a wholesome and whole-souled interest in the genuine and profitable problems of nature and of life, and in the cultivation with it of a steadfast adherence to common sense and to a true logical perspective of the significance and value of things. These qualities, fortunately for our forefathers, are not the prerogative of the modern; and, fortunately for posterity, are likely to remain characteristic of the scientific and antagonistic to the occult.

BIRDS AS FLYING MACHINES.

By FREDERICK A. LUCAS.

FROM the day of Solomon onward the way of a bird in the air has been a subject of general interest, and the attention given to the problem of aerial navigation of late years has caused the flight of birds to be carefully studied in the hope that it might throw some light on the subject. There have been many conceptions, not to say misconceptions, regarding the flight of birds; it has been assumed that their muscles exerted a power quite beyond that of other animals, that the air sacs of some birds and the hollow bones of others gave them a degree of lightness quite unattainable by the use of ordinary materials, while some have even gone so far as to suggest the presence of some mysterious power, something like Stockton's negative gravity, whereby birds could set at naught the law of gravitation and rise at will like a balloon. The strength of a bird's muscles, of some birds' muscles at least, is not to be underrated; a hawk will plant its talons in a bird of nearly its own size and weight and bear the victim bodily away, and an osprey will carry a fish for a long distance. But a tiger has been known to fell a bullock with a single blow of the paw, to carry a man as a cat would carry a rat and to drag an Indian buffalo heavier than himself. On the other hand, some of the petrels, birds which can pass a day or so on the wing with ease, cannot rise from the water after a hearty meal, and the humming-bird, unsurpassed in aerial evolutions, may be trapped in a spider's web. This shows no great power, and long ago Marey found that the pulling force of a hawk's great breast muscle, applied through the humerus, amounted to 1,298 grams per square centimeter, something like seven pounds to the square inch; not a very heavy pull. So it seems fair to assume that while the power exerted by a bird is great, it is very far from marvelous, probably far less in proportion to size than the engine of Maxim's great aeroplane, or the naphtha motor of Professor Langley, which weighs less than ten pounds per horse power. We may get a fair idea of what this means by remembering that a bald eagle weighs from nine to fifteen pounds and that he exerts but a small fraction of a horse power.

Turning to the question of the part played by the air sacs it may be said that their value is not proved; some of the fastest birds get along without them, while birds of the most labored flight are sometimes well provided. In birds like the gannet and brown pelican the air sacs and cellular tissue about the body undoubtedly serve as buffers to break the shock of a headlong plunge into the water from a height of a hundred to a hundred and fifty feet. Or, again, they equalize the internal and external pressure when a soaring bird drops suddenly from a great height, or still more often aid in oxygenating the blood.

The hollow bones of birds are frequently cited as beautiful instances of providential mechanics in building the strongest and largest possible limb with the least expenditure of material, and this is largely true. And yet birds like ducks, which cleave the air with the speed of an express train, have the long bones filled with marrow or saturated with fat, while the lumbering hornbill that fairly hurtles over the tree tops has one of the most completely pneumatic skeletons imaginable, permeated with air to the very toe tips; and the ungainly pelican is nearly as well off. Still it is but fair to say that the frigate bird and turkey buzzards, creatures which are most at ease when on the wing, have extremely light and hollow bones, but comparing one bird with another the paramount importance of a pneumatic skeleton to a bird is not as evident as that of a pneumatic tire to a bicycle.

While it may not be easy to disprove Herr Gätke’s assertion that birds sustain themselves in the air by the exercise of some power beyond our own, it is pretty safe to assume that they do not, and it would seem that the burden of proof should lie with those who take the affirmative side of the question.

If we have nothing to learn from birds in the way of building an engine that shall exert great power for its size and weight we may still have something to gain in the matter of speed, although here the popular idea is apt to be exaggerated. We often read that ducks fly at the rate of a mile a minute, or that the swallow has a speed of two hundred miles an hour, but it is very difficult to lay hands upon any facts that will sustain these assertions.

So, too, homing pigeons are frequently stated to have travelled for long distances at the rate of sixty miles an hour, but some of the published records show that one hundred and twenty miles in two hours and a quarter is unusually fast traveling, and this is at the rate of only nine-tenths of a mile per minute, a speed not unusual for express trains. However, it may be said that actual observations show that ducks do travel from forty to fifty miles an hour, and any sportsman will readily believe that under some conditions they attain a velocity of a mile and a quarter a minute, although a confession of faith is not a demonstration of an assured fact.

So far the lesson taught by the bird is that a machine of low power may attain a very considerable speed and it remains to be seen if there is anything to be learned concerning methods of flight. Broadly speaking, there are two, possibly three, distinct modes of flight, by repeated strokes of the wings and by soaring or sailing, although we find every intermediate stage between the two, or combinations of flapping and sailing, and as a matter of fact no bird can entirely dispense with strokes of the wing.

The humming-bird represents the perfection of one method, the frigate bird of the other, and in his own line each is unrivaled. These two modes of flight are associated with equally distinct modifications of structure, and just as we have every intermediate state of flight between flapping and soaring so the two structural extremes are merged into one another. The humming-bird flies as the Irishman played the fiddle, by main strength, the frigate bird relies on his skill in taking advantage of every varying current of air, and the skeleton of the one indicates great muscular power while that of the other shows its absence. No other bird has such proportionately great muscles as the humming-bird, the keel of the sternum or breast bone from which these muscles arise runs from one end of the body to the other while at the same time it projects downward like the keel of a modern racing yacht. These muscles drive at the rate of several hundred strokes a minute a pair of small, rigid wings, the outermost bones of which are very long while the innermost are very short, a feature calculated to give the greatest amount of motion at the tip of the wing with the least movement of the bones of the upper arm, to which the driving muscles are attached. Another peculiar feature is that the outermost feathers, the flight feathers or primaries, are long and strong, while the innermost, those attached to the forearm, are few and weak; so far as flight is concerned the bird could dispense with these secondaries and not feel their loss. Finally the heart, which we may look upon as the boiler that supplies steam for this machinery, is large and powerful, as is necessary for such a high-pressure engine as the little humming-bird. It is hardly to him that we would look for aid in constructing a flying machine, the expenditure of force is too great for the results attained, the space required for boiler and engine leaves no room for carrying freight.

As just intimated the frigate bird is exactly the reverse of his tiny relative; the body is a mere appendage to a pair of wings, while the breast muscles are so small as to show at a glance that of all flying creatures the frigate bird is the one which has most successfully solved the problem of the conservation of energy and can obtain the greatest amount of power with the least expenditure of muscle.

There is also a great difference between the hummer and the frigate bird, or between flapping and sailing birds generally, in the complexity of what may be termed the muscles of adjustment, the little muscles that run from the shoulder to the elbow and forearm and, among other duties, are concerned in keeping free from wrinkles that portion of the wing which lies between the shoulder and the wrist, forming a triangular flap with the base forming the front edge of the wing and the apex lying in the elbow joint.

The wing of the frigate bird, too, is quite the opposite of that of the hummer, for it is the inner portion of the wing, the upper arm and forearm, which is elongated, and instead of the six feeble secondaries of the humming-bird there are no less than twenty-four; instead of a short, stiff, rounded wing we have one that is long, flexible and pointed. Instead of a wing driven at the rate of several hundred strokes a minute there is a wing that may be held outstretched and apparently motionless for minutes at a time, the muscles of the frigate bird being almost as constantly in repose as those of the other are perpetually in motion.

If the frigate bird represents the highest type of soaring flight two more familiar birds, the turkey buzzard and albatross, are not far behind, and these represent two methods of sailing flight and two distinct modifications in the type of wings. The albatross is continually on the move, ever quartering the water as a well-trained setter does the ground, and yet with all this movement rarely mounting higher than fifty feet above the water and never wheeling in great circles in mid-air. This bird has that type of wing which best fulfills the conditions necessary for an aeroplane, being long and narrow, so that while a fully grown albatross may spread from ten to twelve feet from tip to tip, this wing is not more than nine inches wide. This spread of wing, like that of the frigate bird, is gained by the elongation of the inner bones of the wing and by increasing the number of secondaries, there being about forty of these feathers in the wing of the albatross.

The turkey buzzard is emphatically a high flyer, wheeling slowly about, half a mile or a mile above the earth, while his cousin, the condor, so Humboldt tells us, has been seen above the summit of Chimborazo. If any bird knows how to utilize every breath of wind to the utmost that bird is the albatross, and it is equally a delight and a marvel to see this bird apparently setting at naught all natural laws as he sails with outstretched pinions almost into the eye of the wind or hangs just off the lee quarter of a ship reeling off ten or twelve knots an hour. In this last trick, however, the gull is almost equally expert, evidently making use of the draft from the sails as well as of the eddies caused by the passage of the vessel.

It has long been evident that if man is to navigate the air it must be done after the method of the albatross rather than that of the humming-bird, by the aeroplane and not by any device to imitate the strokes of a bird’s wings, for not only do the largest birds and those of the longest flight for the most part sail or soar, but it is apparent that the limit of size in a vibrating wing must soon be reached, since in a strong wind with its varying eddies it would be quite out of the question to manipulate such a piece of mechanism.

But in spite of the fact that sailing flight calls for the exercise of comparatively little muscular power, the structure of the skeleton suggests that the wing of a soaring or sailing bird needs a particularly strong point of support, for birds which sail or soar have the bones which sustain the direct pull of the wing strengthened or braced as other birds do not. The shoulder joint of a bird is formed by the shoulder blade and coracoid, this last being the bone which is attached to the breast bone and on which comes the direct pull of the wing, and in front of the coracoids, running downwards towards the sternum, is the wishbone or furcula, corresponding to our collar bones or clavicles. It is evident that the greater the length of the coracoid the less able would it be to resist the strain brought upon it, and it is also evident that the simultaneous downward stroke of the wings must have a tendency to force the coracoids inwards, or towards one another. Obviously the greater the strain the greater the need of strengthening or bracing the coracoid to resist it, and there are in the shoulder girdle of a bird various devices looking towards this end. In some birds, the albatross, for example, the coracoid is short and stout, while in others extra bracing is obtained from the wishbone.

In the humming-bird the wishbone is light and weak and so short that it does not come near the sternum; the pigeon, a bird of powerful flight, is little better off, for the wishbone is so long and slender that it does little or nothing towards strengthening the shoulder joint, and in both these birds which fly by rapid wing strokes the entire pull of the wing is taken by the coracoid. In the frigate bird, on the contrary, the wishbone is not only strong, but it rests upon and is firmly soldered to the breastbone, while at its upper end it fuses with the coracoid, thus making the firmest possible support to the wing. The cranes, which soar well, also have the wishbone united with sternum, and in the albatrosses and petrels the wishbone touches the breastbone and is so curved forward as to gain strength in this way while, as previously noted, the strength of the coracoid is increased by its shortness. The turkey buzzard and birds of prey, some of which both soar and flap, have the wishbone strengthened by having more material added to make the furcula thick and strong while at the same time it is shaped like a wide U instead of a V.

Either there is more force exerted in sailing than is at first sight apparent or else extra strength is called for in making sudden turns, or when it becomes necessary, as it does more or less frequently, to take a sudden wing stroke. As wings are levers of the third order the longer the wing the more force is required to move it and more strength is needed at the fulcrum or shoulder joint, and since sailing birds have long wings the need of strength is evident.

Neither birds nor any creatures that live or have lived afford us any criterion as to the limit of size that must be placed on an aeroplane. The largest of whales is weak and insignificant beside an ocean liner, and the condor and albatross, with their spread of ten or twelve feet and weight of ten to twenty pounds, tell us nothing of what may be the possibilities of size and weight.

Among the various problems confronting the would-be navigator of the air is that of at times making headway against a medium moving at the rate of ten, twenty, or thirty miles an hour, sometimes even more, a difficulty that neither locomotive nor steamer is called upon to meet. True, an aeroplane would, to use a technical term, probably lie within two and one-half points of the wind and could thus advantageously beat to windward, but any deviation from a straight course means loss of time, and nowadays time is everything.

The mode of propulsion may be, undoubtedly will be, as entirely different from a wing as the propeller is unlike the tail of a fish, and as the study of fish has thrown little or no light on the problems of the proper form or best motor for a ship, it is doubtful if the study of birds will do more for the aerodrome. Nor does it seem likely that a study of the bird will suggest any new devices in the way of joints, braces, or rudders, for what must be discouraging to those engaged in solving the problems of flight is the utter inadequacy of the bird's wing, from a mechanical standpoint, for the work it is called upon to do, for in all its articulations there is a freedom of movement, an amount of play that would be inadmissible in any machine. The shoulder, elbow and wrist joints are but loose affairs, depending for their efficiency on the pull of the muscles; subtract the element of life from the wing of a bird and it becomes at once limp and useless. And herein is the key to the bird's success as a flying machine; it has *life*, and while the wing may reveal certain principles of balancing, it cannot teach us all the art, for it is done instinctively. The bird has back of it untold ages of experience and its actions during flight demand no thought; the muscles respond instinctively to each change in the pressure and direction of the wind, and the bird need take no thought as to how it shall fly.

Mr. Chanute has taken the greatest step yet made towards overcoming the difficulty of responding to changes in the velocity of the fickle air, but whether or not it will be possible to construct apparatus that will not only adjust itself to changes in the force of the wind, but to eddies and changes in direction as well, remains to be seen, the more that it must act not on planes six feet in length, but on surfaces infinitely larger. The proper method of constructing the wings of an aeroplane so as to insure stability and utilize the power of the wind to the best advantage, and some hints as to balancing and steering are the main assistance that we seem likely to gain from a study of the structure and flight of birds.

ELECTRIC AUTOMOBILES.

By WM. BAXTER, Jr.

AS electricity has been so successful in the street railway, where it has superseded all other forms of motive power, it might naturally be supposed that it would do equally well in the automobile; but when the difference in the conditions is taken into consideration it will be found that such a conclusion is not justified. In the street railway systems the cars run continuously over the same route, and on that account the electric current required to operate the motors can be conveyed to them from a central power station by means of wires. With the automobile the case is very different; the vehicle has no fixed course, but is required to go everywhere, and the current must be supplied from a source carried by it. If primary batteries could be made so as to furnish electric currents at a low cost, then the electric carriage would be in the same position as those operated by steam or gasoline, and it could go wherever the proper chemicals to renew the battery could be obtained. But as there are no such primary batteries, the only way in which the current can be supplied is by the use of storage batteries, and these cannot give out any more energy than is put into them, and in practice cannot give quite as much. Thus if the capacity of the battery is sufficient to run the vehicle forty miles when this distance has been traversed the propelling power will be exhausted, and the batteries will have to be recharged before the carriage can go any further. If the recharging could be done in a few minutes, the storage battery would be as good as a primary battery that would generate electricity economically; but as it requires three or more hours, the electrical vehicle cannot be used for long runs, unless the user is willing to make long stops each time the battery has to be recharged. Even then an electric vehicle could not go everywhere, for it would be compelled to follow routes along which facilities for recharging the batteries could be found. From this fact it can be seen that the electric automobile carriage cannot cover the same field as the steam or the gasoline (in the present state of electrical development). Within the limits to which it is applicable, however, it can perform its work in the most satisfactory manner, and, in fact, no possible objection can be raised against it. Its operation is noiseless and vibration of the vehicle is impossible. There is no heat to inconvenience the passengers, no disagreeable smell, no escaping steam. Any desired speed can be obtained, although, of course, a heavy delivery wagon cannot be used also as a racer. The power can be made sufficient to propel any desired load up any grade, including grades far steeper than any to be encountered on streets or highways.

The only point in which the electric vehicle suffers in comparison with the others is in the weight. The capacity of a storage battery is proportional to its weight, and if it is made light, the power derived from it will be small or the time during which it is furnished will be short. To furnish one horse power for one hour requires about one hundred pounds of battery, so that if the average consumption of energy is at the rate of two horse power, one thousand pounds of battery will keep the vehicle in motion for five hours. The weight of batteries used in automobiles ranges from four or five hundred to about two thousand pounds, and the distance traversed without recharging varies from twenty-five to ninety miles, so that the radius of action of electric vehicles can be said to vary from about twelve to forty-five miles from the charging station.

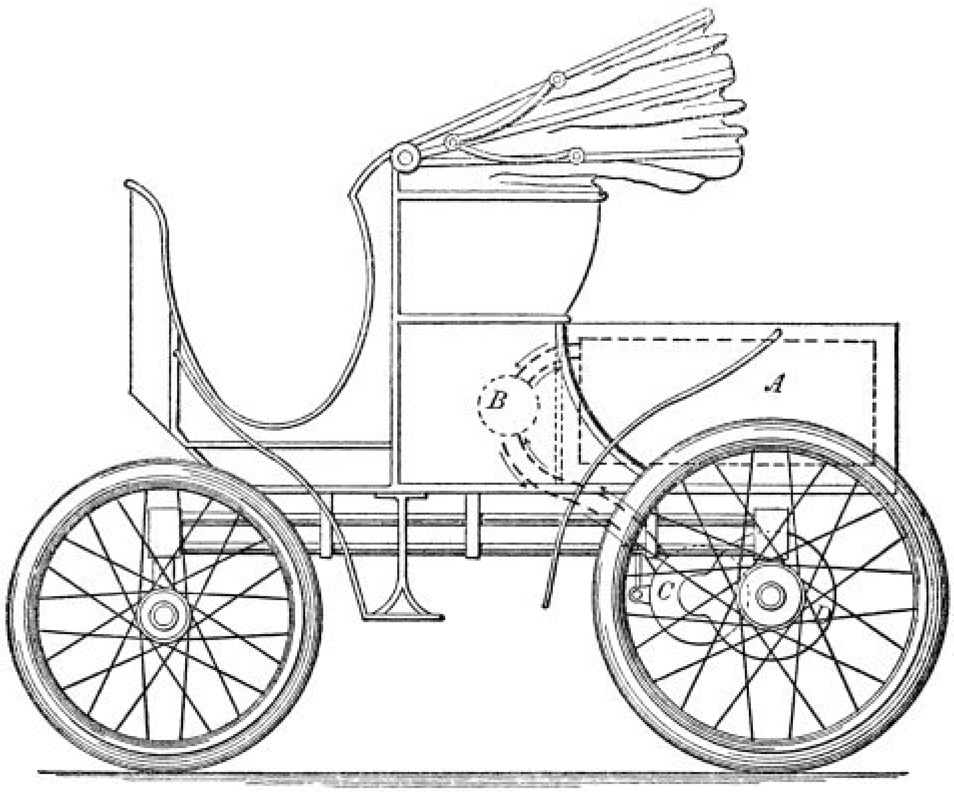

FIG. 1. GENERAL ARRANGEMENT OF AN ELECTRIC CARRIAGE.

The general arrangement of an electric carriage can be understood from Fig. 1. The rectangle shown in broken lines at *A* represents the storage battery. The circle *B* under the seat represents the controlling switch. The motor is at *C* and imparts motion to the axle or wheels through the gearing contained in the casing *D*. When the carriage is stopped the controller *B* is turned into such a position that all electrical connections between the battery and the motor *C* are broken. To start the vehicle the controller *B* is turned so as to make the necessary electrical connections between the battery *A* and the motor *C*, and then the electric current passes from the battery through the controlling switch to the motor, and thence back to the controller and the battery. The heavy broken lines indicate the path of the current and the arrows show the direction. The velocity of the motor and the speed of the carriage are varied by varying the strength of the current, and this is effected by the movement of the controlling switch *B*. There are many ways in which the movement of this switch can vary the strength of the current, but an explanation of any one of them would be dry and rather technical; hence it is sufficient to say that whatever the arrangement of the connections of the controller with the other parts of the system, their relation is such that by the movement of the switch handle the speed of the motor is changed from zero to the maximum velocity.

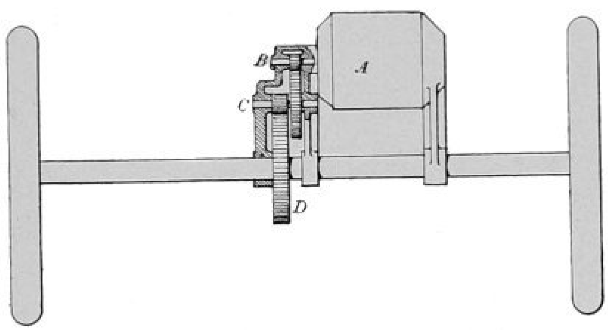

Fig. 2. Double Reduction.

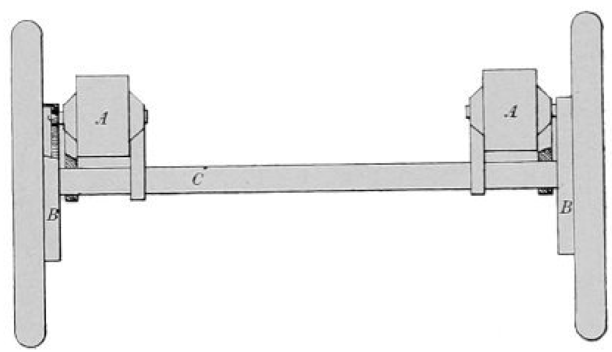

Fig. 3. Single Reduction.

In the majority of American vehicles the motion of the motor is transmitted to the wheels by means of spur gearing. In some cases a single motor is used, in others two; and in one or two designs that have come to public notice, four motors are employed, one for each wheel of the carriage. Fig. 2 illustrates what is commonly called a double reduction gear for single motor equipment. The outline *A* represents the motor, *B* being the shaft. Upon this shaft is mounted a small pinion which meshes into a larger wheel on the intermediate shaft *C*. This shaft carries a pinion which meshes into the wheel *D* mounted upon the axle of the vehicle.

Fig. 3 illustrates a single reduction double motor equipment, the motors being located at *AA*. In this arrangement the pinion on the end of the motor shaft meshes directly into a large gear secured to the carriage wheel, thus dispensing with the intermediate shaft *C* of the previous figure. The single reduction gear is the more simple in construction, but the motors run at a lower velocity, and on that account must be larger for the same capacity. With the double motor construction each wheel is driven independently and the axle *C*, in Fig. 3, remains stationary, as in any ordinary vehicle; but in a single motor equipment, arranged as in Fig. 2, the wheels are fastened to the axle and the latter rotates. When a carriage runs round a short curve the outer wheels will revolve faster than the inner ones, if free to move independently, as in Fig. 3. If they are rigidly attached to the axle, as in Fig. 2, one or the other will have to slide over the ground, and this is decidedly objectionable with rubber tires. To prevent this slipping of the wheels in rounding curves, the axles, in designs following the construction of Fig. 2, are made in two parts, and the gear *D* is arranged so as to drive the two halves, imparting to each one the proper velocity. Gear wheels of this kind are called compensating gears; they are made in many designs, but the most common form is that illustrated in Fig. 4. In this drawing *A* is the gear *D* of Fig. 2, and *BB* are bevel gears which are mounted upon studs *C*, which are virtually the spokes of wheel *AA*. Large bevel gears *E* and *F* are placed on either side of *A E*, being secured to *G*, which is one-half of the axle, and *F* and *H*, which is the other half. If the carriage is running in a straight line, the two parts of the axle *G* and *H* will revolve at the same velocity and the gears *BB* will not revolve around the studs *C*, but in rounding a curve one of the halves of the axle will revolve faster than the other and then the gears *B* will rotate round the studs *C*. The compensating gear is not a feature peculiar to electric vehicles; it is used on all kinds of automobiles when the construction is such as to require it.

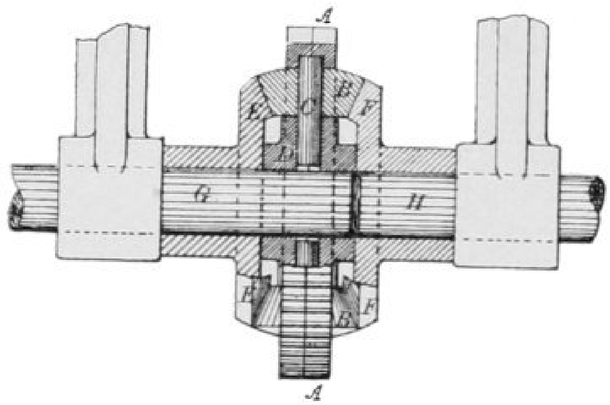

Fig. 4. Compensating Gears.

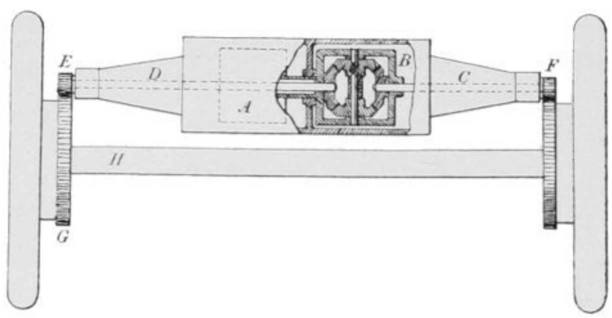

Fig. 5. Single Motor Equipments.

If a compensating gear is placed upon the axle the latter, instead of supporting its end of the vehicle, will itself have to be supported, for as it is cut in two at the center, it has no supporting strength. By placing the compensating gear on another shaft this difficulty can be overcome. Fig. 5 shows the construction used by the Columbia Company in its single motor equipment. In this arrangement the motor casing is made of sufficient length to reach from one side of the vehicle to the other. The armature and field magnets of the motor, which are the parts that develop the power, are located at *A* and the compensating gear is placed at *B*. The motor armature is mounted upon a hollow shaft, which is connected with the compensating gear. The shafts *D* and *C*, upon which are mounted the pinions *E* and *F*, are turned by the side wheels of the compensating gear, and therefore will run at such velocities as the motion of the carriage wheels may require.

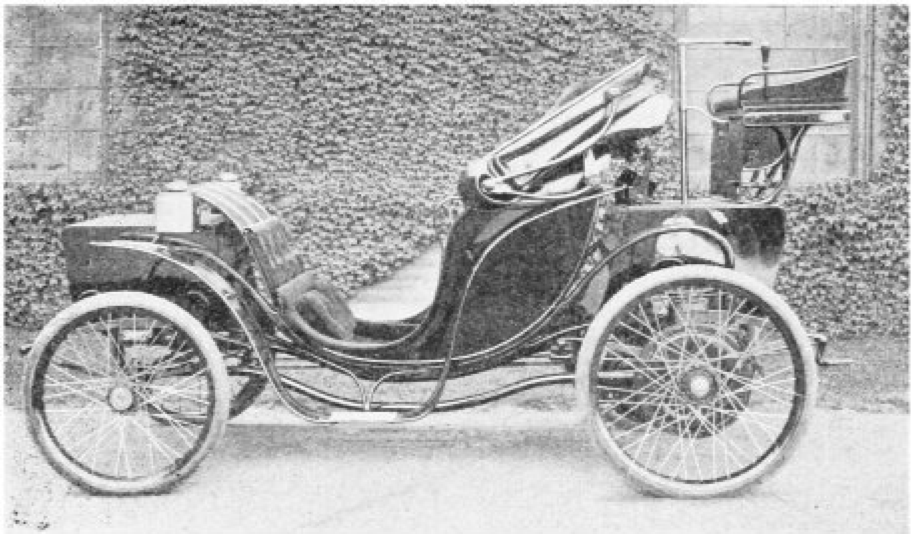

FIG. 6. A COLUMBIA VICTORIA.

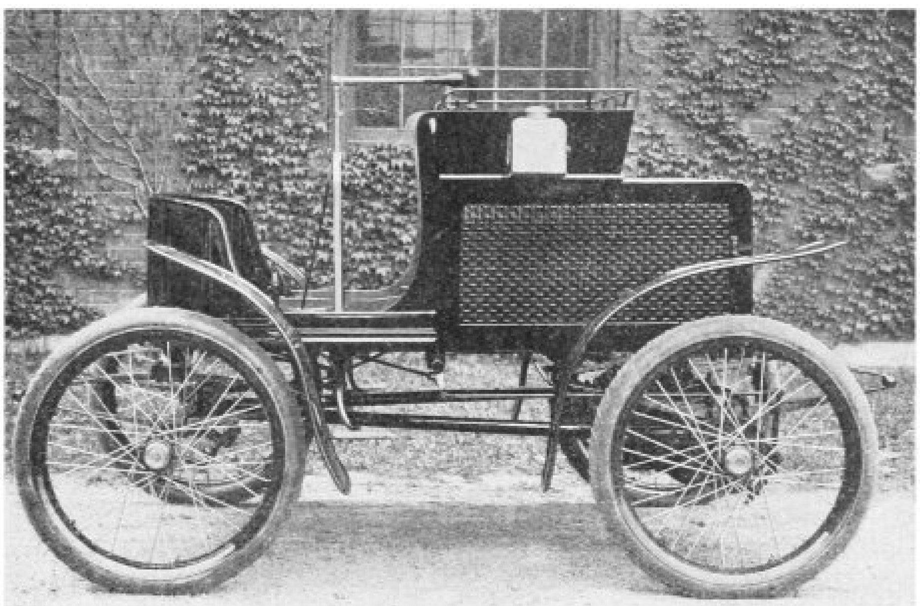

FIG. 7. COLUMBIA VEHICLE WITH DOUBLE MOTOR EQUIPMENT.

Fig. 6 shows a Columbia Victoria provided with a single motor equipment arranged in accordance with the diagram, Fig. 5. Fig. 7 shows another Columbia vehicle in which a double motor equipment is employed. The position of the motor, with reference to the carriage wheel, in the single motor design, is shown in Fig. 8. The gear attached to the carriage wheel is used also as a brake wheel, a friction band being located so as to bear against the periphery, while the pinion on the end of the motor shaft meshes into teeth on the inner side of the rim. This single motor design is also used in the omnibus made by the Columbia Company, a number of which are now in regular service on Fifth avenue, New York. These omnibuses, which are illustrated in Fig. 9, seat eight passengers, and are able to carry as many as are willing to crowd into them. One feature of the electric motor which fits it admirably for automobile service is the fact that for a short time it can put forth an effort far greater than its normal capacity, and it can do this at all times, without any special preparation. Owing to this feature it is practically impossible to stall the vehicle. If the wheels run into a rut or sink into a mud hole, the motor will be able to turn them around, and if they do not slip the carriage will be moved ahead.

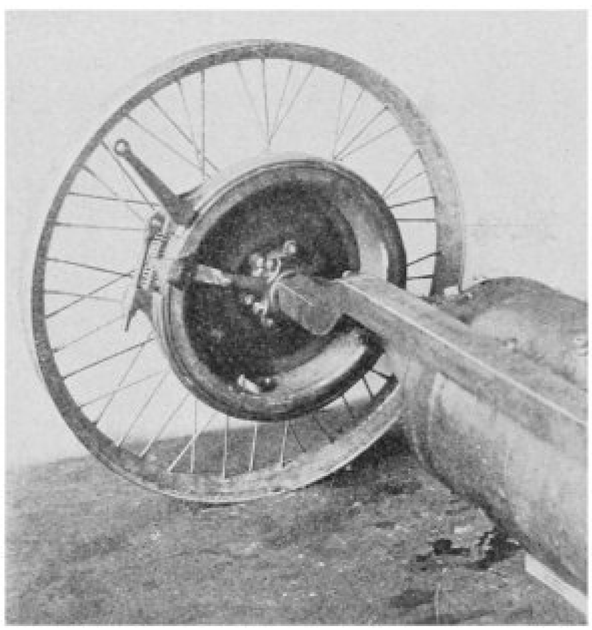

Fig. 8. Position of Motor in the Single Motor Design.

The management of the vehicle is exceedingly simple and entirely free from care, the driver having nothing to tax his mind but the steering lever and the handle of the controlling switch. As the moving parts all have a rotatory motion and are perfectly balanced, there is no possibility of vibration, and there is an entire absence of heat or disagreeable odors.

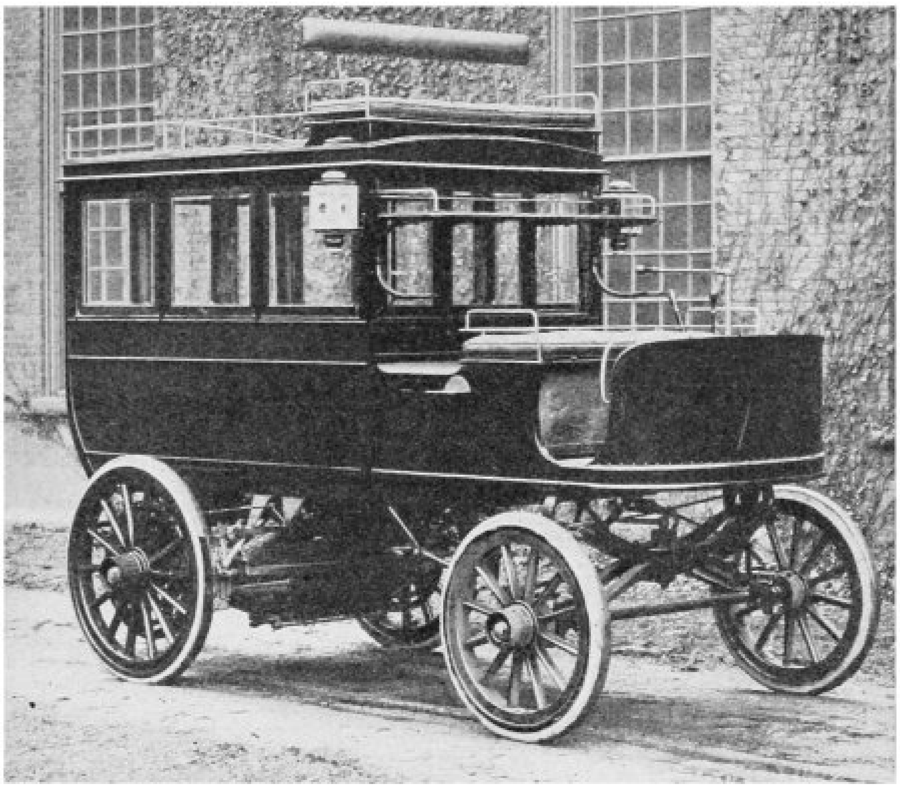

FIG. 9. A COLUMBIA OMNIBUS.

Any one who has observed the action of a two-horse team will have noticed that, unless the pavement is very smooth, the tongue continually swings from side to side, and occasionally with a considerable amount of violence. It will be evident from this fact that if the front axle of an automobile were the same as that of a horse vehicle, the driver would have an unpleasant task, to say the least, in holding the steering lever in position, and should one of the wheels drop into a rut, the handle would be jerked violently out of his hand and the vehicle would sheer off to one side, possibly with serious results. To avoid this difficulty the front wheels of horseless carriages are arranged so as to swing round on a center close to the hub, if not actually within it. The most common construction is illustrated in Fig. 10, the first being a view of the axle and wheels as seen from the front, and the second a view from above. On the left-hand side of Fig. 10, *A* is the axle proper, and *BB* are the portions upon which the wheels are placed. The central part *A* is held rigidly to the body of the vehicle or to the truck which carries it, and the ends *BB* are swung round the studs *PP* in a manner more clearly indicated on the right-hand side. Here the levers *CC* are shown, and these extend from the side of *BB*. The right-angle lever *E* is connected with the steering lever *G* by means of rod *F*, hence, when *G* is moved, rod *D* moves, and thus levers *CC* are rotated round the studs *PP*, and in that way the supporting studs *BB* which carry the wheels are turned. As the studs *PP* are not exactly in line with the plane passing through the center of the rim of the wheel, there is a slight tendency to jerk the steering handle round when a wheel drops into a hole in the pavement, but the leverage of *B* being very short, this tendency is so small as to be hardly noticeable.

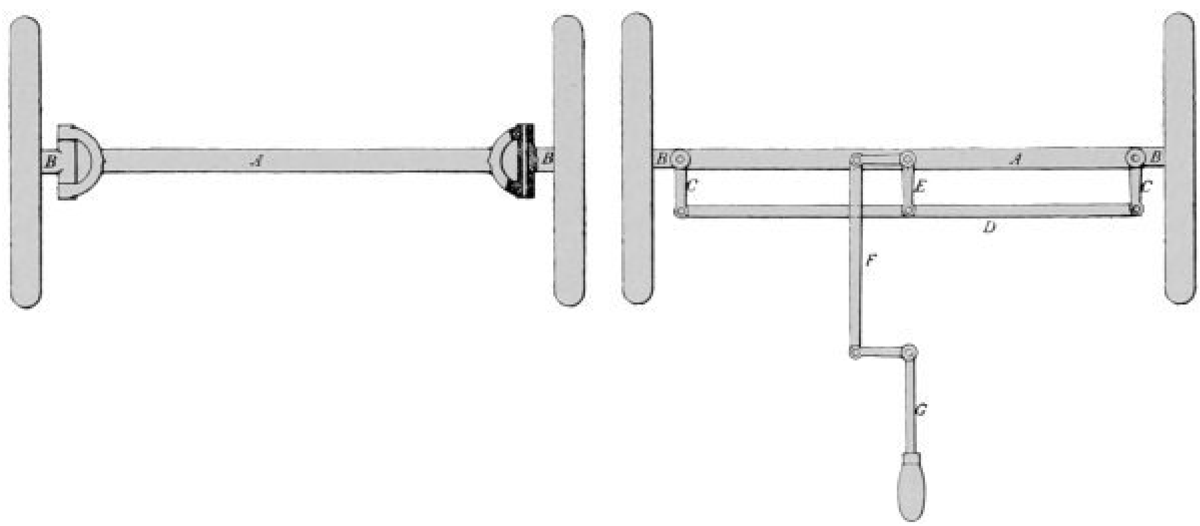

FIG. 10. ARRANGEMENT OF AXLES AND WHEELS.

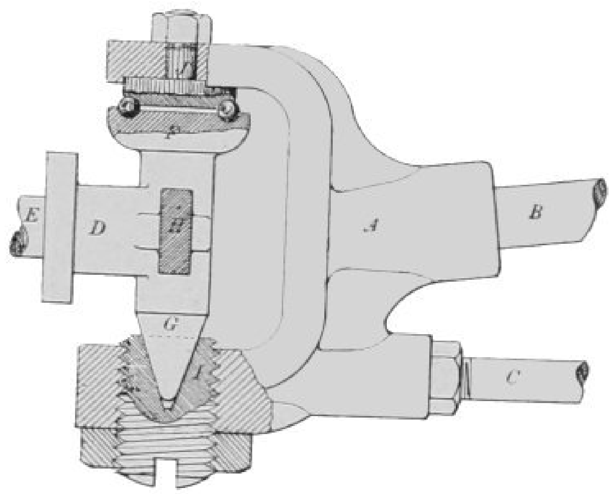

FIG. 11. FRONT AXLE.

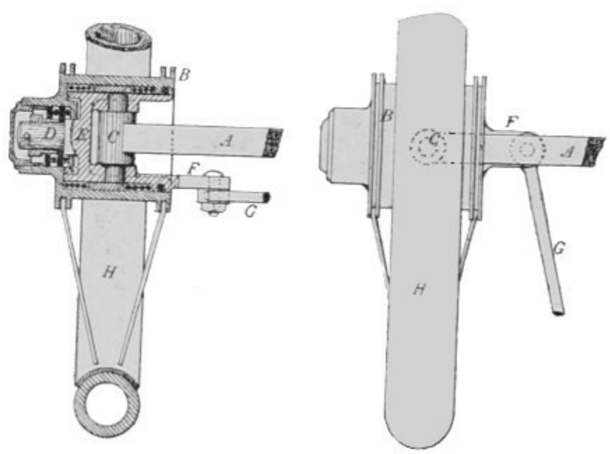

FIG. 12. FRONT AXLE WHEELS.

Fig. 10 illustrates the general principle upon which the front axle is designed, but the construction of the swivel joints *P* is far more elaborate, as can be seen from Fig. 11, which illustrates the actual design employed in the vehicles just described. Looking at Fig. 9, it will be noticed that the front axle consists of two bars, one of which runs in a straight line from side to side, while the other is curved with the convex side upward. In Fig. 11 *B* is the end of the upper curved rod and *C* is the lower straight one. These two rods are secured into the casting *A*, which holds the part *D* upon which the wheel is carried, *D* being the part *B* at the left side of Fig. 10. The end *E* which is broken off in the drawing extends through the hub of the wheel and is provided with ball bearings so as to run without friction. The upper end *F*, of *D*, is arranged so as to be held by a ball bearing, as shown, against the end of *J*. By means of an adjusting screw *I* at the lower end, the parts are brought into proper position with reference to each other. The shaded portion *H* is the lever *C* at the right side of Fig. 10.

The left-hand end of Fig. 12 shows a design for front axle wheels which is one of the many modifications of the general arrangement just described. In this construction the wheel swings round the stud *C*, which is placed within the hub, and in a line, or nearly so, with the center of the rim. The rod *A* is the axle and *F* is the lever extending from the inner part of the wheel hub by means of which the steering is effected. The left-hand side of Fig. 12 is a view as seen from the front and the right-hand side shows the device as seen from above. In this last drawing it will be observed that as the lever *F* is attached to the inner portion of the wheel hub, if it is moved to one side or the other of axle *A*, by pulling or pushing on rod *G*, the wheel will be swung round. The advantage of designs of this type is that there is no strain whatever brought to bear upon the steering handle, and the objection is that the wheel hub is made much larger and the whole construction is somewhat more complicated.

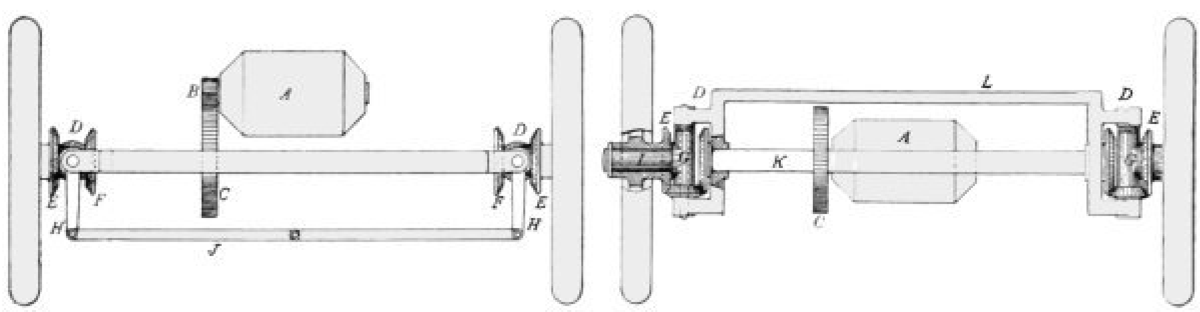

Fig. 13. Constructions Showing Power Applied to Front Wheels.

The arrangement of the front axle, so as to swing the wheels round a center close to the hub or within it, as described in the foregoing paragraphs, is used on all types of automobiles and is not a distinguishing feature of the electric carriage. In some of the lighter vehicles the front wheels are held in forks of a design substantially the same as that of the front wheel of the ordinary bicycle, the tops of the forks being connected with each other by means of a rod, as in the lower part of Fig. 10, so as to obtain simultaneous movement of the two wheels by the movement of a single steering handle.

In the majority of electric vehicles the power is applied to the rear axle, but some are made with the motors geared to the front axle. In a few of these designs the wheels and axle are made the same as in an ordinary carriage, so as to swing round a pivot or king bolt located at the center of the axle and reinforced by a fifth wheel. When this construction is used the steering gear is made so as to hold the axle in position more firmly than in the other designs; but even with this assistance the driver has a harder task than with the independently swinging wheels. The advantage derived from swinging the whole axle is that the carriage can be turned round in a very small space, and on that account the construction is well adapted to cabs.

Several arrangements have been devised by means of which the power can be applied to the front wheels, while these may at the same time swing round independent centers. One of these constructions is illustrated in Fig. 13, the first drawing presenting the appearance when seen from above, the second being a view from the front. In the first diagram the motor is shown at *A*, and by means of pinion *B* and gear *C*, motion is transmitted to the axle, which is shown more clearly in the right-hand figure. On the ends of the axle are bevel gears *FF*, and these mesh into other bevel gears which revolve round the vertical studs *D*. Through this train of gearing the bevel wheels *E* are driven, and these are attached to the hubs of the carriage wheels. From the first diagram of Fig. 13 it can be seen at once that the gears *EE* can swing round *D* in either direction without in any way interfering with the transmission of motion from gears *FF*. The levers *HH* are secured to the sleeves *GG* which swing round the studs *DD*, hence, by connecting these with the rod *J* and moving the latter to one side or the other by means of the steering handle, the wheels are turned in any direction desired.

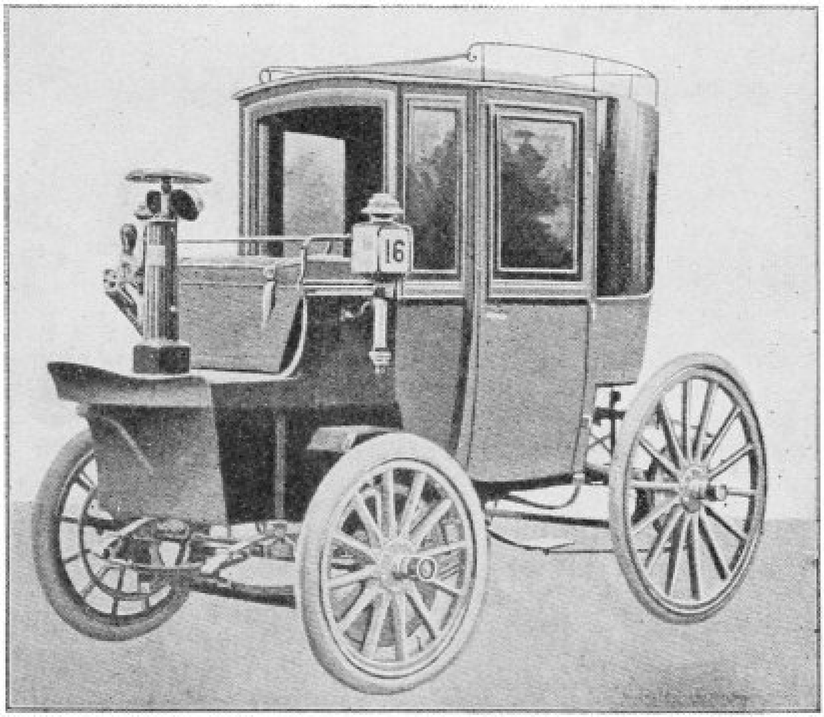

Fig. 14. Krieger Coupé.

While this construction renders the carriage as easy to steer as those in which the motors are connected with the rear axle, it sacrifices the advantage derived from applying the power to the front wheels, namely, the ability to turn round in a small space.

Another design for driving the front wheels which allows them to swing round independent pivots, is shown in Fig. 14, which is a coupé made by Krieger in France. The power is supplied by two motors, one being mounted on each swivel point. The construction can be understood by considering that in the lower part of Fig. 13 the motor would be secured to a suitable support at the end of the frame *L*, being held in such a position that the shaft would replace pivot *D* and a pinion mounted thereon would gear into wheel *E*. What the advantage of this construction may be, the writer is not able to point out; it certainly shows, however, that there are many ways in which the object sought may be accomplished.

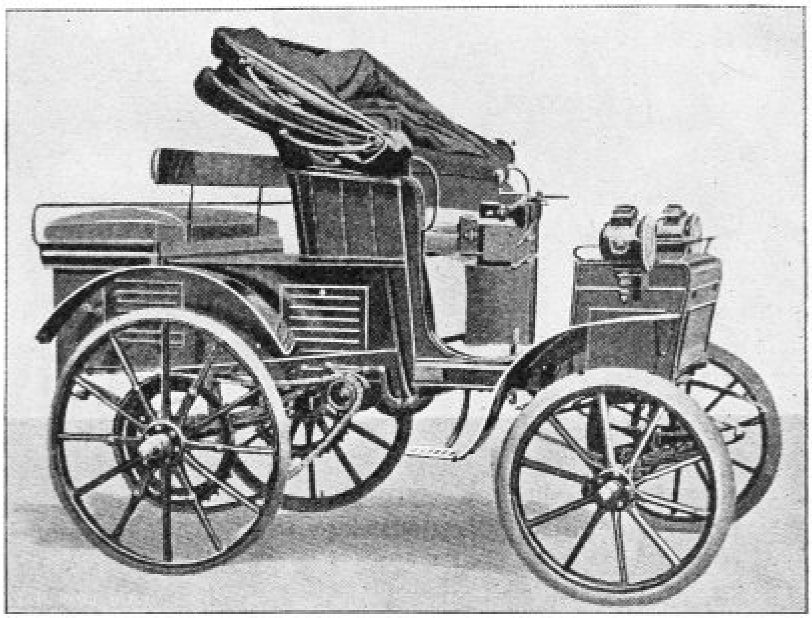

FIG. 15. JENATZY DOG-PHAETON.

American manufacturers of electric vehicles, at least the great majority of them, resort to spur-gearing to transmit the motion of the motor to the driving wheels, but with the French designers the chain and sprocket appears to be in great favor. Fig. 15 shows a Jenatzy vehicle (French), in which the chain is used. This construction would not be received with favor by Americans, who as a rule desire to have the mechanical part of the apparatus hidden from view as much as possible. In the Jenatzy vehicle two chain gears are used, one on each side of the body, and from the engineering point of view this is the most desirable arrangement, as with it the driving wheels are independently operated and a compensating gear need not be placed upon the axle. The American designer, however, would in most cases be controlled more by the artistic appearance and would use a single chain which would be placed under the body of the carriage, and thus as much out of sight as possible.

Fig. 16 shows an English design of electric dog cart. The mechanism consists of a single motor which is connected with the axle by means of spur gearing, this being so arranged that several different speeds can be obtained for the vehicle with the same velocity for the motor. To obtain variable speeds by means of gearing it is necessary to introduce a considerable amount of complication, and in this country the opinion of most designers appears to be that the gain effected thereby is not sufficient to compensate for the increased complication, and differential speed gearing is not often used.

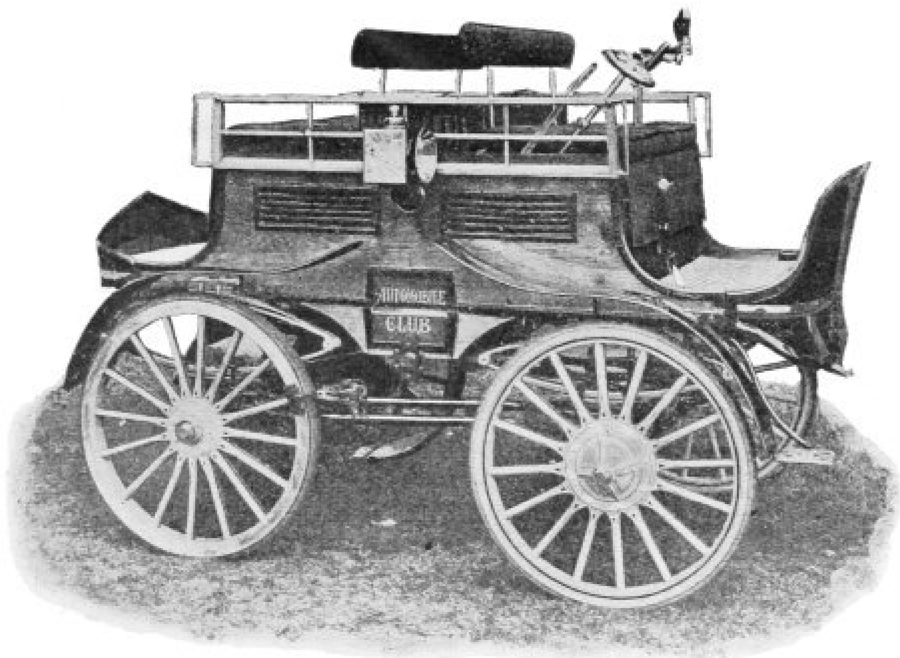

Fig. 16. The Electric Motive Power Company's Dog Cart.

A comparison of Figs. 14 and 16 with 6 and 9 will clearly show that in so far as artistic effect is concerned, our manufacturers of electric vehicles have little to learn from Europeans, although the industry here is much younger than abroad. As to the operative merits, all that can be said is that the American carriages run so well and possess such endurance that it is probable that they are not second to any in these respects.

THE HUMAN BODY AS AN ENGINE.

By Professor E. B. ROSA.

THERE is no more interesting subject for scientific investigation than the structure and operation, the anatomy and physiology of the human body. That it is an amazingly complex and delicate mechanism, performing a multitude of functions in a wonderfully perfect manner, is, of course, an old story. That in the assimilation of its nourishment and in the growth and repair of its tissue the body obeys the laws of chemistry has long been understood. But that the body obeys in everything the fundamental law of physics, namely, the law of the conservation of energy, has not been so generally recognized. For some years the writer was engaged in some investigations upon this subject.[B] The development of the complex apparatus and unique methods of the research required years of patient labor and study. One of the features of the apparatus was an air-tight chamber, in which a man, as the subject of the experiment, could be confined for any desired period, eating, sleeping, working and living while under minute observation. The experiments usually continued four or five days, but were sometimes prolonged to eight or ten days, and the observations were made and recorded day and night continuously for the entire period.

[B] The work was done at Wesleyan University, in collaboration with Prof. W. O. Atwater, under the patronage of the University and the U. S. Department of Agriculture.

The atmosphere within the chamber was maintained sufficiently pure to make a prolonged sojourn within its walls entirely comfortable. A current of fresh air, displacing as it entered an equal quantity of air which contained the products of respiration, was maintained continuously. The respired air was analyzed and measured, and the products of respiration from lungs and skin accurately determined. The ventilating air current was maintained by a pair of measuring air pumps, driven by an electric motor. The air was dried, both before entering and after leaving the chamber by freezing out its moisture. This was done by passing it through a refrigerator where its temperature was reduced far below the freezing point. The refrigerator was operated by an ammonia machine, driven by an electric motor. The quantity of air was automatically recorded by the pumps.

The chamber was so constructed and fitted with electrical and other devices as to afford the means of measuring the quantity of heat which the subject of the experiment gave off from his body. And in order to keep the temperature of the room constant this heat was absorbed and carried away by a stream of cold water, the latter flowing through a series of copper pipes within the chamber, and coming out considerably warmer than it entered. So delicate were the regulating devices that the temperature could be maintained constant, hour after hour, to within one or two hundredths of a degree. In some cases the man under investigation worked regularly eight hours a day, the work done being measured by apparatus designed for the purpose.

Food and drink were passed into the chamber three times a day through an air-tight trap. Both were accurately weighed, their temperature recorded and samples reserved for chemical analysis. Solid and liquid excreta were likewise weighed and analyzed. The observations, analyses and computations of a single experiment thus involved a vast amount of labor and expense, which was only justified by the importance of the question under investigation. In order to be able to understand just what this question is, let us see what is meant by the conservation of matter and energy in the physical world.

The impossibility of creating or destroying matter is very generally recognized. Its forms or properties may be altered, chemical and physical changes may be effected, it may, indeed, vanish from sight, but its quantity remains unchanged. Thus ice may turn to water and water to invisible steam, but the total quantity or mass of the substance remains constant; and if by refrigeration the steam be brought to the condition of ice again, there will be precisely the same amount as before. These are physical changes and are easily effected. We simply apply heat to melt the ice and then more heat to vaporize the water. Conversely, withdrawing heat will condense the vapor to water, when a further subtraction of heat will change the water into ice.

Again, wood disappears when burned and seems to be destroyed. And yet we know that the weight of the resulting smoke and ashes is exactly equal to that of the wood. The matter has been changed in form and composition, but its mass cannot be altered. It is not so easy to bring the smoke and ashes into combination again and so restore the matter to its original form as in the case of ice and steam. But this is done by nature. Ashes go to the soil, smoke into the atmosphere. The forces of nature bring these elements together again in plant and tree, and so it comes about that the materials resulting from the burning of wood again become wood, and over and over again the cycle is repeated as time rolls on. Many other examples might be cited to show what is meant by the indestructibility of matter, or the conservation of matter; but these will suffice to show that the one essential fact is that the matter or stuff of a body cannot be destroyed.

Although matter is protean and its transformations limitless, there are certain changes which cannot be made. Iron cannot be turned into silver, nor silver into gold, nor oxygen into nitrogen. There appear to be indeed about seventy or eighty distinct kinds of matter, and so far as we know one cannot be converted into another. They may be united in countless combinations, but each is itself not only indestructible but unchangeable. Why this is so is an interesting subject of speculation. We do not positively know.

That energy is also something which cannot be created or destroyed is not so generally recognized. Transformations of energy from one form to another are constantly occurring before our very eyes; and yet we seldom stop to think what the conservation of energy means in any given case. Energy itself is often defined as that which has the capacity for doing work, and work is done when force or resistance is overcome. A hod carrier does work when, overcoming the force of gravity upon his body and his hod of brick, he climbs to the top of a ladder; and the work done is a measure of the energy expended. Energy stored up in his body has been transferred to the brick in their elevated position, and if they are allowed to fall to the ground their energy is turned into heat, developed by their impact upon the ground. Again, work is done by a windmill in pumping water up into an elevated reservoir, and the so-called ‘potential’ energy which the water possesses in its elevated position has all been transferred to the water from the wind which drove the mill. If the water be allowed to flow down to the ground again through a water motor the latter could drive machinery and so do work; and the work it could do plus the heat produced by friction would exactly equal the work done in pumping the water up to its elevated position. Thus is the energy conserved, and not destroyed. More or less of it is dissipated by friction, and lost, so far as useful effect may go. But it all remains in existence, somewhere.

Again, coal is burned under the boiler of a steam engine. Heat is produced, steam is generated, the engine does work. The coal possessed a store of energy, potentially. That is, the coal had the capacity of uniting with the oxygen of the air and setting free a store of energy. This energy, potential or latent in the coal, becomes kinetic and evident in the heat of the boiler and the work of the engine. Moreover, the work done by the engine added to the heat given off by the boiler and engine is exactly equal to the total store of energy possessed by the coal. And if from a store of energy, either in the body of a man or horse, or in a pile of wood or coal, a certain portion is expended in doing work, the amount remaining is exactly the difference between that expended and the original amount. In short, energy can be measured, stored up and expended, just as truly as merchandise or money.

Thus the conservation of energy means that energy cannot be created or destroyed; but it may be transferred from one body to another or transformed from one form to another. Heat may be converted into work and work into heat. The chemical energy of a zinc rod may be expended to generate an electric current, and the latter passing through a coil of wire or the filament of a lamp gives up its energy to produce heat and light. The last form of this energy is equal in quantity to the first.

Niagara represents a vast store of energy. Millions of tons of water falling 160 feet could do a vast amount of mechanical work if properly applied through water wheels. More than 50,000 horse power of useful work is actually derived from Niagara’s waters, but this is only a small fraction of the total. The energy is, however, given up in falling, even though no useful work is done. In fact, the water is slightly heated by the impact, and the amount of heat produced is exactly equivalent to the mechanical energy lost by the water.

A cannon ball receives a large amount of kinetic energy from the exploded powder as it leaves the muzzle of a great gun. If it be suddenly stopped by a rigid target its mechanical or mass energy is at once converted into heat; that is, into the vibratory motion of the molecules. Ball and target are highly heated. Indeed, lead bullets are often melted by the heat of impact. Meteors flying through space come into our atmosphere and their speed is checked by its resistance. Part or all of their kinetic energy is thus converted into heat. Both air and meteor are heated; heated to so high a temperature that the meteor becomes brilliantly luminous, and we call it a shooting star. The idea of heat due to frictional resistance is common enough. The *exact equivalence* between the mechanical energy lost and the heat produced is the thing to be especially noticed here.

Let us now take as a final example a locomotive engine. It takes on a store of fuel and water and, directed by its engineer, sets out for a day's duty. The coal to be burned possesses a definite amount of energy. Let us say every pound has one unit of energy, and suppose 5,000 pounds of coal are taken. What becomes of these 5,000 units of energy, appearing as heat when the coal is burned?

1. A large amount of heat is required to keep the boiler and engine hot, due to the loss of heat to the atmosphere. The engine cylinders, as well as fire box and boiler, must be kept very hot; other parts of the engine become more or less heated. All parts therefore continually give off heat, and a large part of the heat produced by the burning coal is thus expended.

2. A second portion is expended in doing work. If our locomotive hauls a 500-ton train up a one-per cent. grade for 100 miles it would be doing 2,640,000 foot-tons of work in addition to that required to overcome the friction of the rails and the resistance of the atmosphere. This would require nearly 500 units of energy which would come from the heat of the coal. The work is done through the agency of steam, but the energy of the steam comes from the burning coal. A small amount of work is also done in pumping water from the tank on the tender into the boiler and in pumping air into the reservoir for the use of the air brakes. This may be called the

internal work of the engine. A second portion of the heat is therefore expended in internal and external work.

3. The steam after expanding in the cylinders of the engine escapes into the atmosphere. Although it has been cooled somewhat by expansion, it is still hot, and carries a large amount of heat away with it. Moreover, the smoke and hot air which pass out through the smokestack carry away a large quantity of heat. Hot ashes likewise carry away heat. Hence a third portion of heat is lost through smoke and steam and ashes. And this is the largest portion of the total quantity of heat generated by the burning coal.

When coal is burned, oxygen of the air unites chemically with the carbon and hydrogen of the coal to form carbonic acid, or carbon dioxid, as it is technically called, and water vapor. The incombustible mineral matter of the coal remains as ashes. Hence smoke contains carbonic acid gas and water vapor in addition to fine particles of unburned coal carried away in the draft of air.

When the grade is steep a great deal of work must be done by the locomotive, much steam is required, and the quantity of fuel burned is large in proportion. When the road is level fuel burns less rapidly, and when the train stops, still more slowly. At night the locomotive rests, fires are banked and combustion is very slow. This process so briefly and incompletely sketched, is more interesting as one examines it closer, and a locomotive seems almost living when one considers minutely its wonderful performance.

But interesting and instructive though the operation of the locomotive may be, it is not for its own sake that I have mentioned it. It is rather in order to point out a remarkable parallel between its operation and that of a human body. A parallel, indeed, between the operation of a complex inanimate engine of iron and steel, and a still more complex living engine of flesh and bone and blood; both obeying the law of the conservation of energy, as well as the other laws of physics and chemistry.

Consider now a human body as a living engine. That man is more than matter is, of course, conceded. But we here regard only the animal body, guided by the brain as its engineer. The day begins, as with the locomotive, by taking a store of fuel and water, namely, food and drink. Food is not, however, burned in the body in a confined receptacle, like coal in the fire box of an engine, but is digested, assimilated and distributed through the body by means of the circulating blood. And while some of it goes to repair bodily waste, becoming tissue, other portions are oxidized or burned to produce heat. Non-digestible parts of the food pass away from the body as refuse, like ashes from the fire box of the engine. That the body fat and muscular tissue are also burned, producing heat, is literally true. A hibernating animal keeps his body warm all winter by burning up his autumnal store of body fat. Even a well-fed body is constantly wearing away, or burning away, and hence requires constant repair. Thus we see two distinct functions for food, which should be carefully distinguished.

In the first place, as already indicated, food repairs waste and builds up the body. It makes blood, bone and muscular tissue. Herein we see a departure from the parallel with the steam engine. A locomotive is a machine which runs in a way determined by its builder. But it cannot grow nor repair wear and tear. It requires a whole machine shop plus skilful mechanics to do that. The body, on the other hand, not only runs like a complex mechanism when supplied with energy, but also builds itself up and repairs waste. We express this by saying that it possesses vital force or life, but in just what vital force consists is a matter of speculation and controversy. The raw material which is employed in this work of repairing and building up is found in the food. But not all food can be so utilized. Only those materials which contain nitrogen, the so-called proteids, as lean meat, the casein of milk and gluten of wheat, can be made use of in this most important work of growth and repair.

In the second place, food is the fuel of the body and is just as truly burned as is coal in a furnace. Moreover, the quantity of heat which a piece of meat or a slice of bread yields when burned in the body is just the same as if it had been burned in a stove. Complete combustion yields a definite amount of heat wherever and whatever may be the place and manner of burning. Any kind of food may serve as fuel for the body, but those which consist mainly of sugar, starch and fat, which contain no nitrogen and so cannot build up the body, are used chiefly as fuel. These fuel foods form the bulk of our daily ration, comparatively little being required for purposes of growth and repair.

We are hearing a good deal recently about alcohol as a food. When it is remembered that alcohol contains no nitrogen it will be seen that it cannot serve the first function of food, namely, the purpose of growth and repair. It can, however, serve as fuel food, for when taken into the body in small quantities it is assimilated and burned up, producing the same amount of heat as if burned in a lamp. In sickness this may be beneficial, at times when the body cannot assimilate other foods. But the injurious effects of alcohol upon the digestive and nervous systems are so important and far-reaching that its value as a fuel food sinks into insignificance in comparison.

The process of combustion or burning in the fire box of our locomotive consists, as has been said, in oxygen of the air uniting with the carbon and hydrogen of the coal, forming carbonic acid and water, and setting free a definite quantity of heat for every pound of fuel so burned. So, in exactly the same way, oxygen, which has been taken up by the blood from the air in the lungs, unites with carbon and hydrogen in the tissues of the body and forms carbonic acid and water, yielding precisely the same amount of heat as though the combustion had occurred in a furnace. This idea of food, that it is literally fuel, is a very suggestive one. And as fuels differ in the quantity of ash contained and the amount of heat produced, so food materials differ in the quantity of undigestible residue and in their heat-producing power.

Remembering the analogy of the steam engine, let us now inquire what becomes of the energy supplied to the body in the fuel foods eaten, and which is turned into heat by this process of combustion constantly going on.

1. A large amount of heat is constantly being expended in keeping the body warm. Like the locomotive, the body is warmer than the surrounding air, and is constantly losing heat to the atmosphere. Unlike the locomotive, however, the body has a nearly uniform temperature throughout, namely, 98 degrees Fahr. The delicate regulation of temperature which is automatically maintained in the animal body is one of the wonders of physiology.

2. A second portion of energy is required to do the mechanical work of the body. When a locomotive hauls a loaded train up grade, or steams up grade alone, it is doing work in proportion to the total weight and the height to which it is carried. So when a man walks up hill or climbs a ladder he is lifting his body against the force of gravity, and hence doing work. If his weight be 200 pounds he is doing twice as much work as though he weighed only 100 pounds. If a man weighing 150 pounds climbs Bunker Hill Monument (220 feet), 33,000 foot-pounds of work will then be done;

and if he succeeds in making the ascent in one minute, he would be doing work at the rate of one full horse power for that minute. If he climbs a mountain two miles high in three hours and twelve minutes he would be doing work in so lifting his body at the rate of one quarter of a horse power. This is, of course, a faster rate of work than an average man could maintain. In all the functions of daily life the body is necessarily doing some mechanical work. Even dressing and eating require a certain expenditure of energy, and in ordinary business and manual labor the amount of mechanical work done is considerable. Moreover, a large amount of work is done by the heart in pumping the blood through the circulatory system, and by the chest in respiration. This, then, the internal and external work done, as in the locomotive, represents the second portion of energy derived from the food eaten.

3. The warm air, carbonic acid gas and water vapor passing away from the lungs in respiration carry with them a large amount of heat. This corresponds to the loss of heat in the locomotive through the smoke passing out the smokestack, and in both cases the loss is greater when work is being done and less during inaction. The refuse products of the body (as the ashes of the locomotive) also carry away heat. This is the third portion of heat and is a large one.

Work is done in the locomotive by the expanding steam in the cylinders of the engine. The steam is cooled as it expands. Hence heat disappears when work is done; that is, is converted into mechanical energy, and a steam engine is hence called a heat engine; an engine for converting heat into work, according to the law of the conservation of energy. As the pistons are pushed to and fro by the tremendous pressure of the expanding steam, the reciprocating motion is communicated to the great drivers of the engine by strong arms of steel. But how is work done in the body? That is a question of prime importance and of surpassing interest. When muscle contracts and force is exerted, as when the body is lifted or an oar is pulled, muscular tissue (or material stored in muscular tissue) is oxidized; that is, burned, and heat is produced; yet not as much heat appears as would have appeared on the combustion of the same amount of body material if no work had been done. Apparently, then, heat has been converted into work. But we cannot trace the process with the same clearness as in the cylinder of a steam engine. Whether the potential energy of the body material is directly converted into work, or whether combustion first produces heat and a part of this heat is then converted into work, we do not know. In other words, we do not know whether the animal body as a machine for doing mechanical work is a heat engine or some other kind of engine. This is a fundamental question, as well as a very difficult one, and to a student of thermodynamics and physiology it prompts all sorts of speculation.

When one tries to picture to himself how the potential energy of food or body tissue can be directly converted into mechanical work, he is apt to turn to the other alternative and imagine that in some way the body is a heat engine. For we know that heat results from the oxidation of tissue, and we also know how heat can be converted into mechanical work. But we are at once confronted with a difficulty. One of the fundamental laws of thermodynamics requires that when heat is converted into work there shall be a difference of temperature between the source of heat and the place to which the heated material employed passes after doing the work. In other words, in a heat engine, whatever the mechanism, there must be a fall of temperature, which is greater as the relative amount of work, or efficiency, is greater. In the human body the efficiency perhaps surpasses that of the best steam engines; hence there should be a fall of temperature comparable with that between the boiler and condenser of a steam engine. This may be 100 degrees or more, and we do not know of any such difference of temperature in the body. Indeed, we know, on the contrary, that the temperature of the body is remarkably uniform, as already stated. It is possible, however, that there are molecular differences of large amount. In other words, if we could make an ultra-microscopic survey of temperature in a muscle during contraction, there might be found places of high temperature where combustion was occurring, and all the requirements of a heat engine of molecular dimensions fulfilled. But this is a matter of speculation. The process may yet be found to be electrical, or something else quite different from that of a steam engine.

We thus find between the animal body and a locomotive engine a striking parallel. In many particulars the chemical and physical processes going on in the latter are found also in the former. In both, the fundamental law of the conservation of energy is strictly observed. Nevertheless, the animal body considered simply as a machine is far more complex in its structure and operation than the engine, and far more of mystery envelops its working. Much remains for the chemist and physicist and physiologist to reveal, and no more fascinating field of research exists.

CHAPTERS ON THE STARS.

By Professor SIMON NEWCOMB, U. S. N.

THE SPECTRA OF THE STARS.

THE principles on which spectrum analysis rests can be stated so concisely that I shall set them forth for the special use of such readers as may not be entirely familiar with the subject. Every one knows that when the rays of the sun pass through a triangular prism of glass or other transparent substance they are unequally refracted, and thus separated into rays of different colors. These colors are not distinct, but each runs into the other by insensible gradations, from deep red through orange, yellow, green and blue to a faint violet.

This result is due to the fact that the light of the sun is composed of rays of an infinite number of wave-lengths, or, as we might express it, of an infinite number of shades of color, since to every wave-length corresponds a definite shade. Such a spreading out of elementary colors in the form of a visible sheet is called a spectrum. By the spectrum of an incandescent object is meant the spectrum formed by the light emitted by the object when passed through a refracting prism, or otherwise separated into its elementary colors. The interest and value which attach to the study of spectra arise from the fact that different bodies give different kinds of spectra, according to their constitution, their temperature and the substances of which they are composed. In this manner it is possible, by a study of the spectrum of a body, to reach certain inferences respecting its constitution.

In order that such a study should lead to a definite conclusion, we must recall that to each special shade of color corresponds a definite position in the spectrum. That is to say, there is a special kind of light having a certain wave-length and therefore a certain shade which will be refracted through a certain fixed angle, and will therefore fall into a definite position in the spectrum. This position is, for every possible kind of light, expressed by a number indicating its wave-length.

If we form a spectrum with the light emitted by an ordinary incandescent body, a gaslight for example, we shall find the series of colors to be unbroken from one end of the spectrum to the other. That is to say, there will be light in every part of the spectrum. Such a spectrum is said to be continuous. But if we form the spectrum by means of sunlight, we shall find the spectrum to be crossed by a great number of more or less dark lines. This shows that in the spectrum of the sun light of certain definite wave-lengths is wholly or partly wanting. This fact has been observed for more than a century, but its true significance was not seen until a comparatively recent time.

If, instead of using the light of the sun, we form a spectrum with the light emitted by an incandescent gas, say hydrogen made luminous by the electric spark, we shall find that the spectrum consists only of a limited number of separate bright lines, of various colors. This shows that such a gas, instead of emitting light of all wave-lengths, as an incandescent solid body does, principally emits light of certain definite wave-lengths.

It is also found that if we pass the light of a luminous solid through a sufficiently large mass of gas, cooler than the body, the spectrum, instead of being entirely continuous, will be crossed with dark lines like that of the sun. This shows that light of certain wave-lengths is absorbed by the gas. A comparison of these dark lines with the bright lines emitted by an incandescent gas led Kirchhoff to the discovery of the following fundamental principle:

Every gas, when cold, absorbs the same rays of light which it emits when incandescent.

An immediate inference from this law is that the dark lines seen in the spectrum of the sun are caused by the passage of the light through gases either existing on the sun or forming the atmosphere of the earth. A second inference is that we can determine what these gases are by comparing the position of the dark lines with that of the bright lines produced by different gases when they are made incandescent. Hence arose the possibility of spectrum analysis, a method which has been applied with such success to the study of the heavenly bodies.

So far as the general constitution of bodies is concerned, the canons of spectrum analysis are these:

Firstly, when a spectrum is formed of distinct bright lines, the light which forms it is emitted by a transparent mass of glowing gas.

Secondly, when a spectrum is entirely continuous the light emanates from an incandescent solid, from a body composed of solid particles, which may be ever so small, or from a mass of incandescent gas so large and dense as not to be transparent through and through.

Thirdly, when the spectrum is continuous, except that it is crossed by fine dark lines, the body emitting the light is surrounded by a gas cooler than itself. The chemical constitution of this gas can be determined by the position of the lines.

Fourthly, if, as is frequently the case, a spectrum is composed of an irregular row of bright and shaded portions, the body is a compound one, partly gaseous and partly solid.

It will be seen from the preceding statement that, in reality, a mass of gas so large as not to be transparent cannot be distinguished from a solid. It is therefore not strictly correct to say, as is sometimes done, that an incandescent gas always gives a spectrum of bright lines. It will give such a spectrum only when it is transparent through and through.[C]

[C] As this principle is not universally understood, it may be well to remark that it results immediately from Kirchoff’s law of the proportionality between the radiating and absorbing powers of all bodies for light of each separate wave-length. When a body, even if gaseous in form, is of such great size and density that light of no color can pass entirely through it, then the consequent absorption by the body of light of all colors shows that throughout the region where the absorption occurs there must be an emission of light of these same colors. Thus light from all parts of the spectrum will be emitted by the entire mass.

A gaseous mass, so large as to be opaque, would, if it were of the same temperature inside and out, give a continuous spectrum, without any dark lines. But the laws of temperature in such a mass show that it will be cooler at the surface than in the interior. This cooler envelope will absorb the rays emanating from the interior as in the case when the latter is solid. We conclude, therefore, that the fact that the great majority of stars show a spectrum like that of the sun, namely, a continuous one crossed by dark lines, does not throw any light on the question whether the matter composing the body of the star is in a solid, liquid or gaseous state. The fact is that the most plausible theories of the constitution of the sun lead to the conclusion that its interior mass is really gaseous. Only the photosphere may be to a greater or less extent solid or liquid. The dark lines that we see in the solar spectrum are produced by the absorption of a comparatively thin and cool layer of gas resting upon the photosphere. Analogy as well as the general similarity of the spectra lead us to believe that the constitution of most of the stars is similar to that of the sun.

CLASSIFICATION OF STELLAR SPECTRA.

When the spectra of thousands of stars were recorded for study, such a variety was found that some system of classification was necessary. The commencement of such a system was made by Secchi in 1863. It was based on the observed relation between the color of a star and the general character of its spectrum.

Arranging the stars in a regular series, from blue in tint through white to red, it was found that the number and character of the spectral lines varied in a corresponding way. The blue stars, like Sirius, Vega and α Aquilæ, though they had the F lines strong, as well as the two violet lines H, had otherwise only extremely fine lines. On the other hand, the red stars, like α Orionis and α Scorpii, show spectra with several broad bands. Secchi was thus led to recognize three types of spectra, as follows:

The first type is that of the white or slightly blue stars, like Sirius, Vega, Altair, Rigel, etc. The typical spectrum of these stars shows all seven spectral colors, interrupted by four strong, dark lines, one in the red, one in the bluish green, and the two others in the violet. All four of these lines belong to hydrogen. Their marked peculiarity is their breadth, which tends to show that the absorbing layer is of considerable thickness or is subjected to a great pressure. Besides these broad rays, fine metallic rays are found in the brighter stars of this type. Secchi considers that this is the most numerous type of all, half the stars which he studied belonging to it.

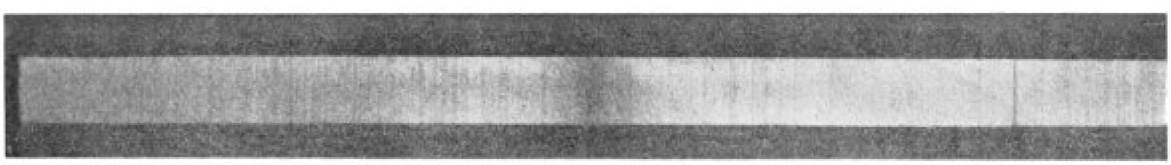

FIG. 1. SPECTRUM OF SIRIUS.

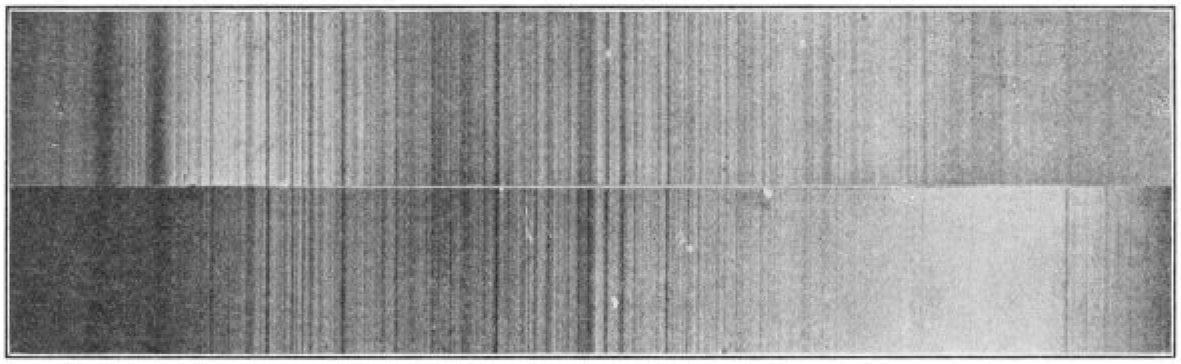

Fig. 2. Spectra of α Aurigæ and Sun.

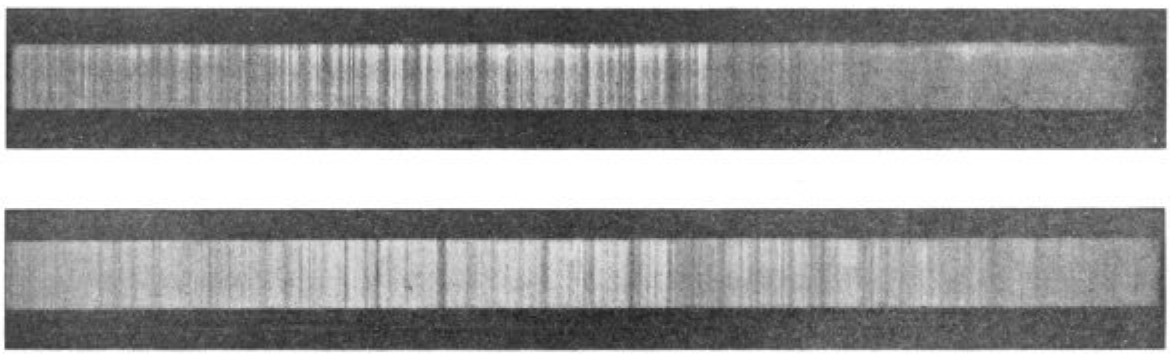

Fig. 3. Spectra of α Bootis and β Geminorum.

The second type is that of the somewhat yellow stars, like Capella, Pollux, Arcturus, Procyon, etc. The most striking feature of the spectrum of these stars is its resemblance to that of our sun. Like the latter, it is crossed by very fine and close black rays. It would seem that the more the star inclines toward red, the broader these rays become and the easier it is to distinguish them. We give a figure showing the remarkable agreement between the spectrum of Capella, which may be taken as an example of the type, and that of the sun.

The spectra of the third type, belonging mostly to the red stars, are composed of a double system of nebulous bands and dark lines. The latter are fundamentally the same as in the second type, the broad nebulous bands being an addition to the spectrum. α Herculis may be taken as an example of this type.

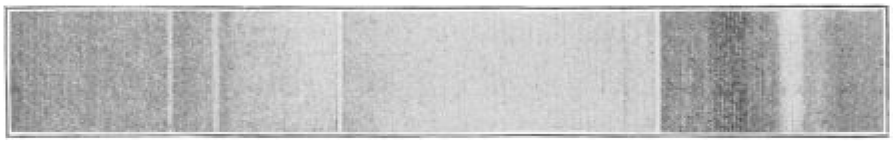

FIG. 7. SPECTRUM WITH BOTH BRIGHT AND DARK LINES.

It is to be remarked that, in these progressive types, the brilliancy of the more refrangible end of the spectrum continually diminishes relatively to that of the red end. To this is due the gradations of color in the stars.

To these three types Secchi subsequently added a fourth, given by comparatively few stars of a deep red color. The spectra of this class consist principally of three bright bands, which are separated by dark intervals. The brightest is in the green; a very faint one is in the blue; the third is in the yellow and red, and is divided up into a number of others.

To these types a fifth was subsequently added by Wolf and Rayet, of the Paris Observatory. The spectra of this class show a singular mixture of bright lines and dark bands, as if three different spectra were combined, one continuous, one an absorption spectrum, and one an emission spectrum from glowing gas. Less than a hundred stars of this type have been discovered. A very remarkable peculiarity, which we shall discuss hereafter, is that they are nearly all situated very near the central line of the Milky Way.

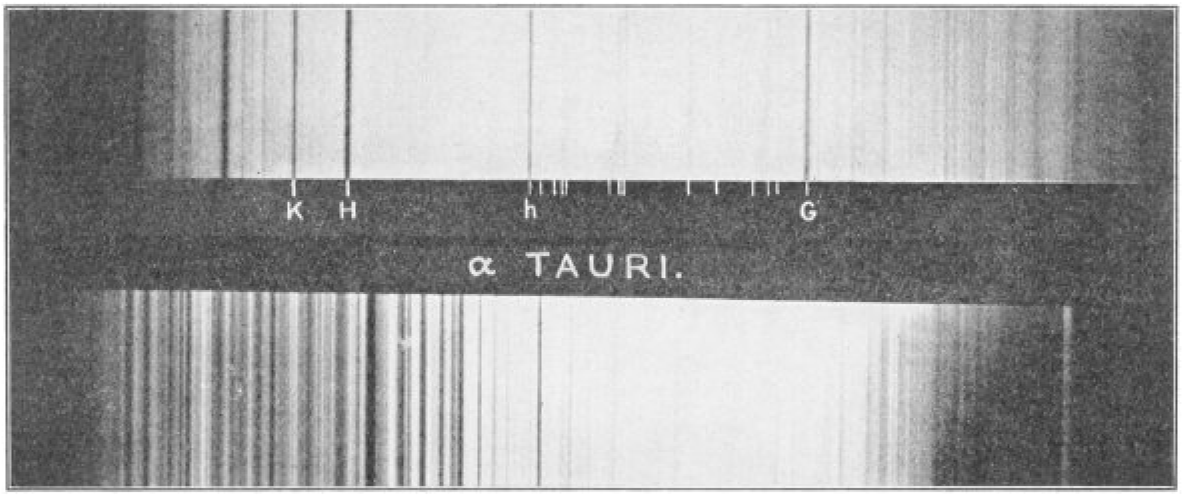

Fig. 4. Spectra of α Cygni and α Tauri.

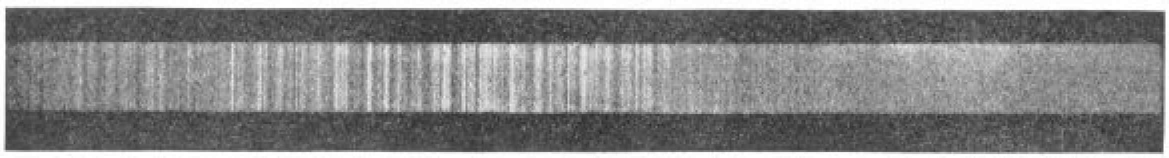

Fig. 5. Spectrum of α Orionis.

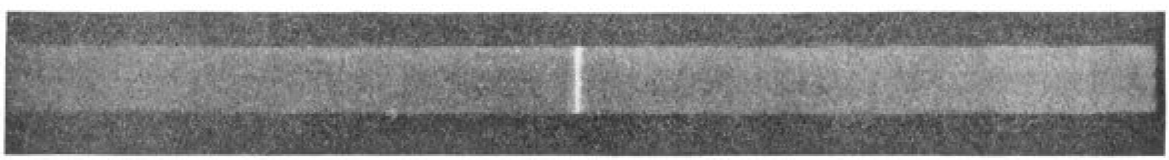

Fig. 6. Spectrum of γ Cassiopeiæ

Vogel proposed a modification of Secchi's classification, by subdividing each of his three types into two or three others, and including the Wolf-Rayet stars under the second type. His definitions are as follows:

Type I is distinguished by the intensity of the light in the more refrangible end of the spectrum, the blue and violet. The type may be divided into three subdivisions, designated *a*, *b* and *c*:

In I*a* the metallic lines are very faint, while the hydrogen lines are distinguished by their breadth and strength.

In I*b* the hydrogen lines are wanting.

In I*c* the lines of hydrogen and helium both show as bright lines. Stars showing this spectrum are now known as helium stars.

According to Vogel, the spectra of type II are distinguished by having the metallic lines well-marked and the more refrangible end of the spectrum much fainter than in the case of type I. He recognizes two subdivisions:

In II*a* the metallic lines are very numerous, especially in the yellow and green. The hydrogen lines are strong, but not so striking as in I*a*.

In II*b* are found dark lines, bright lines and faint bands. In this subdivision he includes the Wolf-Rayet stars, more generally classified as of the fifth type.

The distinguishing mark of the third type is that, besides dark lines, there are numerous dark bands in all parts of the spectrum, and the more refrangible end of the latter is almost wanting. There are two subdivisions of this type:

In III*a* the broad bands nearest the violet end are sharp, dark and well-defined, while those near the red end are ill-defined and faint. In III*b* the bands near the red end are sharp and well-defined; those toward the violet faint and ill-defined. The character of the bands is therefore the reverse of that in subdivision *a*.

This classification of Vogel is still generally followed in Germany and elsewhere. It is found, however, that there are star spectra of types intermediate to all these defined. Moreover, in each type the individual differences are so considerable that there is no well-defined limit to the number of classes that may be recognized. At the Harvard Observatory a classification quite different from that of Vogel has been used, but it is too detailed for presentation here. The stars of type II are frequently termed Capellan stars, or Solar stars. Certain stars of type I are termed Orion stars, owing to the number of stars of the type found in that constellation. The stars which show the lines of helium are known as helium stars. We mention these designations because they frequently occur in literature. It would, however, be outside the object of the present work to describe all these classifications in detail. We therefore confine ourselves to a few illustrations of spectra of the familiar types described by Secchi and Vogel.

There are many star spectra which cannot be included in any of the classes we have described. Up to the present time these are generally described as stars of peculiar spectra.

As the present chapter is confined to the more general side of the subject, we shall not attempt any description of special spectra. These, especially the peculiar spectra of the nebulæ, of new stars, of variable stars, etc., will be referred to, so far as necessary, in the chapters relating to those objects.

The most interesting conclusion drawn from observations with the spectroscope is that the stars are composed, in the main, of elements similar to those found in our sun. As the latter contains most of the elements found on the earth and few or none not found there, we may say that earth and stars seem to be all made out of like matter. It is, however, not yet easy to say that no elements unknown on the earth exist in the heavens. It would scarcely be safe to assume that, because the line of some terrestrial substance is found in the spectrum of a star, it is produced by that substance. It is quite possible that an unknown substance might show a line in appreciably the same position as that of some substance known to us. The evidence becomes conclusive only in the case of those elements of which the spectral lines are so numerous that when they all coincide with lines given by a star, there can be no doubt of the identity.

PROPER MOTIONS OF THE STARS.

We may assume that the stars are all in motion. It is true that only a comparatively small number of stars have been actually seen to be in motion; but as some motion exists in nearly every case where observations would permit of its being determined, we may assume the rule to be universal. Moreover, if a star were at rest at one time it would be set in motion by the attraction of other stars.

Statements of the motion from different points of view illustrate in a striking way the vast distance of the stars and the power of modern telescopic research. If Hipparchus or Ptolemy should rise from their sleep of 2,000 years—nay, if the earliest priests of Babylon should come to life again and view the heavens, they would not perceive any change to have taken place in the relative positions of the stars. The general configurations of the constellations would be exactly that to which they were accustomed. Had they been very exact observers they might notice a slight difference in the position of Arcturus; but as a general rule the unchangeability would have been manifest.

In dealing with the subject, the astronomer commonly expresses the motion in angular measurement as so many seconds per year or per century. The keenest eye would not, without telescopic aid, be able to distinguish between two stars whose apparent distance is less than 2′ or 120″ of arc. The pair of stars known as (ε) Lyræ are 3′ apart; yet, to ordinary vision they appear simply as a single star. To appreciate what 1″ of arc means we must conceive that the distance between these two stars is divided by 200. Yet this minute space is easily distinguished and accurately measured by the aid of a telescope of ordinary power.

On the other hand, if we measure the motions by terrestrial standards they are swift indeed. Arcturus has been moving ever since the time of Job at the rate of probably more than 200 miles per second—possibly 300 miles. Generally, however, the motion is much smaller, ranging from an imperceptible quantity up to 5, 10 or 20 miles a second. Slow as the angular motion is, our telescopic power is such that the motion in the course of a very few years (with Arcturus the motion in a few days) can be detected. As accurate determinations of positions of the stars have been made only during a century and a half, no motions can be positively determined except those which would become evident to telescopic vision in that period. Only about 3,000 stars have been accurately observed so long as this. In the large majority of cases the interval of observation is so short or the motion so slow that nothing can be asserted respecting the law of the motion.

The great mass of stars seem to move only a few seconds per century, but there are some whose motions are exceptionally rapid. The general rule is that the brighter stars have the largest proper motions. This is what we should expect, because in the general average they are nearer to us, and therefore their motion will subtend the greatest angle to the eye. But this rule is only one of majorities. As a matter of fact, the stars of largest proper motion happen to be low in the scale of magnitude. It happens thus because the number of stars of smaller magnitudes is so much greater than that of the brighter ones that the very small proportion of large proper motions which they offer over-balances those of the brighter stars.

The discovery of the star of greatest known proper motion was made by Kapteyn, of Groningen, in 1897, coöperating with Gill and Innes, of the Cape Observatory. While examining the photographs of the stars made at this institution, Kapteyn was surprised to notice the impression of a star of the eighth magnitude which at first could not be found in any catalogue. But on comparing different star lists and different photographs it soon became evident that the star had been previously seen or photographed, but always in slightly varying positions. An examination of the observed positions at various times showed that the star had a more rapid proper motion than any other yet known. Yet, great though this motion is, it would require nearly 150,000 years for the star to make a complete circuit of the heavens if it moved round the sun uniformly at its present rate.

The fact that the stars move suggests a very natural analogy to the solar system. In the latter a number of planets revolve round the sun as their center, each planet continually describing the same orbit, while the various planets have different velocities. Around several of the planets revolve one or more satellites. Were civilized men ephemeral, observing the planets and satellites only for a few minutes, these bodies would be described as having proper motions of their own, as we find the stars to have. May it not then be that the stars also form a system; that each star is moving in a fixed orbit performing a revolution around some far-distant center in a period which may be hundreds of thousands or hundreds of millions of years? May it not be that there are systems of stars in which each star revolves around a center of its own while all these systems are in revolution around a single center?

This thought has been entertained by more than one contemplative astronomer. Lambert’s magnificent conception of system upon system will be described hereafter. Mädler thought that he had obtained evidence of the revolution of the stars around Alcyone, the brightest of the Pleiades, as a center. But, as the proper motions of the stars are more carefully studied and their motion and direction more exactly ascertained, it becomes very clear that when considered on a large scale these conceptions are never realized in the actual universe as a whole. But there are isolated cases of systems of stars which are shown to be in some way connected by their having a common proper motion. We shall mention some of the more notable cases.

The Pleiades are found to move together with such exactness that up to the present time no difference in their proper motions has been detected. This is true not only of the six stars which we readily see with the naked eye, but of a much larger number of fainter ones made known by the telescope. It is an interesting fact, however, that a few stars apparently within the group do not partake of this motion, from which it may be inferred that they do not belong to the system. But there must be some motion among themselves, else the stars would ultimately fall together by their mutual attraction. The amount and nature of this motion cannot, however, be ascertained except by centuries of observation.

Another example of the same sort is seen in five out of the seven stars of Ursæ Major, or The Dipper. The stars are those lettered β, γ, δ, ε and θ. All five have a proper motion in R. A. of nearly 8″ per century, while in declination the movements are sometimes positive and sometimes negative; that is to say, some of the stars are apparently lessening their distance from the pole, while others are increasing it. But when we project the motions on a map we find that the actual direction is very nearly the same for all five stars, and the reason why some move slightly to the north and others slightly to the south is due to the divergence of the circles of right ascension. It is worthy of remark that the community of motion is also shown by spectroscopic observations of the radial motions described below.

The five stars in question are all of the second magnitude except δ, which is of the third. It is a curious fact that no fainter stars than these five have been found to belong to the system.

From a study of these motions Höffler has concluded that the five stars lie nearly in the same plane and have an equal motion in one and the same direction. From this hypothesis he has attempted to make a determination of their relative and actual distances. The result reached in this way cannot yet, however, be regarded as conclusive.

There are three stars in Cassiopeia, β, η and μ each having a large proper motion in so nearly the same direction that it is difficult to avoid at least a suspicion of some relation between them. The angular motions are, however, so far from equal that we cannot regard the relation as established.

In the constellation Taurus, between Aldebaran and the Pleiades, most of the stars which have been accurately determined seem to have a common motion. But these motions are not yet so well ascertained that we can base anything definite upon them. They show a phenomenon which Proctor very aptly designated as star-drift.

The systems we have just described comprise stars situated so far apart that, but for their common motion, we should not have suspected any relation between them. The community of origin which their connection indicates is of great interest and importance, but the question belongs to a later chapter.

MOTIONS IN THE LINE OF SIGHT, OR RADIAL MOTIONS.

No achievement of modern science is more remarkable than the measurement of the velocity with which stars are moving to or from us. This is effected by means of the spectroscope through a comparison of the position of the spectral lines produced by the absorption of any substance in the atmosphere of the star with the corresponding lines produced by the same substance on the earth. The principle on which the method depends may be illustrated by the analogous case of sound. It is a familiar fact that if we stand alongside a railway while a locomotive is passing us at full speed, and at the same time blowing a whistle, the pitch of the note which we hear from the whistle is higher as the engine is approaching than after it passes. The reason is that the pitch of a sound depends upon the number of sound beats per second.

```
A      B                                         X
*      .      .      .      .      .      .      .
       *     .     .     .     .     .    .      .
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Now, we may consider the waves which form light when they strike our apparatus as beats in the ethereal medium which follow each other with extraordinary rapidity, millions of millions in a second, moving forward with a definite velocity of more than 186,000 miles a second. Each spectral line produced by a chemical element shows that that element, when incandescent, beats the ether a certain number of times in a second. These beats are transmitted as waves. Since the velocity is the same whether the number of beats per second is less or greater, it follows that, if the body is in motion in the direction in which it emits the light, the beats will be closer together than if it is at rest; if moving away they will be further apart. The fundamental fact on which this result depends is that the velocity of the light-beat through the ether is independent of the motion of the body causing the beat. To show the result, let A be a luminous body at rest; let the seven dots to the right of A be the crests of seven waves or beats, the first of which, at the end of a certain time, has reached X. The wave-length will then be one seventh the distance A X. Now, suppose A in motion toward X with such speed that, when the first beat has reached X, A has reached the point B. Then the seven beats made by A while the first beat is traveling from A to X, and A traveling from A to B, will be crowded into the space B X, so that each wave will be one seventh shorter than before. In other words, the wave-lengths of the light emitted by any substance will be less or greater than their normal length, according to the motion of the substance in the direction in which its light is transmitted, or in the opposite direction.

The position of a ray in the spectrum depends solely on the wave-length of the light. It follows that the rays produced by any substance will be displaced toward the blue or red end of the spectrum, according as the body emitting or absorbing the rays is moving towards or from us. This method of determining the motions of stars to or from us, or their velocity in the line of sight from us to the star, was first put into practice by Mr.—now Sir William—Huggins, of London. The method has since been perfected by photographing the spectrum of a star, or other heavenly body, side by side with that of a terrestrial substance, rendered incandescent in the tube of a telescope. The rays of this substance pass through the same spectroscope as those from the star, so that, if the wave-lengths of the lines produced by the substance were the same as those found in the star spectrum, the two lines would correspond in position. The minute difference found on the photographic plate is the measure of the velocity of the star in the line of sight.

It will be seen that the conclusion depends on the hypothesis that the position of any ray produced by a substance is affected by no cause but the motion of the substance. How and when this hypothesis may fail is a very important question. It is found, for example, that the position of a spectral ray may be altered by compressing the gas emitting or absorbing the ray, and it may be inquired whether the results for motion in the line of sight may not be vitiated by the absorbing atmosphere of the star being under heavy pressure, thus displacing the absorption line.

To this it may be replied that, in any case, the outer layers of the atmosphere, through which the light must last pass, are not under pressure. How far inner portions may produce an absorption spectrum we cannot discuss at present, but it does not seem likely that serious errors are thus introduced in many cases.

These measures require apparatus and manipulation of extraordinary delicacy, in order to avoid every possible source of error. The displacement of the lines produced by motion is in fact so minute that great skill is required to make it evident, unless in exceptional cases. The Mills spectrograph of the Lick Observatory in the hands of Professor Campbell has, notwithstanding these difficulties, yielded results of extraordinary precision. Quite a number of investigators at some leading observatories of Europe and America are pursuing the work of determining these motions. The determinations have almost necessarily been limited to the brighter stars, because, owing to the light of the star being spread over so broad a space in the spectrum, instead of being concentrated on a point, a far longer exposure is necessary to photograph the spectrum of a star than to photograph the star itself. The larger the telescope the fainter the star whose spectrum can be photographed. Vogel, of Potsdam, who has made the most systematic sets of these measures that have yet appeared, included few stars fainter than the second magnitude. With the largest telescopes the spectro of stars down to about the fifth magnitude may be photographed; beyond this it is extremely difficult to go. The limit will probably be reached by the spectrograph of the Yerkes Observatory, which is now being put into operation by Professors Hale and Frost.

THE MOTION OF THE SUN.

When a star is found to be seemingly in motion, as described in the last section, we may ascribe the phenomenon to a motion either of the star itself or of the observer. In fact no motion can be determined or defined except by reference to some body supposed to be at rest. In the case of any one star, we may equally well suppose the star to be at rest and the observer in motion, or the contrary. Or we may suppose both to have such motions that the difference of the two shall represent the apparent movement of the star. Hence our actual result in the case of each separate star is a relation between the motion of the star and the motion of the sun.

I say the motion of the sun and not of the earth, because although the observer is actually on the earth, yet the latter never leaves the neighborhood of the sun, and, as a matter of fact, the ultimate result in the long run must be a motion relative to the sun itself as if we made our observations from that body. The question then arises whether there is any criterion for determining how much of the apparent motion of any given star should be attributed to the star itself and how much to a motion of the sun in the opposite direction.

If we should find that the stars, in consequence of their proper motions, all appeared to move in the same direction, we would naturally assume that they were at rest and the sun in motion. A conclusion of this sort was first reached by Herschel, who observed that among the stars having notable proper motions there was a general tendency to move from the direction of the constellation Hercules, which is in the northern hemisphere, towards the opposite constellation Argo, in the southern hemisphere.

Acting on this suggestion, subsequent astronomers have adopted the practice of considering the general average of all the stars, or a position which we may regard as their common center of gravity, to be at rest, and then determining the motion of the sun with respect to this center. Here we encounter the difficulty that we cannot make any absolute determination of the position of any such center. The latter will vary according to what particular stars we are able to include in our estimate. What we can do is to take all the stars which appear to have a proper motion, and determine the general direction of that motion. This gives us a certain point in the heavens toward which the solar system is traveling, and which is now called the *solar apex*, or the apex of the solar way.

The apparent motion of the stars due to this motion of the solar system is now called their *parallactic motion*, to distinguish it from the actual motion of the star itself.

The interest which attaches to the determination of the solar apex has led a great number of investigators to attempt it. Owing to the rather indefinite character of the material of investigation, the uncertainty of the proper motions, and the additions constantly made to the number of stars which are available for the purpose in view, different investigators have reached different results. Until quite recently, the general conclusion was that the solar apex was situated somewhere in the constellation Hercules. But the general trend of recent research has been to place it in or near the adjoining constellation Lyra. This change has arisen mainly from including a larger number of stars, whose motions were determined with greater accuracy.

Former investigators based their conclusions entirely on stars having considerable proper motions, these being, in general, the nearer to us. The fact is, however, that it is better to include stars having a small proper motion, because the advantage of their great number more than counterbalances the disadvantage of their distance.

The conclusions reached by some recent investigators of the position of the solar apex will now be given. We call A the right ascension of the apex; D its declination.

Prof. Lewis Boss, from 273 stars of large proper motion found

$$A = 283^\circ.3;\ D = 44^\circ.1.$$

If he excluded the motions of 26 stars which exceeded 40″ per century the result was

$$A = 288^\circ.7;\ D = 51^\circ.5.$$

A comparison of these numbers shows how much the result depends on the special stars selected. By leaving out 26 stars the apex is changed by 5° in R. A., and 7° in declination.

It is to be remarked that the stars used by Boss are all contained in a belt four degrees wide, extending from 1° to 5° north of the equator.

Dr. Oscar Stumpe, of Berlin, made a list of 996 stars having proper motions between 16″ and 128″ per century. He divided them into three groups, the first including those between 16″ and 32″; the second between 32″ and 64″; the third between 64″ and 128″. The number of stars in each group and the position of the apex derived from them are as follows:

Gr.	I, 551 stars; A =	287°.4;	D = × 45°.0
	II, 339	282°.2	43°.5
	III, 106	280°.2	33°.5

Porter, of Cincinnati, made a determination from a yet larger list of stars with results of the same general character.

These determinations have the advantage that the stars are scattered over the entire heavens, the southern as well as the northern ones. The difference of more than 10° between the position derived from stars with the largest proper motions, and from the other stars, is remarkable.

The present writer, in a determination of the precessional motion, incidentally determined the solar motion from 2,527 stars contained in Bradley's Catalogue which had small proper motions, and from about 600 more having larger proper motions. Of the latter the declinations only were used. The results were:

From small motions:	A = 274°.2;	D = × 31°.2
From large motions:	276°.9	31°.4

From all these results it would seem that the most likely apex of the solar motion is toward the point in

Right Ascension,	280°
Declination,	38° north.

This point is situated in the constellation Lyra, about 2° from the first magnitude star Vega. The uncertainty of the result is more than this difference, four or five degrees at least. We may therefore state the conclusion in this form:

The apex of the solar motion is in the general direction of the constellation Lyra, and probably very near the star Vega, the brightest of that constellation.

It must be admitted that the wide difference between the position of the apex from large and from small proper motions, as found by Porter, Boss and Stumpe, require explanation. Since the apparent motions of the stars are less the greater their distance, these results, if accepted as real, would lead to the conclusion that the position of the solar apex derived from stars near to us was much further south than when derived from more distant stars. This again would indicate that our sun is one of a cluster or group of stars, having, in the general average, a different proper motion from the more distant stars. But this conclusion is not to be accepted as real until the subject has been more exhaustively investigated. The result may depend on the selection of the stars; and there is, as yet, no general agreement among investigators as to the best way of making the determination.

The next question which arises is that of the velocity of the solar motion. The data for this determination are more meagre and doubtful than those for the direction of the motion. The most obvious and direct method is to determine the parallactic motion of the stars of known parallax. Regarding any star 90° from the apex of the solar motion as in a state of absolute rest, we have the obvious rule that the quotient of its parallactic motion during any period, say a century, divided by its parallax, gives the solar motion during that period, in units of the earth’s distance from the sun. In fact, by a motion of the sun through one such unit, the star would have an apparent motion in the opposite direction equal to its annual parallax. If the star were not 90° from the apex we can easily reduce its observed parallactic motion by dividing it by the sign of its actual distance from the apex.

Since every star has, presumably, a proper motion of its own, we can draw no conclusion from the apparent motion of any one star, owing to the impossibility of distinguishing its actual from its parallactic motion. We should, therefore, base our conclusion on the mean result from a great number of stars, whose average position or center of mass we might assume to be at rest. Here we meet the difficulty that there are only about 60 stars whose parallaxes can be said to be determined; and one-half of these are too near the apex, or have too small a parallax, to permit of any conclusion being drawn from them.

A second method is based on the spectroscopic measures of the motion of stars in the line of sight, or the line from the earth to the star. A star at rest in the direction of the solar apex would be apparently moving toward us with a velocity equal to that of the solar motion. Assuming the center of mass of all the stars observed to be at rest, we should get the solar motion from the mean of all.

Thus far, however, there are only about 50 stars whose motions in the line of sight have been used for the determination, so that the data are yet more meagre than in the case of the proper motions. From them, however, using a statistical method Kapteyn has derived results which seem to show that the actual velocity of the solar system through space is about 16 kilometres, or 10 statute miles, per second.

THE PSYCHOLOGY OF RED. (II.)

BY HAVELOCK ELLIS.

THE facts and considerations we have passed in review fairly indicate the physiological and psychological preëminence of red among the colors of the spectrum to which we are sensitive. What is the cause of that preëminence?

It seems to me that two orders of causes have coöperated to produce this predominant influence, one physical and depending on the special effects of the long-waved portion of the spectrum on living matter, the other psychological and resulting from the special environmental influences to which man, and to some extent even the higher animals generally, have been subjected. It is possible that these two influences blend together and cannot at any point be disentangled; it is possible that acquired aptitude may be inherited or that what seem to be acquired aptitudes are really perpetuated congenital variations; but on the whole the two influences are so distinct that we may deal with them separately.

On the physical side the influence of the red rays, although there is much evidence showing that it may be traced throughout the whole of organic nature, is certainly most strongly and convincingly exhibited on plants. The characteristic greenness of vegetation alone bears witness to this fact. The red rays are life to the chlorophyll-bearing plant, the violet rays are death. A meadow, it has been justly said, is a vast field of tongues of fire greedily licking up the red rays and vomiting forth the poisonous bile of blue and yellow. An experiment of Flammarion's has beautifully shown the widely different reaction of plants to the red and violet rays. At the climatological station at Juvisy he constructed four greenhouses—one of ordinary transparent glass, another of red glass, another of green, the fourth of dark blue. The glass was monochromatic, as carefully tested by the spectroscope, and dark blue was used instead of violet because it was impossible to obtain a perfect violet glass. These were all placed under uniform meteorological and other conditions, and from certain plants such as the sensitive plant, previously sown on the same day in the same soil, eight of each kind were selected, all measuring 27 millimetres, and placed by two and two in the four greenhouses on the 4th of July. On the 15th of August there were notable differences in height, color and sensitiveness, and these differences continued to become marked; photographs of the plants on the 4th of October showed that while those under blue glass had made no progress, those under red glass had attained extraordinary development, red light acting like a manure. While those under blue glass became insensitive, under red glass the sensitive plants had become excessively sensitive to the least breath. They also flowered, those under transparent glass being vigorous and showing buds, but not flowering. The foliage under red glass was very light, under blue darkest. Similar but less marked effects were found in the case of geraniums, strawberries, etc. The strawberries under blue glass were no more advanced in October than in May; though not growing old their life was little more than a sleep. It appears, however, that the stimulating influence of red light fails to influence favorably the ripening of fruit. Zacharewiez, professor of agriculture at Vaucluse, has found that red, or rather orange, produces the greatest amount of vegetation, while as regards fruit, the finest and earliest was grown under clear glass, violet glass, indeed, causing the amount of fruit to increase but at the expense of the quality.

Moreover, the lowest as well as the highest plants participated in this response to the red rays, and in even a more marked degree, for they perish altogether under the influence of the violet rays. Marshall Ward and others have shown that the blue, violet and ultra-violet rays, but no others, are deleterious to bacteria. Finsen has successfully made use of this fact in the treatment of bacterial skin diseases. Reynolds Green has shown that while the ultra-violet rays have a destructive influence on diastase, the red rays have a powerfully stimulating effect, increasing diastase and converting zymogen into diastase.

While the influence of the red rays on the plant is thus so enormous and easily demonstrated, the physical effects of red on animals seem to be even opposite in character, although results of experiments are somewhat contradictory. Béclard found that the larvæ of the flesh fly raised under violet glass were three fourths larger than those raised under green glass; the order was violet, blue, red, yellow, white, green. In the case of tadpoles, Yung found that violet or blue was especially favorable to the growth of frogs; he also found that fish hatch most rapidly under violet light. Thus the influence that is practically death to plants is that most favorable to life in animals. Both effects, however, as Davenport truly remarks in his ‘Experimental Morphology,’ when summing up the results of investigations, are due to the same chemical metabolic changes, but while plants succumb to the influence of the violet rays, animals, being more highly organized, are able to take advantage of them and flourish.

At the same time the influence of violet rays on animal tissue is by no means invariably beneficial; they are often too powerful a stimulant. That the violet rays have an influence on the human skin which in the first place, at all events, is destructive and harmful in a high degree, is now clearly established by the observations and experiments of Charcot, Unna, Hammer, Bowles and others, while Finsen has made an important advance in the treatment of disease based on this fact. The conditions called ‘sun-burn,’ ‘snow-burn,’ ‘snow-blindness,’ for instance, which may affect even travelers on snow-fields and Arctic explorers, are now known to be wholly due to the violet and not to the red rays. Unna’s device of wearing a yellow veil, and Bowles’s plan of painting the skin brown, thus shutting off the violet rays, suffice to prevent sun-burn. The same effect is also obtained by nature, which under the stress of sunlight, and largely through the irritation of the violet rays themselves, weaves a pigmentary veil of yellow and brown on the skin, which thus protects from the further injurious influence of the violet rays and renders the sunlight a source of less alloyed joy and health.

That the presence of the red rays, or at all events the exclusion of the violet, is of great benefit in many skin diseases seems to be now beyond doubt. This has been shown by Finsen in his treatment of smallpox in red rooms; it appears that it was also known in the Middle Ages as well as in Japan, Tonquin and Roumania, red bed-covers, curtains or carpets being used to obtain the effect. Under the treatment by red light not only is the skin enabled to heal healthfully without scarring, but the whole course of the disease is beneficially affected and abbreviated, the fever is diminished and also the risk of complications. Another physician has discovered that a similar beneficial effect is produced by red light in measles. A child with a severe attack of measles was put into a room with red blinds and a photographic lamp. The rash speedily disappeared and the fever subsided, the child complaining only of the absence of light; the blinds were consequently removed, and the fever, rash and prostration returned, to disappear again when the blinds were resumed.

Whether red light, or the exclusion of violet, exerts a beneficial influence on the hæmoglobin of the blood and on metabolism generally has not been distinctly proved, but it seems to me to be indicated by such experiments as those of Marti published a few years ago in the *Atti dei Lincei*. This investigator found that while feeble irritation of the skin promotes the formation of blood corpuscles, strong irritation diminishes the blood corpuscles and also the hæmoglobin; at the same time he found that darkness also diminishes the number of red corpuscles, while continued exposure to intense light (even at night the electric light, which, however, is rich in violet rays) favors increased formation of red corpuscles, and in some degree of hæmoglobin. Finsen has shown that inflammation of the skin caused by chemical or violet light leads to contraction of the red corpuscles.

This brings us to the consideration of the influence of the red rays on the nervous system. From time to time experiments have been made as to the influence of various colored lights, chiefly on the insane, as first suggested by Father Secchi in 1895. Even yet, however, the specific mental influences of the various colors are not quite clear. It has been found by some that the red rays are far more soothing and comfortable, less irritating, than the total rays of uncolored light, and Garbini found that angry infants were soothed by the light through red glass, only slightly by that through green and not at all by other colored light. On the other hand, it is stated that a well-known dry plate manufacturer at Lyons was obliged to substitute green-colored glass in the windows of his large room for the usual red because the work people sang and gesticulated all day and the men made love to the women, while under the influence of green glass (which also allows yellow rays to pass) they became quiet and silent and seemed less fatigued when they left off work. We need not attach much value to these statements, but in this connection it is interesting to refer to the results obtained some years ago by Féré and recorded in his ‘Sensation et

Mouvement.’ Experimenting on normal subjects as well as on nervous subjects, who were found more sensitive, with colored light passed through glass or sheets of gelatine, he found notable differences in muscular power, measured by the dynamometer, and in the circulation as measured by plethysmographic tracings of the forearm under the influence of different colors. He found in this manner with one subject whose normal muscular power was represented by 23 that blue light increased his power to 24, green to 28, yellow to 30, orange to 35 and red to 42. The dynamogenic powers of the different colors were thus found to rank in the spectral order, red representing the climax of energy, or, as Féré puts it, “the intensity of the visual sensation varies as the vibrations.” Féré found that colors need not be perceived in order to show their influence, thus proving the purely physical nature of that influence, for in a subject who was unable to see colors with one eye, the color stimulus had the same dynamogenic effect whether applied to the seeing or the defective eye. Increase of volume of blood in the limbs, measured by the plethysmograph, so far as we can rely on Féré’s experiments, ran parallel with the influence on muscular power, culminating with red, so that no metaphor is involved, Féré remarks, when we speak of red as a ‘warm’ color. On the insane the results attained by the use of colored glass do not seem to be quite coherent. Some of the earlier observers described the beneficial effects of blue glass in soothing maniacs. Pritchard Davies, however, was not able to find that red light had any beneficial effect, though on some cases blue had, while Roffegean found that, in the case of a somber and taciturn maniac who could rarely be persuaded to eat, three hours in a red-lighted room produced a markedly beneficial effect, and a man with delusions of persecution became quite rational and was even in a condition to be sent home after a few days in the same room. He also found that a violent maniac wearing a strait jacket, after a few hours in a room with blue glass windows became quite calm and gave no further trouble. Osburne has found, after many years’ experience, that in the absence of structural disease violet light (for from three to six hours) is most useful in the treatment of excitement, sleeplessness and acute mania; red he has found of some benefit, though to a much less degree in such cases (it must be remarked that violet light as usually applied is not free from red), while he has not found any color with which he has tried experiments (red, orange or violet) of benefit in melancholia. The significance of these facts is not altogether clear; the influence, as Pritchard

Davies concluded, seems to be largely moral, though it may be that the colors of long wave-length are tonic and those of short wave-length sedative.

So far I have been chiefly concerned to point out that the immense emotional impressiveness of red has a basis in physical laws, being by no means altogether a matter of environmental associations. It is true that the two groups of influences overlap, and that we can not always distinguish them. We can not be sure that the greater sensitiveness to the red rays may not have been emphasized in the organism, not necessarily as the result of inherited acquirement, but probably as the perpetuation of a variation of sensibility, found beneficial in an environment where red was liable to be especially associated with objects that were to be avoided as terrible or sought as useful. In this way the physical and environmental factors would run in a circle.

We have to bear this consideration in mind when we take into account the susceptibilities of animals, especially of the higher animals, to red. The color sense, it is well known, is widely diffused among animals; indeed this fact has been brought forward, especially by Pouchet, to prove that there can have been no color evolution in man; this it can scarcely be said to show, since evolution does not run in a straight line, and it is quite conceivable and even probable that the ancestors of man were less dependent than many lower animals, for the means of living, on a highly developed color sense. Thus a color sense that among some creatures is so highly developed as to include even the ultra-violet rays, was among our own ape-like ancestors either never developed or partially lost.

Graber, in his important investigation into the color sense of animals, showed that of fifty animals studied by him forty showed strong color preferences in their places of abode. In general he found, without being able to explain the fact, that animals which prefer the dark are red lovers, those which prefer the light are blue lovers. The common worm, with head and tail cut off, still preferred red to blue nearly as much as when uninjured. (This would seem to indicate the same kind of susceptibility to unaccustomed violet rays which we have already encountered in the phenomenon of sun-burn.) The triton and cochineal, with eyes removed and heads covered with wax, still had delicate sense for color and brightness. The flea infesting the dog had a finer color sense than the bee, while nearly all the animals Graber investigated were more or less sensitive to the ultra-red rays.

Among insects it scarcely appears, nor should we expect that there would be any peculiarly marked predilection or aversion for red. Cockerell and F. W. Anderson, from observations in various parts of the United States, believe that yellow (*i. e.*, the brightest color) is the most attractive to insects, and the former doubts whether insects can distinguish red from yellow. Among the higher animals, and even among fishes and birds, there is not only a color sense, but a highly emotionalized color sense, and red appears to be usually the color that arouses the emotion. There is a proverb, ‘Women and mackerel are caught by red,’ and perch is also said to be caught by red bait. Sparrows appear to be repelled by red; the case is reported of a hen sparrow, kept in captivity for ten years, which though otherwise a fearless bird ‘would on seeing scarlet show painful signs of distress and faint away.’ The lady who records this observation has noted the same repugnance to red, though in a less marked degree, in other sparrows, one of which showed a predilection for blue objects, and she remarks that when feeding outdoor sparrows from the window they flew away when she wore a red jacket, while a blue jacket inspired them with confidence; other birds, she found, except a cockatoo, were unaffected by colors. Red, it is well known, is very obnoxious to turkey cocks, while the fury aroused in various quadrupeds by red was known at a very early period; Seneca referred to it in the case of the bull, the most familiar example; it is seen in buffaloes, sometimes in horses, and also, it is said, in the hippopotamus.

The phenomena of color aversion and color predilection among insects may possibly be in some degree a matter of physical sensibility, varying according to the creature’s tissues, habitat and needs, but as we approach the vertebrates and especially the mammals there can be little doubt that it is mainly a matter of environment and association; in other words, that it is accounted for by the color of food, the color of blood and the color of the chief secondary sexual characters.

Let us, however, confine ourselves to man, and consider what are the chief colored objects that are of most vital concern to the human and most closely allied species.

One of the earliest groups of such objects—some would say the most important group in this connection—is that of ripe fruits. Certainly among the frugivorous apes and among many races of primitive man, the color of fruits must be a powerful factor in developing a sensibility for red rays, and in associating such sensibility with emotional satisfaction. The color of fruits is most generally red, orange or purple, and since purple is largely made up of red, it is clear that the influence of fruits will almost exclusively bear on the rays of long wave-length. We may reasonably suppose that the search for fruits acted as an important factor in the development of a special sensibility for red.

A later factor in the predilection for the red, orange and yellow rays, though scarcely a factor in their discrimination, lies in the fact that these are the colors of fire. Flame, apart from its beauty, on which certain poets, Shelley especially, have often insisted, is a source of massive physical satisfaction. Even under the conditions of civilization we are often acutely sensitive to this fact, while under the conditions of primitive life, in imperfect shelters, caves or tents, where no other source of artificial light and heat is known, the satisfaction is immensely greater. At the same time fire is associated with food, it is a protection from wild beasts and the accompaniment of the festival. It may even take on a sacred and symbolic character, and the Roman goddess Vesta was, as Ovid said, simply 'living flame.'

While fruit or fire would tend to make the emotional tone of red pleasant, another very powerful factor in its emotional influences, though this time as much by causing terror as pleasure, is the fact that it is the color of blood. That 'the blood is the life' is a belief instinctively stamped even on the emotions of animals, and it has not died even in civilized man, for the sight of blood produces on many persons a sickening and terrifying sensation which is only overcome by habit and experience or by a very strong effort of will. It is not surprising that in some parts of the world, and even in our own Indo-European group of languages, the name for red is 'blood-color.'

It is evident, however, that at a very early period of primitive culture the blood had ceased to be merely a source of terror, or even of the joy of battle. We find everywhere that blood is blended into complex ritual customs, and thus associated with complex emotional states. Among the ancient Arabians blood was smeared on the body on various occasions, and in modern Arabia blood is still so used. Everywhere, even in the folk-lore of modern Europe, we find that blood is a medicine, as it is also among the primitive aborigines of Australia, so carefully investigated by Baldwin Spencer and Gillen. Among these latter primitive people we meet with a phenomenon of very great significance. We find, that is, that blood is the earliest pigment. There can be little doubt that the earliest paint used by man—no doubt by man when in a much more primitive condition than even the Australians—was blood. In the initiation rites of the Arunta tribes, as described by Spencer and Gillen, the chief performer is elaborately decorated with patterns in eagle-hawk down stuck to his body with blood drawn from some member of the tribe. It was estimated that one man alone, on one of these occasions, allowed five half-pints to be taken from him during a single day; at the same time the blood is not regarded as sufficient pigment and the down is also colored red and yellow with ochre. Red ochre, Spencer and Gillen remark, is frequently a substitute for blood or is used with it. Blood is a medicine, and when any one is ill he is first rubbed over with red ochre, it being obvious to the primitive mind that the ochre will share the remedial properties of blood; in the same way ceremonial objects may sometimes be rubbed over with ochre instead of blood. They associate this red ochre especially with women's blood; and it is said that once some women after long walking were so exhausted that hemorrhage came on and this gave rise to deposits of red ochre. Other red ochre pits, also, they attribute to blood which flowed from women. It appears also that the blood with which sacred implements used in the ritual ceremonies of these Central Australians were smeared must be drawn from women.

Far from Australia, among the hill tribes of the Central Indian hills, we find the same blood ritual and the same tendency to substitute pigments for blood. Among some of the Bengal tribes, says Crooke, blood is drawn from the husband's little finger, mixed with betel and eaten by the bride. A further stage is seen among the allied Kurmis who mix the blood with lac dye. Lastly come the rites, common to all these tribes, in which the bridegroom, often in secrecy, covered by a sheet, rubs vermilion on the parting of the girl's hair, while the women relations smear their toes with lac dye. It is a sacramental rite, and after the husband's death the widow solemnly washes off the red from her hair, or flings the little box in which she keeps the coloring matter into running water.

Some of the foregoing facts, both in Australia and India, suggest the transition to another factor in the emotional potency possessed by red. Red is not only the color of fire and of war and of ritual pigment; it is the color of love. This is certainly an ancient and powerful factor in the emotional attitude towards red. Secondary sexual characters, even among birds, are often red; many fishes, also, at the epoch of the oviposit show a red tint on the orifice of the sexual apparatus; patches of red, sometimes very brilliant, but only appearing when the animal is mature, are perhaps the commonest adornments of monkeys. In man the color of the hair and beard, the most conspicuous of the secondary sexual characters, is most usually brown, or some other variety of red. The lips are crimson, the mucous membrane generally a dark red; the scarlet of the blush, among all fair races, whatever other sources it may have, is always regarded as especially the ensign of love. The rose is the flower of love, as the pale lily is of virtue. This association is quite inapt, and many people who are sensitive in such matters feel that the lily and many white flowers are far more symbolical of rapture and voluptuousness than the rose. It is, however, the color and not the scent or other qualities that has exerted decisive influence on the choice of the symbol. In the Teutonic symbolism of fourteenth century Europe red was the color of love, as also, with yellow, it was the favorite color for garments. In more modern times this last tendency has survived. Sardou decides, it is reported, the color of the dresses to be worn in his plays, on the ground that if he did not the actresses would all wear red to attract attention to themselves, as once occurred at the Odéon. Eighteen hundred years earlier, Clement of Alexandria had written: “Would it were possible to abolish purple in dress, so as not to turn the eyes of the spectators on the faces of those that wear it!” He proceeds to lament that women make all their garments of purple (the classic purple was really a red) in order to inflame lust—those ‘stupid and luxurious purples’ which have caused Tyre and Sidon and the Lacedæmonian Sea to be so much in demand for their purple fishes. Similar phenomena are noted on the other side of the world. Thus the Japanese, as the Rev. Walter Weston informs us, have a proverb: ‘Love flies with a red petticoat.’ Married women are not there supposed to wear red petticoats, for they are too attractive, and a married woman should be attractive only to her husband. The æsthetic Japanese may be thought to be specially sensitive to color, but in Africa also, in Loango, as Pechuel-Loesche mentions, pregnant women are forbidden to wear red, and it would doubtless be possible to find many similar indications of this feeling in other parts of the world.

We have now passed in review all the influences which, by force of their powerful attraction or repulsion, have during countless ages impressed on man, and often on his ancestors, the strong and poignant emotions which accompany the sensation of the most vividly and persistently seen of all colors. We find evidence of the reality of the influences we have traced—especially those of fire, blood and love—in Christian ecclesiastical symbolism, according to which red variously signifies ardent love, burning zeal, energy, courage, cruelty and bloodthirstiness. To the antagonism and complexity of these influences we must doubtless attribute the disturbing nature of the emotion aroused by the group of red sensations and the fluctuations in the predilection felt towards it. It is at once the most attractive and the most repulsive of colors. To enjoy it we must use it economically. The vision of poppies on a background of golden corn, the glint of roses embowered in green leaves, the sudden flash of a scarlet flower on a southern woman's dark hair—it is in such visions as these that red gives us its emotional thrill altogether untouched by pain. If the 'multitudinous seas' were indeed 'incarnadined' for us in 'one red,' if the sky were scarlet, or all vegetation crimson, the horror of the world would be painful to contemplate for nervous systems moulded to our vision of nature. Our eyes have developed in a world where the green and blue rays meet us at every step, and where we have in consequence been almost as dulled to them as we are to the weight of the atmosphere that presses in on us on every side. It is under the clouded skies of northern lands that blue is counted the loveliest of colors; it is in the desert that green becomes supremely beautiful and sacred.

THE EXPENDITURE OF THE WORKING CLASSES[D].

By HENRY HIGGS.

[D] Address by the President to the Economic Science and Statistics Section at the Dover meeting of the British Association for the Advancement of Science.

THE prime concern of the economist and of the statistician is the condition of the people. Other matters which engage their attention—particular problems, questions of history, discussions of method, developments of theory—all derive their ultimate importance from their bearing upon this central subject. The statistician measures the changing phenomena of the production, distribution and consumption of wealth, which to a large extent reflect and determine the material condition of the people. The economist analyzes the motives of these phenomena, and endeavors to trace the connection between cause and effect. He is unable to push his analysis far without a firm mastery of the theory of value, the perfecting of which has been the chief stride made by economic science in the nineteenth century. When we read the 'Wealth of Nations' we are forced to admit that in sheer sagacity Adam Smith is unsurpassed by any of his successors. It is only when we come to his imperfect and unconnected views upon value that we feel the power of increased knowledge. J. S. Mill supposed in 1848 that the last word had been said on the theory of value. In his third book he writes: "In a state of society in which the industrial system is entirely founded on purchase and sale ... the question of value is fundamental. Almost every speculation respecting the economical interests of a society thus constituted implies some theory of value; the smallest error on this subject infects with corresponding error all our other conclusions, and anything vague or misty in our conception of it creates confusion and uncertainty in everything else." And he adds: "Happily, there is nothing in the laws of value which remains for the present or any future writer to clear up; the theory of the subject is complete."

We know now that he was wrong. Thanks in the main to economists still alive, and especially to the mathematical economists, we have at length a theory of value so formally exact that, whatever may be added to it in the future, time can take nothing from it; while it is sufficiently flexible to lend itself as well to a *régime* of monopoly as to one of competition. Yet our confidence in this instrument of analysis is far from inspiring us with the assurance which has done so much to discredit economics by provoking its professors to dogmatize upon problems with the whole facts of which they were imperfectly acquainted. Given certain conditions of supply and certain conditions of demand, the economist should have no doubt as to the resulting determination of value; but he is more than ever alert to make sure that he has all the material factors of the case before him; that he understands the facts and their mutual relation before he ventures to pronounce an opinion upon any mixed question. He must have the facts before he can analyze them. A small array of syllogisms, which, as Bacon says, "master the assent and not the subject," are not an adequate equipment for him. He sees more and more the need for careful and industrious investigation, and prominent among the subjects which await his trained observation are the condition of the people and the related subject of the consumption of wealth. Training is, indeed, indispensable. Every social question has its purely economic elements for the skilled economist to unravel, and when this part of his task has been achieved, he is at an advantage in approaching the other parts of it, while his habit of mind helps him to know what to look out for and what to expect.

It is a curious paradox that, busying ourselves as we do with the condition of the people, we are lamentably lacking in precision in our knowledge of the economic life and state of the British people in the present day. Political economy has, however, followed the lines of development of political power. At one time it was, as the Germans say, cameralistic—an affair of the council chamber, a question of the power and resources of the king. Taking a wider but still restricted view of society, it became capitalistic, identifying the economic interests of the community to a too great extent with those of the capitalist class. It has at length become frankly democratic, looking consciously and directly to the prosperity of the people at large.

Thus, then, we have at once a more accurate theory, a livelier sense of caution as to its limitations in practice, and the widest possible field of study. So far as most of us are concerned, we might as well spend our time in verifying the ready reckoner as in tracing and retracing the lines of pure theory. These tools are made for use. Economic science is likely to make the most satisfactory progress if we watch the social forces that surround us, detecting the operation of economic law in all its manifestations, and in observing, coördinating and recording the facts of economic life. It is not enough, to borrow the language of the biologist (part of which he himself borrowed from the old economist), to talk of the struggle for existence, the survival of the fittest and of evolution. We want, above all, his spirit and his method—the careful, minute, systematic observation of life as affected by environment, heredity and habit. Different problems are brought to the front by different circumstances and appeal to different minds; but at all times and to all economists the condition of the people is of chief interest, and the consumption of wealth is so closely connected with it that it might seem superfluous to plead for its study. Yet some such plea is necessary. The arts of production improve apace. The victories of science are rapidly utilized by manufacturers anxious to make a fortune. Even here the descriptive study of the subject is hampered by the trade secrets involved in many processes, and by a feeling that production may safely be left to the unresting intelligence of captains of industry, so that the onlooker is chiefly concerned in this branch of the subject with solicitude for the health and safety of the workmen employed. The departments of distribution and exchange appeal especially to the pride of intellect. The delicate theorems of value in all their branches—wages, rent, interest, profits, the problems of taxation, the alluring study of currency, the mechanism of banking and exchange—have attracted the greatest share of the economist’s attention. On the practical side of distribution the growth of trade unions, the spread of education, the improved standard of living, have increased the bargaining power of the working classes and combined with other causes to effect a gratifying improvement in the distribution of wealth, so that they receive a growing share of the growing national dividend. The practical and the speculative aspects alike of the consumption of wealth have received less consideration. Nobody sees his way to a fortune through the spread of more knowledge of domestic economy in workmen’s homes; and the scientific observer has curbed his curiosity before what might seem an inquisitorial investigation into the question, what becomes of wages? Economists long ago discovered the necessity of distinguishing between money wages and real wages. It is now necessary for us to distinguish between real wages and utilities—not to stop at the fact that so many shillings a week *might* procure such and such necessaries, comforts, or luxuries, but to ascertain how they *are* expended. From the first we can deduce what the economic condition of the people might be; from the second we shall know what it is. And when we know what it is we shall see more clearly what with more wisdom it might become. Wealth, after all, is a means to an end. It is not enough to maximize wealth; we must strive to maximize utilities. And we can no more judge of the condition of a people from its receipts alone, than we can judge of the financial condition of a nation from a mere statement of its revenues.

The condition of the people has, of course, improved, and is improving. Public hygiene has made great progress, and houses are better and more sanitary, though for this and other reasons rents have risen. Wages are higher. Commodities are cheaper. Coöperation and the better organization of retail business, giving no credit, have saved some of the profits of middlemen for the benefit of the consumer, while authority fights without ceasing against frauds in weights and measures, and adulteration. Free libraries, museums, picture galleries, parks, public gardens and promenades have multiplied, and it is almost sufficient to observe that no one seems to be too poor to command the use of a bicycle. But with all this progress it is to be feared that housekeeping is no better understood than it was two centuries ago—perhaps not even so well. In the interval it has become enormously simplified. The complete housewife is no longer a brewer, a baker, a dyer, a tailor and a host of other specialists rolled into one. But among the working classes the advent of the factory system has increased the employment of women and girls away from home to such an extent that many of them now marry with a minimum of domestic experience, and are with the best intentions the innocent agents of inefficiency and waste, even in this simplified household.

If we were suddenly swallowed up by the ocean, it appears probable that the foreign student would find it easier to describe from existing documents the life and home of the British craftsman in the middle ages than of his descendant of to-day. In part, no doubt, our fiscal system, with its few taxes upon articles of food and its light pressure on the working classes, is responsible for this neglect. During the Napoleonic war Pitt sent for Arthur Young to ask him what were the ordinary and necessary expenses of a workman's family, and the question would again become one of practical politics if any large addition were required in the proceeds of indirect taxation. Taxation has the one advantage of providing us with statistics. We know tolerably well the facts in the mass about the consumption of tea and coffee, dried fruits and tobacco, and of alcohol, while the income tax (aided in the near future by returns of the death duties) may give us some idea of the stratification of the wealthier classes. But the details of consumption are still obscure. It has already been suggested that some restraint may arise from the sentiment that individuals are likely to resent what they may regard as a prying into their affairs. But when we travel abroad we are curious to notice, and do notice without giving offence, the dress, the habits and the food of peasants and workmen; and it is difficult to resist the conclusion that we are less observant at home because these common and trivial details appear to us unworthy of attention. In his 'Principles of Economics' Professor Marshall says: "Perhaps £100,000,000 annually are spent, even by the working classes, and £400,000,000 by the rest of the population of England, in ways that do little or nothing towards making life nobler or truly happier." And, again, speaking before the Royal Statistical Society in 1893: "Something like the whole imperial revenue, say 100 millions a year, might be saved if a sufficient number of able women went about the country and induced the other women to manage their households as they did themselves." These figures show, at any rate, the possibilities of greatness in the economic progress which may result from attention to the humblest details of domestic life.

Economics, like other sciences, lies under a great debt of obligation to French pioneers. The physiocrats, or *économistes*, of the eighteenth century, were the first school of writers to make it worthy of the name of a science. In Cournot, France gave us a giant of originality in pure theory. In Comte, we have a philosopher fruitful in suggestion to the narrower economist. In Le Play, we have a writer as yet little known in England, but to whom recognition and respect are gradually coming for his early perception of the importance of ascertaining the facts of consumption, and it is to Le Play's 'family budgets,' the receipts and expenses of workmen's families, that I desire especially to call attention. I have given elsewhere an account of his life and work.[E] Broadly speaking, he sets himself by the comparative study of workmen's families in different countries of Europe to arrive at the causes of well-being and of misery among the laboring classes. The subject was too large to lead him in many directions to very precise conclusions. We are reminded in reading him of an incident at a dinner of the Political Economy Club in 1876, when Mr. Robert Lowe propounded the question: "What are the more important results which have followed from the publication of the 'Wealth of Nations' just one hundred years ago?" Some of the most enthusiastic admirers of Adam Smith were present, Mr. Gladstone and M. Léon Say among the number; and Mr. Lowe trenchantly declared that it all came to this: "The causes of wealth are two, industry and thrift; the causes of poverty are two, idleness and waste." It was left to Mr. W. E. Forster to make the rugged remark: "You don't want to go to Adam Smith for that—you can get that out of the Proverbs of Solomon." And Le Play's conclusions frequently go still further back, to the Decalogue. There are, however, many observations, suggestive and original, upon the material facts, the economic life, of the families he brought under review. And we are now concerned rather with his method than with his conclusions. Given half a dozen Le Plays applying their minds to the study of the consumption of wealth among the working classes of England, we might expect soon to see a greater advance in comfort, a greater rise in the standard of life, than improved arts of production alone are likely to yield in a generation. Certain English writers had, indeed, prepared family budgets before Le Play arose. But their method was usually incomplete except for the specific purpose they had before them. David Davies and Sir F. Eden were chiefly concerned with the poor law, Arthur Young and Cobbett with agricultural politics, Dudley Baxter and Leone Levi with taxation. Le Play may fairly be called the father of the scientific family budget. His studies of four English families[F] are the most complete economic pictures of English popular life to be found in literature. With the aid of some local authority he chose what was thought a fairly typical family, and then, frankly explaining his scientific object and securing confidence, he set himself to study it. Nothing of economic interest is too unimportant for him to record. A minute inventory and valuation of clothes, furniture and household goods; a detailed account, item by item, of income from all sources and of expenditure upon all objects for a year, with the quantities and prices of foods, &c.; a description of the family, member by member, their past history, their environment, how they came to be where they are and to earn their living as they do; their resources in the present, their provision for the future; their meals, hygiene and recreations; their social, moral, political and religious observances—nothing escapes him. And the whole is organized, classified, fitted into a framework identical for all cases, with the painstaking and methodical industry of the naturalist. Contrasted with this the realism of novelists, the occasional excursions of journalists, the observations of professed economists, are pitiably incomplete. As early as 1857 Le Play found one ardent admirer in England, Mr. W. L. Sargant, whose "Economy of the Laboring Classes," avowedly inspired by Le Play, is a valuable and interesting piece of work. Since then, however, with the magnificent exception of Mr. Charles Booth, little has been done to throw light upon the mode of life of the wage-earners of England. The Board of Trade heralded the formation of its Labor Department by issuing a blue book—unhappily without sequel—entitled "Returns of Expenditure by Working Men," in 1889, and the Economic Club has published a useful collection of studies in 'Family Budgets,' 1896. But we shall probably still depend very much upon foreign observers for fuller knowledge of the subject. M. René Lavollée, an adherent who may almost be called a colleague of Le Play, has devoted to England a whole volume of his important work 'Les Classes Ouvrières en Europe: études sur leur situation matérielle et morale.'[G] M. Urbain Guérin, another member of the Société d'Economie Sociale, founded by Le Play to carry on his work, has recently added a study of a tanner's family in Nottingham to Le Play's gallery of portraits; and some of the young members of the Musée Social and the Ecole Libre des Sciences Politiques have come among us animated with the same scientific curiosity. A vivid (and, so far as Newcastle is concerned, a trustworthy) sketch by a German miner, "How the English Workman Lives," just translated into English, is our latest debt to foreign observers. It may be hoped that the British Association, largely attended as it is by persons who would shrink from more ambitious scientific labors, will furnish some workers ready to do their country the very real service of recording such facts as they can collect about the economic habits of our own people, and so helping us to know ourselves.

[E] *Harvard Quarterly Journal of Economics*, vol. iv., 1890; *Journal of Royal Statistical Society*, March, 1893; *Palgrave's Dictionary of Political Economy*, *s. v.* Le Play, 1896.

[F] *Les Ouvriers Europeens*, Paris, folio, 1855.

[G] Paris, 1896, tom. iii., 656 pp., large 8vo.

Consider, for a moment, the consumption of food. To the ordinary English workman life would seem unendurable without white wheaten bread. Other forms of bread he knows there are, but he has unreasoning prejudices against wholemeal bread—the food of workhouses and prisons—and against rye bread or other kinds of bread, the food of foreigners. But in many parts of Europe the working classes have no bread. Cereals of some sort, prepared in some way, they of course employ. Wheat, oats, rye, barley, maize, buckwheat, even chestnuts, are used indifferently in different places, and rice and potatoes are among the substitutes. What is the relative value of these as foodstuffs, and what is the best mode of preparing them? The reasons which induced men in the middle ages to consume the cereals of their own neighborhood have been so much weakened by the cheapening of transport and the international specialization of industries, that the conservatism of food habits is brought into strong relief when we find neighboring peoples abandoning, first in town and then in country, marked distinctions of national costumes, but clinging everywhere to national differences of food. We are perhaps on the eve of considerable changes here. Two years ago an American economist told me in Boston that fruit had been the great ally of the workmen in a recent severe strike. There had been an exceptionally large crop of bananas, which were sold at one cent apiece, and the strikers had sustained themselves and their families almost entirely upon bananas at a trifling cost—very greatly below their usual expense for food. Returning to London I found bananas on sale in the streets for a halfpenny. No doubt they were consumed here in addition to, and not in substitution for, ordinary food; but they illustrate the fact that the foods of other latitudes are no longer the sole luxury of the rich, but are brought within the reach of all classes, and that our popular food habits need no longer be made to conform to the narrow range of former days, but may be put upon a wider rational basis. The vegetarians, largely dependent upon other countries, have recognized this. The chemist and the physiologist might give us great assistance in these matters. Most of the calculations which I have seen as to the constituents of foods, their heat-giving and nutritive properties, appear to ignore the greater or less facility with which the different foods are assimilated. It is surprising that rice, in some respects the most economical of all grains, needing no milling, easily cooked and easily digested, is not more largely consumed by the poorer families in this country.

The effect upon our food habits of the introduction of railways and the supply of comparatively cheap fuel to every household is almost incalculable. But for this the consumption of tea, perhaps even of potatoes where there is no peat, would be very small. The preference of the French for liquid, and of the English for solid, food, has been attributed to the greater relative facilities which the French once enjoyed for making a fire, though the persistence (if not the origin) of our popular habits in this respect probably lies rather in the fact that a Frenchwoman's cookery makes greater demands upon her time and attention. One result of this preference is that the essential juices of meat preserved by the French in soups and ragouts are with us to a large extent absolutely wasted. Owners of small house properties complain that, however well trapped their sinks may be, the pipes are constantly choked, and that the mysterious mischief is almost invariably cured by liberal doses of boiling water, which melt the solidified fats cast away in a state of solution. The number of persons who died of starvation in the administrative county of London in 1898, or whose death was accelerated by privation, amounted to 48; and we shall be pretty safe in estimating the total number in the United Kingdom at something less than 500. The common and inevitable reflection is that they might have been easily relieved from the superfluities of the rich; but it is true also that their sufficient sustenance was destroyed many times over through the ignorance of the poor. It would be difficult to find an English cookery book which a workman's wife would not reject as too fanciful and ambitious to be practical. A little French treatise, ‘La parfaite Cuisinière, ou l’Art d’utiliser les Restes,’ strikes in its title, at any rate, the keynote of the popular domestic economy of which we stand much in need in England. Housekeeping, even the humblest, is a skilled business. To know what to buy, how to use it and how to utilize waste does not come by the light of nature. If more knowledge and more imagination were devoted to the teaching of cookery in our board schools, the family meal might be made more varied, more appetizing, more attractive and more economical, leaving a larger margin for the comforts, culture and recreations which help to develop the best social qualities. A happy family is a family of good citizens. It would be discourteous to another section of this Association to quote without reserve the *mot* of Brillat-Savarin: “He who discovers a new dish does more for the happiness of mankind than he who discovers a new planet.” We must stipulate that the new dish effects an improvement in the economy of the working classes.

Take, again, the consumption of coal. Mr. Sargant says, “It is impossible to say how much of the superiority of English health and longevity is owing to the use of open fireplaces”; probably a considerable part is owing to it. We all know how close and stifling is the atmosphere of a room heated by a stove, and how much more difficult it is to keep a room perfectly ventilated in summer than it is in winter, when the fire is constantly changing the air. It may be true that three fourths of the heat of our fireplaces passes up the chimney and is lost to us; but we gain far more advantage by the fresh air constantly introduced into the room. Now, with improved grates and improved fireplaces we may retain all the advantages of the open fire without so great a waste either of the substance of the consumer or of the national stock of coal; and attention is already being devoted to this fact in middle-class households, but some time must yet elapse before the advantage is reaped by the working classes. At a former meeting of this Association Mr. Edward Atkinson exhibited a portable oven or cooking-stove, which was a marvel of simplicity and economy. He has described it at length in his ‘Science of Nutrition,’ 1892. He argues that the attempts to combine cooking with the warming of a room or house are absurdly wasteful; that almost the whole of the fuel used in cooking is wasted; and that nine tenths of the time devoted to watching the process of cooking is wasted; and he estimates the waste of food from bad cooking in the United States at $1,000,000,000 a year. I have not, however, heard of his oven being at all extensively used.

Upon the thorny subject of dress it is perilous to venture; but it is impossible to be in the neighborhood of a London park on a Sunday afternoon without feeling that the efforts of domestic servants to follow the rapidly changing vagaries of fashion are carried to a pernicious degree of waste. The blouse of the French workman and the bare head of the Parisian factory-girl or flower-girl are infinitely more pleasing than the soiled and frowsy woolens or the dowdy hats of their English fellows, nor does the difference of climate afford an adequate explanation of the difference of habit. We must perhaps admit a greater dislike in England to any external indication of a difference in wealth by a costume different in kind. M. Lavollée, after referring to the low price of the ready-made suits which the English factories "fling by the million on the markets of the world, including their own," adds: "This extraordinary cheapness is, however, not always without inconvenience to the consumer. If the clothes he buys cost little, they are not lasting, and their renewal becomes in the long run very burdensome. This renewal is, too, the more frequent in that the wife of the English workman is in general far from skillful in sewing and mending. Whether she lacks inclination, or the necessary training, or whether the fatigues of a too frequent maternity make her *rôle* as a housewife too difficult for her to support, the woman of the people is generally, on the other side of the Channel, a rather poor cook, an indifferent needle-woman and a still more indifferent hand at repairs." As a consequence, he says, the English workman has often no alternative but to wear his garments in holes or to replace them by others. Given an equal income, there is probably no doubt that a French working-class family will be better fed and better clad than a corresponding English family dealing in the same market, and will lay up a larger stock of the household goods, and especially linen, which are the pride of the French peasant.

The waste resulting from the immoderate use of alcohol and from the widespread habit of betting, serious as it is, need not detain me, as I wish to confine myself more particularly to waste which can hardly be called intentional. It is not suggested that every man should confine his expenditure to what is strictly necessary to maintain his social position. The great German writer on finance, Professor Wagner, is accustomed to say that “parsimony is not a principle.” It is sometimes, indeed, a bad policy and a wasteful policy; and life would be a very dull business if its monotony were not relieved by amusement and variety, even at the occasional expense of thrift. Le Play refers to tobacco as “the most economical of all recreations.” How else, he asks, could the Hartz miner “give himself an agreeable sensation” a thousand times in a year at so low a cost as 10 francs? But nobody would wish to see a free man using his tobacco like the Russian prisoners described to me by Prince Krapotkin, as chewing it, drying and smoking it, and finally snuffing the ashes! Nor should we desire to eradicate from society the impulses of hospitality, and even of a certain measure of display. An austere and selfish avarice, if generally diffused, may strike at the very existence of a nation.

Another respect in which French example may be profitable to us is the municipal management of funerals (*pompes funébres*). Many a struggling family of the working classes has been seriously crippled by launching out into exaggerated expenses at the death of one of its members, and especially of a bread-winner. The French system, while preserving the highest respect for the dead, has some respect for the living, who are frequently unable and unwilling at a time of bereavement to resist any suggestion for expensive display, which seems to them a last token of affection as well as a proof of self-respect.

As regards housing the English cottage or artisan’s house is regarded on the Continent rather as a model for imitation than as a subject for criticism; but the pressure of population upon space in our large cities, joined with a love of life in the town, may possibly prove too strong for the individualist’s desire for a house to himself. If we should be driven to what Mrs. Leonard Courtney has proposed to call Associated Homes, the *famillistère* founded by M. Godin at Guise, and rooted in the idea of Fourier’s *phalanstère* will show us what has already been achieved in this direction. Dissociated from industrial enterprise it might easily become popular in England. Some of its collective economies are certainly deserving of imitation, and the experience not only of the Continent, but also of America, may soon bring us face to face with the question whether the preparation of dinners, in large towns, should not—at least for the working classes—be left to the outside specialist like the old-home industries of baking and brewing. An excellent example of scientific observation is ‘Les Maisons types,’ by M. de Foville, the well-known master of the French Mint. He describes in detail the various forms of huts, cottages and houses scattered over France in such a fashion that it is said the traveler in a railway train may tell, by reading the book, through what part of the country he is passing; and he gives the reasons, founded upon history or local circumstances, for the peculiarities in architecture to be observed. The book is a useful warning against rash generalizations as to the best type of house for a working man.

A well-informed writer shows, in a recent article in the ‘Times,’ that not less than about fifty million gallons of water a day might be saved in London, “without withdrawing a drop from any legitimate purpose, public or private, including the watering of plants.” He says: “The detection of waste is carried out by means of meters placed on the mains, which record automatically the quantity of water passing hour by hour throughout the day and night. The whole area served by a given water supply is mapped out into small districts, each of which is controlled by one of these detective meters. The chart traced by the apparatus shows precisely how much water is used in each of the twenty-four hours. It records in a graphic form and with singular fidelity the daily life of the people. It shows when they get up in the morning, when they go to bed at night, when they wash the tea-things, the children and the clothes; it shows in a suburban district when the head of the household comes from the city and starts watering his flowers; it shows when the watering-cart goes round; but, above all, it shows when the water is running away to waste, and how much.”

I quote this not to multiply examples of the waste of wealth, but to illustrate the insight which a few figures, such as those recorded by this meter, give us into the lives of the people. How much more does the account-book, a detective meter of every economic action, give us an animated photograph of the family life. Nothing is so calculated to stimulate social sympathy or to suggest questions for consideration. Like a doctor’s notes of his patients the facts are not for publication in any form which will reveal the identity of the subject; but when we have enough of them they will be of the highest scientific value. We have at present too few to offer any useful generalizations. All that can be done is to serve as a finger-post to point the road along which there is work to be done.

If nothing has been said about the waste and extravagance of the wealthier classes, it is because economy is with them of less moment. They suffer little or no privation from extravagance, and derive less advantage from checking it than those to whom every little is a help. And so far as much of this waste is concerned, they sin against the light. It is one thing to point out a more excellent way to the unwary, another to preach to those who, seeing the better, follow the worse.

But the expenditure of the working classes is also, from a scientific point of view, vastly more important. Their expenses are more uniform, less disturbed by fantasy, or hospitality, or expensive travel, and will give us more insight into the hitherto inscrutable laws of demand. The time is far removed when any reduction in the cost of living could be successfully made the pretext for a reduction in the rate of wages. The Committee on the Aged Deserving Poor recommends under certain conditions pensions varying with the 'cost of living in the locality.' The same factor, we are told, enters into the adjustment of postmen's wages as between town and town. How are we to know the comparative cost of living without these details of expenditure? How else can we measure with any exactness the progress of civilization itself? How else can we discover the cohesive force of the family in holding together the structure of society, the mutual succor of young and old, the strong and the infirm or sick, the well-to-do and the victim of accident or ill-luck? To what department soever of economic life we turn our eyes we find live men and women, born into families, living in families, their social happiness and efficiency largely dependent on their family lives, and when we consider how greatly our knowledge and insight into society will be increased by a more intimate acquaintance with the economics of the family, we may well cherish the highest hopes for the future progress of our science. The theory of this subject, at any rate, is not 'complete.' It has not even been begun.

Upon certain aspects of the spending or using of wealth as opposed to the getting of wealth, like the expenditure of central and local governments, it would hardly be proper for me to enlarge. The first is subject to the watchful control of the tax-payer, of Parliament, and of a highly trained civil service; the second to the jealous criticism of the rate-payer and his representative. But there is some social expenditure, like the scandalous multiplication of advertisements (which by a refinement of cruelty gives us no rest night or day), which is wicked to a degree. In all these matters of the consumption of wealth, individually and collectively, we are as yet, it must be again repeated, too ignorant of the facts. An unimaginative people as we are, we are fortunately fond enough of travel to have suggestions constantly forced upon us by the different experiences and habits of foreign countries. And we are happy in a neighbor like France, with her literary and social charms and graces, her scientific lucidity and inventiveness, and the contrasts of her social genius to inspire comparisons, and in many respects to set us examples. I have singled out one of her many writers for attention, precisely because of this quality of suggestiveness. Other investigators have, of course, attacked the subject. In Belgium and Switzerland, Germany, Italy and Austria, and the United States, governments and individuals have recently undertaken the preparation of family budgets; but in many respects Le Play's monographs are the first and greatest of all. They yield excellent material, upon which science, in its various branches, has yet to do work which will benefit mankind in general; and promises especially to benefit the people of this country. The cosmopolitan attitude of the older economists was largely due to their centering their attention upon the problems of exchange. To them the globe was peopled by men like ourselves, producing the fruits of the earth, anxious to exchange them to the greatest mutual advantage, but hindered from doing so by the perversity of national governments. The facts of consumption, at any rate, are local. They are often determined by geology, geography, climate and occupation; and, however fully we may admit the economic solidarity of the world, and the advantage which one part of it derives from the prosperity of another, yet we may be easily forgiven for thinking that our first duty lies to our own brethren; that our natural work is that which lies at our own doors; that, as the old proverb says, 'the skin is nearer than the shirt.' And we may fairly be excused if we attempt to make our contribution to the welfare of the human family through the improvement of the consumption of wealth and the condition of the people in our own land.

THE CONQUEST OF THE TROPICS.

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THE most beautiful and the most fruitful portions of the earth are at the present time in the possession of partially civilized, or barbarous and savage races, to the exclusion of the more enlightened Caucasian. Shall he ever remain unable to possess and occupy tropical lands to the exclusion of dark-skinned and inferior races? Will the time never come when he can rear a family of strong and vigorous children, of pure blood, under the equatorial sun? Is it true that the white man removing to the Tropics necessarily deteriorates?

The almost universal belief is that these questions must be answered in the affirmative. That, owing to the great heat, and to evil influences operating through the air, the water and the soil, it will always be impossible for white people to live in hot countries permanently, and, at the same time, to retain the physical vigor of temperate latitudes, and to rear healthy children. But these persons do not take into account certain recent great discoveries in the domain of science, medicine and hygiene. In the light of these discoveries, it is not wise to say that the white man will never conquer the Tropics.

White races have, in the past, reached a high degree of civilization in hot countries. Egypt, where the first civilization arose, is a land of tropical heat. The valley of the Euphrates, where arose the civilization of Babylon, and much of Persia, are both tropical or sub-tropical in temperature. The people of Egypt, Babylon and Persia were white. It would seem that to originate a civilization is more difficult than to maintain it.

Many countries, now most salubrious, were once considered very unhealthful. Health conditions were so bad in England, after the withdrawal of the Romans, that for nearly a thousand years there was absolutely no increase in the population, and the most dismal accounts of the reign of disease have come down to our times. What was true in England, was in great part true of all of Europe throughout the Dark Ages. Scurvy, rheumatism, fevers and plagues held high carnival in recurring epidemics every few years. If we can believe the reports, it was fully as dangerous then to dwell in the most favorable portion of Europe as it is now in the most dreaded tropical regions.

New England was at first thought to be a very unhealthful land. The early settlers in Massachusetts wrote to their friends in England imploring shipments of ale and beer, because the water was 'wholly unfit to drink'. What held concerning New England was doubtless maintained about every other portion of the Continent settled by the English, and, in some cases, these views prevailed until recent times.

It is well known that our ancestors thought it would never be possible for white people, or, indeed, for any people, to live on the treeless prairies of the great West. The earliest settlers always occupied the wooded belts, and only seventy years ago the prairies, which now sustain millions of happy and healthy whites, were looked upon probably in much the same way as we regard the plains of the Amazon, of the Orinoco, or even the Sahara of Africa.

Many persons yet living can recall the terrible struggles with disease which the first settlers passed through in Ohio, Indiana, Illinois, Missouri, and even in salubrious California. The early settlers in these States were doubtless as sallow, as cadaverous looking, and with as little prospect of leaving vigorous descendants as the present white inhabitants in Cuba, Porto Rico or the Philippines. The reputations of Florida, Louisiana and Texas were no better.

Adults can live without deterioration in the Tropics. This has been proven by English and Dutch officers in India, Ceylon, Java, Sumatra and elsewhere. In the West Indies are men from the United States and from all the countries of Europe, who have been in the islands twenty, thirty, forty, and in some cases even fifty, years, who are to-day the picture of good health, active and vigorous in their work. The same is true in all parts of the tropical world. Adults can live in good health there.

Children born in the Tropics, if educated in temperate latitudes, can return to the Tropics, and this can continue indefinitely in the same families without deterioration. This has been found true in India, Java, the Sandwich Islands and in the West Indies.

It has been assumed, heretofore, that the bracing climate of the north-lands has produced vigorous constitutions in the children sent from the Tropics. That this was of some value will not be denied, but it is insisted that of greater value is the education in the higher ideals of the temperate latitudes. In the Tropics, ideas of morality, of sanitation, of correct living, are very crude. A child born and reared in the midst of low ideals unconsciously absorbs them, and assimilates readily with the population around him. The Spanish idea that everyone born in a tropical colony is necessarily a 'degenerate' is practically true, if he is also reared among 'degenerates.' The custom which exists in these colonies of giving each child born a 'degenerate' native child as a companion and playfellow, only makes more sure the outcome. Isolated families exist in Cuba and Porto Rico, where, high ideals having been maintained and inculcated in the children, we now find vigorous descendants to the third and fourth generations. There are many such families in Porto Rico, and the same is true in the Sandwich Islands.

It is here maintained that it is the letting go, little by little, the correct views of living, which causes the white race to deteriorate, and not the climate. The necessities of life are fewer and easier to obtain in hot countries than in cold ones; and this makes it easy for men to become indolent, to lose ambition and to sink to a low level of living and thinking.

The Tropics, contrary to the usual view, are healthful regions. Malaria exists in hot countries, but so it does in temperate ones. Typhoid fever and contagious diseases are no worse than in cold climes. Smallpox is regarded as a mild disease. Scarlet fever is said not to exist at all. Where filth is allowed to accumulate disease prevails, but in lands well drained and free from decaying matter and filth, there is, under ordinary care, no more to be feared from disease than in the most favored portions of the earth. At present, in hot countries the people pay little attention to sanitation. As a rule, they are unutterably dirty. They live in their own filth, and seem to enjoy it. The germs of disease from one body are promptly taken into another before they have time to die, or are cultivated in filth deposits until the whole community is affected.

The Tropics, in themselves, are no more and no less healthful than temperate regions. But the people in cold countries have some respect for sanitation, while those in hot countries have very little or no respect for decent cleanliness. This is the whole explanation of this matter. People who have the latrine in the kitchen and uncleaned for a century, who sleep in rooms into which a breath of fresh air cannot enter; who seldom wash their bodies; who use rum and tobacco instead of food; who permit children to cohabit promiscuously, can scarcely hope to escape disease, if any prevails in their neighborhood. Such conditions are the rule with the masses in hot countries.

Those who become ‘acclimatized’ will be able to live in hot countries. It is doubtful whether or not there is any actual condition known as ‘acclimatization,’ although if the term means becoming accustomed to filth, and to certain germs which live in filth, there may be something in the term.

Instead of a bodily change, the individual gradually becomes educated to his new environment. He learns what to eat and drink, what to wear and where to sleep, when and how much to work, to come in out of the shower and to change his wet clothes, to avoid the midday sun and the damp air of the night. When a man new to the Tropics has learned these things, he is ‘acclimatized.’ Some learn them at once; others are years in learning, and meanwhile suffer from sickness and distress.

New conditions must be met in every country new to the pioneer, whether the country is in temperate or hot latitudes. In opening up a new country to settlement, it is the severe labors, the exposure, the meagre diet, the anxiety, the general hard conditions of life, which undermine the general system and make the body an almost unresisting victim to the germs of malaria and other diseases. It is not the climate in new countries, but the hard conditions of life, which kill the settlers.

So, in the recent war with Spain, bad conditions in the northern camps, uncleanliness of person due to lack of water, over-exertion in practice marches, sleeping on the ground, change of food, overcrowding in tents preventing restful sleep, unsanitary conditions on transports, caused the men to be landed in the Tropics in an extremely bad condition of body.

Landing in the rainy season, opening the earth to form trenches for defence and about their tents, sleeping upon the damp ground, with a deficient and unbalanced ration, with no change of clothes for nearly three months, it is no wonder that many became sick. But the sickness was not due to the climate at all. It was due to the hard conditions in the home camps, and to hard conditions during the campaigns in the islands.

It is said that the heat, the rains and the insects of the Tropics are certainly unbearable by a white person from the temperate latitudes. But these things are magnified by the distance from which they are viewed. So far as the tropical lands recently acquired by the United States are concerned, they are not elements to be dreaded.

These lands are all Oceanic Islands. Surrounded by immense areas of water, they have an unvarying, or slightly varying, temperature. They are warm the whole year round, while never hot. In all these Islands the midday temperature is about 80° Fahrenheit. At night it falls to 75° or even to 70°; in the mountains still lower, depending upon the elevation.

But this heat is moderated by sea breezes. Except for about an hour in the morning, there is a breeze the whole day long, which tempers the heat. Sunstroke is unknown. No bad conditions arising from the heat have been seen in Porto Rico. The nights are always so cool that refreshing sleep may be obtained, and the effect of the sun is tempered by clouds, which shade the earth nearly all summer.

All the islands have mountains which may be reached in a few hours, where the climate of the temperate latitudes may be enjoyed by those desiring the change.

The tropical rains are no serious drawback. They fall at a fixed time each day, usually from two to four o'clock in the afternoon. They are much like heavy June showers in the States, unaccompanied by thunder and lightning. The ground soon dries off, and the rain has occasioned no inconvenience of consequence to anyone. The absence of thunder and lightning is remarkable. This is certainly true in Porto Rico.

The hurricanes and other great wind storms are probably no more frequent nor more destructive than are cyclones in the States. In Porto Rico there is a belief that a single severe hurricane occurs about once in each hundred years.

Insects are strangely few. The mosquito is grown in the cisterns, and is abundant in the towns. It is practically absent in the country. The flea is found only in the towns, where it is a sort of domestic animal. A little attention to cleanliness would diminish the numbers. The bedbug has not been seen in a year in Porto Rico, though there is no reason why it should not be here. Centipedes, spiders and tarantulas are so scarce that the natives expect about fifty centavos for each large specimen which they catch. Indeed, instead of an abundance of insects, these islands are remarkable for the small number of species and individuals indigenous to them.

Recent inventions and discoveries have made the conquest of the Tropics by the Caucasian race possible. There have been great discoveries made in chemistry, biology, bacteriology and medicine within recent years. Chemical discoveries have produced new and powerful remedies. Biology and bacteriology have brought to light numerous microscopic forms of life, traced their life histories, and shown that beyond a doubt, many, if not all, of the diseases designated communicable (contagious and infectious) are due to living beings called ‘germs.’ The experimental physician has discovered, in some cases, remedies which will destroy these germs after they have been introduced into the body, while the sanitarian has made vast studies in demonstrating how they may be destroyed before entering the body. Thus, sterilized food, water and clothing never convey diseases. Cities which are kept clean and have pure water supplies have little fear of epidemic diseases. The draining of lowlands, the thorough cultivation of the soil, the paving of streets and the use of quinine cause malaria to retreat from its old haunts.

Biologists have shown that a tick conveys the Texas cattle fever; the tsetse fly in Africa spreads the ‘fly disease’ among the cattle in that continent. The house-fly spread typhoid fever among our soldiers last summer, and there is good reason for believing that the mosquito is in large part the disseminator of malaria. Consumption, dysentery, the Asiatic plague, leprosy, typhoid fever, are all germ diseases. Knowing the causes of these diseases, the life history of the germs, and the remedies to apply, it is hoped that in a very few years the biologist, the bacteriologist, the sanitarian, all working together, will make tropical diseases to be no more dreaded than are the diseases of temperate regions. As warm countries become better known, physicians will certainly become more skillful to treat the diseases peculiar to them.

Rapid transportation and rapid communication between the tropic and temperate regions will rob the former of many terrors. When a person can communicate with his family every few days, or by telegraph in a few hours, and when he knows he can reach his old home readily, one element which disturbed former pioneers is removed.

Rapid transportation and the discovery of the process of canning fruits, vegetables and meats, together with the process of manufacturing ice, and of cold storage methods, make it possible for a person in a hot country to enjoy the foods to which he was accustomed in his old home. This will be a great help until he has learned to use native products.

Education and good laws will remove from the Tropics many undesirable features which now repel people from the North. It has been already remarked that the people in these islands have no knowledge of sanitation, and live in utter disregard of all the well-known rules of hygiene. Some of the most striking examples of this are the living in their own excretions, sleeping in air-tight compartments, the lack of a variety of food, working long hours in the hot sun with an empty stomach, using rum, tobacco and coffee in place of food, the utter lack of any restraint of the sexual instinct by either men or women of the lower classes and by the men of all classes, producing a well-nigh universal corruption of blood.

These unsanitary and unhygienic conditions have dwarfed the tropical dwellers in body and in mind. These things cannot be laid to the climate. They are due to ignorance. The same condition would produce similar results in Pennsylvania or Connecticut, and such results were seen a generation ago in New Mexico, California and elsewhere.

The laws under which these people have been living have been monstrously bad. Marriage has in some cases been actually discouraged; there was little opportunity and little inducement to accumulate property. There were few schools, and they were of poor quality. The different races, white, Indian and African, have fully commingled, and the result the world knows is bad. The strongest arguments against the mixing of the Caucasian and the African are to be found in the West India Islands. The mixed races will be much harder to deal with than pure bloods of any race.

The climate in Cuba and Porto Rico—and the same is claimed for the Philippines—is equal to any in the States south of the Carolinas. With the masses educated and with wholesome laws, these islands will all become garden spots, and will ultimately be occupied by pure-blooded Anglo-Saxons, the present inhabitants disappearing before the stronger and purer-blooded race.

DISCUSSION AND CORRESPONDENCE.

POETRY AND SCIENCE.

In spite of the occasional croak of prophets of evil, poetry is not in danger of being crowded out of the hearts of men by the materialism of science. It is true that just now there are no poets of surpassing genius with whom the reading public is popularly acquainted. It is true that the development of our material civilization through the surprisingly rapid advance of scientific discovery is a thing which engages attention to a very great degree. It is true that the necessity of dealing continually with practical, matter-of-fact details, whether of the office, or the factory, or the laboratory, is not in itself distinctly poetical. It is true that planning practical uses for the Röntgen rays or liquid air is not essentially stimulating to a love for poetry, but this is only one aspect of the case.

A great deal of the appeal of poetry comes through what it suggests of the unknown and mysterious, suggestions, not of the strange and the fanciful, but of the beautiful, hints of a something beyond the beauty to which our eyes have yet come, a beauty to which, perhaps, for all our longing, they may never come. A man for whom the problems of existence have ceased to be problems, a man whose theology is a settled thing, who believes certain things definitely and rests with assured ease in his belief, a man for whom the vague anticipations of a world of doubt as yet beyond his ken "make no purple in the distance," such a man can neither have appreciation for a wide range of poetry, nor will he write verse that can take any serious place as poetry for modern readers. The poetry of a primitive people, dealing with primitive emotions, finds in more elementary things, like the boy in Wordsworth's "Intimations of Immortality," hints and suggestions of a "something that is gone," "the glory and the freshness of a dream." These emotions become our emotions sympathetically, and not because they are quite the normal feelings for the mature reader of poetry to-day. The things that were a wonder to the Greek of Homer’s time have ceased to be a wonder to us, and if a poet would excite the same feelings in us he must employ other means. Science, in giving us absolute knowledge in regard to many things which not so long ago were full of strangeness for us, has taken out of them the olden poetry and the trees have nymphs that direct their growth no longer, the streams that were once dæmon-haunted are now merely water courses, and the other spirits of the earth and air have gone far away into the world’s forgetfulness. But while we have been pushing out into the unknown and annexing portions of it to the region of the known, we have been merely enlarging the boundary, not obliterating it. More than this man never can do. Always beyond the farthest vision of his telescope and microscope will lie the unknowable, growing smaller, perhaps, but seeming larger as it gives up some of its secret places for the inhabiting of the dwellers in the known. And this is the significant thing, that, as our knowledge grows, our sense of what lies beyond that knowledge finds an increasing number of things that may excite wonder. Every new scientific discovery, at least in certain departments of science, simply acts as an index finger pointing the way to related phenomena not yet understood. And so it will be ever. The most learned man that the schools, and the fields and the sky aided by the finest instruments human skill can devise, can produce, will only find himself awed by the vast darkness of the unknown into which his eyes cannot pierce.

There is another phase of the question that must not pass unnoticed. As the region of the unknown widens it offers more objects of interest and may thereby more fully absorb attention. When reality is sufficiently rich in experience we do not care to indulge in dreams. When the present satisfies us and answers all our needs we are less inclined to look forward to the future, whether that glows before us with the hues of promise or darkens with the threat of coming storm. But the fullest life may weary at times and wish, for the mere rest of change, to go outside of itself and find in the strangeness of something new and not yet known a relaxation and recreation for the tired hand and brain. And so the strenuousness of modern life with its ceaseless outreaching for new pleasures and new truths will be ready always for the soothing restfulness of a poetry that gives the form of beauty to things just beyond the wonderland of the known.

But how to make poetry of these things is the perplexing problem. Truth, whether of the world of fact or of the world of imagination reaching out into the spiritual realm, is not poetry until in some fashion it is made beautiful in its appeal to our sensibilities. A hundred years ago the things that were fitting subjects for poetic treatment were much more elementary and as emotional stimulus they reached consciousness in a much more immediate and direct fashion than the themes that are fitted for poetry now. The poet who would achieve distinct success in the higher walks of poetry to-day must be master of an art surpassing that of all but a few of his brethren of the craft who have gone before him. The world of the known is so large, comparatively, now, and the individual is so far removed from the boundaries of the unknown, save, perhaps, at one point, that more art is required to induce him to travel the longer distance out of the world of cold fact into the borderland of strangeness where suggestions of new truth and new beauty may come to quicken aspirations.

It is true that there are themes that were new a thousand years ago and will be new a thousand years hence, but a poet to achieve distinct success must strike a note not only individual, but one closely attuned to the thought and feeling of his time. Milton we know rather as a voice of Puritan England than as a poetic genius. We call Wordsworth a great poet and are conscious as we do so, that he deserves the distinction rather because he interpreted to men a new phase of thought and feeling, than because he knew how to make his verse wholly pure poetry rather than bald prose. Even poets of such spiritual elevation as Shelley and Coleridge caught the feeling and the tone of their time, and the revolutionary spirit and the love of nature that was molding Wordsworth finds a distinct voice in them as well. Even Burns, isolated as he was, is not altogether an anomaly, and no one need be told that Byron was in an extreme degree the voice of the reactionary spirit of post-revolutionary Europe. William Morris, retelling old legends of Greek and Saxon, none the less informed his verse with the humanitarian and æsthetic spirit of modern life, and applied his sense of the beautiful to the problems of nineteenth century existence. Swinburne, too, is democratic and in his vision the world moves on to new glories even though the old be not wholly faded from the earth.

Robert Browning is first and fundamentally a painter of character, a student of the more subtle moods that dominate the individual, and toward this the reader of English fiction would hardly fail to see that the development of literature has steadily been advancing for two centuries. Even Mrs. Browning through the somewhat morbid and mawkish sentimentality and the overstrained art of "Aurora Leigh," in the vague and uncertain way of a woman whose contact with reality was necessarily slight, catches at the problems of nineteenth century feeling. Tennyson, as all men know, gave us poetry that was inwrought of the latest word of science, the last aspiration of religious hope, the newest sure conclusion in the field of social endeavor for the betterment of man.

And Tennyson in "In Memoriam," as Browning in "Paracelsus" and Lowell in "The Cathedral," has taught us that abstract truth may be made into poetry and that of the loftiest and most vitalizing kind. And to such poetry the world is ready to give a willing ear, though it will not be satisfied with the mere tricking out in rhyme and meter of scientific truth. The difficulty for the poet to-day is not merely that of new knowledge, but that of a science advancing so rapidly that the poet, whose art is meditative, can hardly avail himself of its latest revelations before their significance has vanished in the light of some new and revolutionary discovery announced from some investigator's laboratory. This is so new a thing that literary conditions have not yet been adjusted to it, as we may fairly hope that they will be some time in the not distant future.

A thing, almost if not quite, as distinctive of our time as the progress of scientific discovery is the growth of the democratic spirit. This latter has been a thing of common observation for over a century, and about that long ago Wordsworth and Shelley, Burns and Byron voiced with glowing enthusiasm the new revolutionary gospel. Since then it has been the theme of other pens and has become a matter of commonplace, and yet, though it has not lost interest because of the fulfilment of the hopes of man, it is not now a vital force in literature of the better class. The reason for this is, perhaps, not far to seek. In the domain of politics the advance in thought and feeling from a hundred years ago is a matter of no great moment. The poet who would voice for the world a message of brotherhood, thrilled with the spirit of a new humanity, inevitably finds himself harking back; he is compelled to repeat the sentiments of Mrs. Browning's perfervid Italian poems, or Whittier's simple songs, or Shelley's vague theorizing: he ceases to be individual. Under present conditions, strenuously vocal as the world is with the voices of those trying to be heard, failure to be distinctly and positively individual is failure to gain attention.

And it is significant that we are approaching the solution of social problems in the scientific way. The development of a better state of society is to come about, as we now realize, through the operation of natural laws, and not by the sensational process of awakening in the hearts of men a flashing enthusiasm for new forms of government.

Benjamin Kidd's 'Social Evolution' indicates quite clearly the new point of view from which all problems of society are to be considered, and perhaps, not less remarkable for a like significance is Henry Drummond's 'Ascent of Man.' As the laboratory gives up its secrets, as the mysteries of biology and processes of growth in the organic world become less mysterious, we are approaching nearer and nearer to a knowledge of the laws that are concerned in all growth, whether of the star fish or of the modern state. Assuming that man is the most vitally concerned in the organization of society here in this present world, and with the problem of another world, whether real or imaginary, whether a perfect state, or state of growth as that of earth, one cannot escape the reflection that both these problems have become in a measure problems of science, rather than problems of intuition or authority or emotional susceptibility.

And when science has come so close to all the inmost convictions and aspirations of man, there must follow a poetry of science, fuller, richer, more vitalizing and more enduring than any that has gone before it. It will appeal to a nobler and loftier sense of beauty, a finer and more perfect conception of truth. It will clothe its utterances in an imagery as much more varied as the knowledge of to-day is fuller than that of yesterday. It will be artistic beyond the dreams of other days, and its art will be something more than that of mere intuition. It will glow with color, but no crudeness of taste will guide the artist's brush, and the intelligent, æsthetic sense of a broadly cultured people will find inspiration in it, as once heroes did in the songs of the bards of old.

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ANTIQUITY OF THE CHEWING GUM HABIT.

In the letter of Columbus on the discovery of America, facsimile edition, 1892, of the four Latin editions belonging to the Lenox Library, the following occurs in the translation (page 11): “Finally, that I may compress in few words the brief account of our departure and quick return, and the gain, I promise this, that if I am supported by our most invincible sovereigns with a little of their help, as much gold can be supplied as they will need, indeed, as much of spices, of cotton, of chewing gum (which is only found in Chios), also as much of aloeswood, and as many slaves for the navy as their majesties will wish to demand.”

The date of this letter is March 14, 1493,—over four hundred years ago. It will be seen by the above that the chewing gum habit is by no means a modern or recent one, and doubtless antedates Columbus’ letter by many years.

The reference to Chios, an island in the Grecian Archipelago, is presumably for the purpose of indicating the character of the ‘gum.’ The Chios ‘gum’ of the ancients has been described as an earth of a compact character, probably argillaceous, and had the reputation of possessing medicinal qualities. Its consistency and appearance may have been such as to have led to its being popularly called ‘gum.’

That the chewing of gum, or some other article or waxy substance suitable for chewing, was in vogue at the time, there can be no doubt, and that the discovery of such a substance would be regarded as an important acquisition is implied by its being specially mentioned and promised by Columbus.

Years ago, more than half a century, shoemakers’ wax, so-called, Burgundy pitch and crude spruce-gum were chewed to a considerable extent, as the writer clearly remembers.

Betel chewing, the leaves and the nut mixed in certain proportions *with lime*, as practiced in Asiatic countries, naturally occurs to the mind in connection with the foregoing, as well as occasional instances of chewing slate pencils and lime mortar, an interesting case of the latter having been brought to my notice several years since by a well-known physician of Newark, N. J. But these are rather exceptional and individual cases, therefore not to be regarded as general or popular habits. From the *chewing* of earthy substances to the *eating* of the same, would appear to be but a natural step. The latter habit, so far as facts are available, is of comparatively infrequent occurrence and restricted to a much smaller number of persons. Beds of white infusorial earth, resembling magnesia in appearance, known as *Bergmehl*, occur in Lapland and Finland. This is, or has been used in seasons of scarcity, mixed with flour made of some kind of grain or ground birch-bark, and *clay-eating* probably, to a greater or less extent, still continues to be a habit in North Carolina as in the past. The effect of this habit, as any intelligent person would suppose, is decidedly injurious to the individual that pursues it. In several cases that have come under my observation the results are exhibited in sallowness of complexion, lack-lustre eyes, distension of the abdomen caused by engorgement or clogging of the liver, and other intestinal derangement, listlessness and general debility.

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SCIENTIFIC LITERATURE.

A GRAMMAR OF SCIENCE.

The increasing specialization of the sciences and the consequent occupation with the details and technical manipulations of a specialty render it possible for many a student to secure the equipment needed for his immediate activity, with but little appreciation of the general principles that give direction and solidarity to his science, or of the more general and fundamental conceptions which the various sciences and the spirit and progress of science as a whole have in common. The student runs the danger of gaining a certain familiarity with the vocabulary and the usage of the language of science, but of ignoring its grammar. One of the purposes met by Prof. Karl Pearson's 'The Grammar of Science' is to give the serious student an opportunity to acquaint himself with these underlying conceptions—cause and effect and probability, space and time, motion and matter and the composition of the physical and organic worlds. It discusses with him and for him the nature of the knowing process, and demonstrates how the sciences stand—not for a literal copy of reality, but represent a special abstraction and construction on the basis of experience, which serve the purposes of intelligibility and logical system. A law of nature is not an objective reality, but "a *résumé* in mental shorthand, which replaces for us a lengthy description of the sequences of our sense-impressions. Law in the scientific sense ... owes its existence to the creative power of his [man's] intellect." Science is thus not the mere reflection of perceptual experience, but is dependent for its advance quite as much upon the formation of appropriate conceptions by the exercise of insight and a keen logical analysis and synthesis. Hence, the importance of the imagination as a requisite for scientific discovery, which leads Professor Pearson to regard Darwin and Faraday as superior in this quality to the best of the poets and novelists. Not only the content of the sciences but the spirit and the means that guide its advance form part of the grammar of science. The nature of the scientific method, the appreciation that the scope of science is really coincident with the scope of verifiable knowledge; that science represents a mode of approach and of inquiry, and that the scientist or the scientifically-minded individual is characterized by a definite logical attitude, by a manner of entering into relation with his surroundings and of dealing with reality; that science discountenances attempted short-cuts and inspired revelations, or guesses of the riddles of existence; that it avoids metaphysic and impractical speculation; that it justifies its existence and the energies which are expended on its behalf by the mental training it provides in education, by its illumination of the problems of life and society, by the practical benefits it confers in the various fields of human activity, as well as by the gratification it yields to some of the most permanent and most worthy of our intellectual and æsthetic impulses—these and other propositions are ably and interestingly presented and constitute an essential portion of this very stimulating and clarifying volume. The success of the work is attested by the appearance of this second edition; the chief addition consists of a discussion of the quantitative method as applied to biological phenomena, which the readers of others of the author’s works will recognize as one of his favorite subjects of investigation.

THE TEACHING OF ELEMENTARY MATHEMATICS.

The book with the above title, by David Eugene Smith, principal of the State Normal School, at Brockport, N. Y., contains much of value, presented in a very readable and attractive manner. The subjects treated are arithmetic, algebra and geometry. About half the book is devoted to the first. The author sketches the history of the teaching of arithmetic from the earliest times, gives a critical examination of the different systems which have been tried and aims to discover the correct general principles upon which the instruction should proceed. He notices the tendency of many of our schools to follow too closely the Grube method, or a modification of it. The chapter on the present teaching of arithmetic is full of valuable suggestions. Algebra and geometry are treated in the same way. Much useless lumber is cleared away, and the whole discussion is marked by strong common-sense, an element not always present in discussions of this kind. The extreme differentiation in the teaching of these three branches which prevails in so many schools is condemned. It is urged that the blending of algebraic method and notation with the higher parts of arithmetic, and the early introduction of the inductive study of geometric form, both contribute to the substantial progress and development of the student. Valuable references are given to other writings for fuller discussions on special topics. These references cover works in English, French, German and Italian.

GEOLOGY.

Professor Suess's great work, 'Das Antlitz der Erde,' has been translated into French with emendations and annotations, and thus becomes accessible to an enlarged number of readers. No strictly geological publication since the time of the first appearance of Sir Charles Lyell's 'Principles of Geology' has brought together so many data concerning the nature of the altitude of the continents in relation to sea level. Geologists have generally assumed that it is the land which rises or sinks when a change of level takes place in relation to the sea. Professor Suess attacks this view and endeavors to show that the ocean has and has had its great movements, now keeping up its waters in the equatorial district, now accumulating about the poles and transgressing the low lands of its borders. An exhaustive review of the geological structure of the known parts of the earth, particularly complete with regard to the borders of the oceans and the the Mediterranean, is presented as a basis for discussing the evidence of such changes as the sinking in modern geological times of lands or islands in what is now the North Atlantic. By the sinking of the ocean floor, it is held that the sea level is lowered around the earth, thus giving rise to emerged lands. Parts of these plateaus have in turn sunk, and so the earth has experienced varied and often sudden changes of the relations of land and sea. The work is entertainingly written, despite the laborious compilation of geological details, which is made evident in its numerous chapters. The geological explanation of the Noachian Deluge is perhaps one of the most interesting sections of the work. Aside from the theory which the work sets forth, it affords the best general survey of the earth’s surface which is at present available in any language. It has been supplied with numerous recent references by M. de Margerie and his able assistants in the work of translation.

A YEARBOOK OF BIOLOGY.

L’Année Biologique for 1897.—Every year the number of biological workers increases, the number of repositories of researches is multiplied and the difficulties of keeping informed of the results obtained in even a restricted department of science are enhanced. Hence, new bibliographical works are ever welcome, especially if they give not only titles but abstracts. *L’Année Biologique* does not only this, but more, for its abstracts are likewise critical reviews indicating the true place in the science of the results given in any paper. It goes still further, in that it summarizes the advance made during the year in each subject, and the contents of the volume are rendered still more accessible by a thorough author-genus subject index. Everything seems to be done that is possible to make the results of general biological studies available. Occasionally figures are reproduced and comprehensive, synoptic articles on the recent advances in one subject are printed. In the present volume there is a report on senile degenerescence, by Elie Metchnikov; on the urinary tubules in vertebrates, with seventeen figures, by P. Vignon; and on the conditions of existence in and the bionomic divisions of fresh waters by G. Prouvot. The reviews are all signed by the authors, the critical remarks being bracketed. Many of the reviews have the dignity of distinct contributions to science, as where a half-page abstract is followed by a two-page discussion. The reviewers, or ‘collaborators,’ are drawn from various countries, America, Austria, Belgium, England, Russia and Scotland being represented in addition to France. This periodical may be commended in the strongest terms to biologists and to others interested in the results of biology. It is surprising that the work is still so little known in this country. Scientific men have a right to take pride in the unremunerative efforts of the chief editor, Professor Delage, to make accessible the literature of the science of general biology in order to facilitate its advancement.

ASTROPHYSICS.

The ‘Atlas of Representative Stellar Spectra, together with a Discussion of the Evolutional Order of the Stars,’ by Sir Wm. Huggins, K. C. B., and Lady Huggins (Wesley & Son), is not only a sumptuous and beautifully illustrated volume, but is also of great scientific value. Sir Wm. Huggins belongs to that group of men in England who, unconnected with any university, devote themselves to research for the pure love of truth. His distinguished services to science received recognition on the occasion of the Queen’s diamond jubilee, when with only two other scientific men he received the order of knighthood. His accomplished wife, who is his constant coadjutor, was the only woman mentioned in the list of Jubilee honors. Sir Wm. Huggins may be said to be the founder of the so-called ‘New Astronomy,’ for scarcely more than a quarter of a century ago his spectroscope, turned upon a newly discovered star, first revealed the cause of the sudden lighting up of these beacons in the heavens, and turned upon the nebula showed them to be of glowing gas. Since that time the telescope of the Tulse Hill Observatory, armed with spectroscope and camera, has been constantly and laboriously analyzing the light of star, comet and nebula, to solve the mystery of their constitution. “We never go anywhere,” said Lady Huggins; “astronomy, at best, is a heart-breaking object of devotion beneath English skies, and we are always at home to catch every gleam between the clouds.”

This book gives, in charming narrative, which would be read with interest by one previously ignorant of the subject, the history of the pioneer work “when nearly every observation revealed a new fact, and almost every night’s work was red-lettered by some discovery.”

There follow full details of later work, especially of the first detection, by the shifting of the lines of their spectra, of the motion of stars towards us or from us in the line of sight. We learn also how terrestrial chemistry has been enriched by this study of the stars, and how the nature of long known elements like hydrogen and the existence of undiscovered elements like helium have been first made out from stellar spectra.

But, as the supreme problem for the biologist is the development of man, so the supreme problem for the astronomer is that of the evolutional order of the stars. This problem, too, is discussed in the light of the discoveries at Tulse Hill. From the simple but beautiful harmonic system of hydrogen lines which characterizes a white star like Vega, we learn how we pass to the more developed star of a solar type, like Capella, and thence to Arcturus, and Belelgueze, which indicate a still later stage of development. At least this is the theory of the author. Aside from its great theme lucidly discussed the book deserves to be upon every library table as a superb specimen of bookmaking. For once, beautiful truth is promulgated in fitting guise. Lady Huggins is an artist and archæologist as well as an astronomer, and the initial letters of the chapters are illuminated with original sketches and designs from quaint old manuscripts, which make the book artistically as well as astronomically worthy of the prize which it received from the Royal Society as the most distinguished contribution to the scientific literature of the year.

EXPERIMENTAL MEDICINE.

Anyone who wishes to gain a fairly adequate idea of what experiments on living animals have accomplished for the welfare of the human race and of other animals as well, can now do so by reading ‘Experiments on Animals,’ by Stephen Paget. Mr. Paget has collected evidence showing the part that animal experiments have played in the progress of physiology, pathology, bacteriology and therapeutics. He has not ventured to offer opinion or even statements unsupported by exact and verifiable facts. A large part of the book’s space is filled by original quotations from scientific workers, from Galen down to the recent students of the malaria parasite. It shows plainly that knowledge of the processes of life in health and disease has throughout depended on experiments on living substances. Mr. Paget’s book is not dependent for its interest solely on the laudable curiosity to know the worth of animal experiments. For these have been so important in the science of medicine that their story is at the same time the history of a great number of medical discoveries. There is, too, a freshness and biographical interest in the quotations from the famous past and present students of medical science which makes them very readable.

ICHTHYOLOGY FOR ANGLERS.

In his “Familiar Fish, their Habits and Capture,” Mr. Eugene McCarthy has put forth a readable volume which doubtless will prove popular among the disciples of Izaak Walton, for it is essentially a book for anglers, written by an angler of experience. A preliminary chapter, devoted to fish-culture, dwells on the destruction of eggs and fry in nature and the necessity for artificial measures. It is a fairly good general outline of the subject, although some of the methods described are obsolete. The many breeders of ornamental fish will wonder whether the author is intentionally facetious in stating that the “famous double-tailed goldfish frequently seen are raised in Japan, and are produced by violently shaking the eggs in a pan.”

About a third of the book is devoted to brief accounts of the distribution, food, habits and peculiarities of the fresh-water fishes most sought by anglers, the salmons, trouts, basses and pikes naturally receiving most attention. The remaining pages deal chiefly with the description of angling paraphernalia and methods, camping, boating and useful data for sportsmen. By far the best chapters are those treating of the ouananiche and its capture, as the author writes from ample experience. He gives it first rank among our game fishes and holds that “pound for pound the ouananiche can greatly outfight the salmon, and none of the freshwater fishes can equal it in this respect; the black bass approaches it the nearest but never equals it.”

The volume is freely illustrated with fishing scenes, angling apparatus and twenty-five full-page figures of fishes, all but one of which are copied, without credit, from the reports of the U.S. Fish Commission.

The author submitted his manuscript to President Jordan “to be justified in advancing the claim” that the descriptions of the different fishes “are absolutely reliable and correct,” and a prefatory note by Dr. Jordan is in that author’s most pleasing style and adds considerably to the literary excellence of the volume; but evidently that distinguished ichthyologist did not believe any responsibility attached to him, for even a cursory glance by him over the manuscript would have eliminated a number of ichthyological incongruities, such as the inclusion of the white bass, one of the Serranidæ, in the same family as the black basses (Centrarchidæ). The author’s conception of zoölogical nomenclature and classification is decidedly novel. In the final chapter, on “scientific names of fish mentioned,” the first species referred to is *Salmo salar*, of which it is stated that “the word *salmo* is used in connection with a large variety of the trouts, to designate the family or descent. It is the first name given, as is the case with all other kinds of fish, being the specific name indicating the species. The other names following are subspecific.” The land-locked salmon of the Saguenay River is by some systematic writers regarded as a variety of the sea salmon, and bears the name *Salmo salar ouananiche* McCarthy. Strange to say, this is the only species in the volume for which the name of the original describer is given, and in explaining his own connection with the fish, Mr. McCarthy says: “McCarthy, so named from his first writing fully regarding the fish!”

To the zoölogist the volume will be of no use, as it embodies few new observations on the fishes considered and is largely a compilation from other well-known works. The author, however, deserves credit for bringing the subject to the attention of anglers in such an attractive form; and, as an attempt to extend the knowledge of the habits, distribution and relationships of our game fishes among this large and influential class of citizens, the volume should be accorded a welcome.

MICROSCOPY OF DRINKING-WATER.

Mr. G. C. Whipple, Director of the Mount Prospect Laboratory of the Brooklyn Waterworks, has prepared a handbook for the water analyst and the waterworks engineer, with the title given above. It deals with the purposes, methods and results of the biological examination of drinking-water, affording means for the identification of the microscopic life found in water supplies and suggesting means for the elimination or control of those organisms which disagreeably affect the color or odor of potable waters. The construction of reservoirs, the storage of surface and of ground waters and the growth of organisms in pipes are also discussed. Though the motive of the book is thus technical, the subject is developed by the author along broad lines in a thoroughly scientific manner, and he has brought together a great deal of information, not only for the sanitary engineer, but also for the physicist, the chemist and the biologist. The problems in limnology, such as the temperature, stagnation and circulation of reservoir waters; the distribution and relative numbers of different organisms and their relation to chemical analyses are discussed in the light of the results of many years' investigation of water-supplies. The seasonal succession of organisms, their movements with respect to light and other stimuli, and their horizontal and vertical distribution, are in like manner fully treated. The scope of the work and the treatment of the subject make the book a valuable one alike for engineering and biological laboratories and for the general library.

THE PROGRESS OF SCIENCE.

The summer laboratories and the scientific expeditions which are employing the vacation period of the men of science in this country would make a long list. A vacation from teaching means to the scientific man a chance to work, and at present there are numerous organized means of enabling him to profit by this chance. The most definite form which such arrangements for summer work have taken is the summer laboratory or experiment station for biologists. Such a station affords conveniently the mechanical appliances for scientific work in a good locality for collecting material to work with. The marine or other forms of life are thus made accessible to those whose professional work during the year keeps them in an unfavorable locality. Besides the laboratory at Woods Holl, which is the nearest American representative of Professor Dohrn’s great laboratory at Naples, there is an important summer station at Cold Spring Harbor, Long Island, under the auspices of the Brooklyn Institute, and others cared for by Leland Stanford, Jr. University, the University of Indiana, the Ohio State University and other institutions. It is common to combine teaching with research at these laboratories and in some cases they become essentially summer schools, though generally giving courses of a higher order than the ordinary summer school for nature study. But research is often the chief and sometimes the sole purpose of these stations, and a vast amount of work is done each year. The most important of these summer stations is the Woods Holl Marine Biological Laboratory, situated on the southern coast of Massachusetts, between Buzzard’s Bay and Vineyard Sound. This laboratory has been fortunate in having been the summer home at one time or another of a majority of the leading zoölogists of the country. It has been usual for the advanced students in universities to take courses or carry on research there, and Woods Holl training has been a valuable recommendation. The reason is not far to seek. The material advantages, the spirit of zeal for concrete fact, the acquaintance with superior men in the science and with a large number of equals, all help to give the best sort of professional training. Such a place also serves as a refinery where opinions and theories may be purified by healthy criticism and by the subtler influence of example. There is a story of three eminent biologists who got involved in a controversy over a disputed question. They argued for a while. Finally one of them said: “Let us get the eggs in question and study them together.” This was done, and the three men spent the afternoon over their microscopes patiently working out the problem together; and they did work it out. One of the great advantages of summer laboratories is that they put fellow-students in a frame of mind in which they can work things out together.

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The Woods Holl Laboratory has a right to claim a large share in the credit for three of the most important developments in biology in the last decade—the study of ‘cell lineage,’ of regeneration of organs and of the influence of abnormal conditions on the development of embryos. Workers there have traced the development of the different cells into which the egg-cell divides and have discovered just what parts of the body arise from each group of cells. They have shown that the way in which the egg divides and redivides is as constant, is as much a part of the nature of the animal, as its adult form and structure are. They have replaced previous vague notions of the development of animals by exact accounts of the cell-origin of different organs of the body. Others have studied the abilities of mutilated animals to reproduce the parts lost and the conditions and limitations of such regeneration. Such studies have greatly broadened our views of the nature of animal tissues. Others have investigated the results of artificial conditions on the development of animals, especially in the earliest stages. For instance, from eggs broken into pieces there have been developed twins, triplets and monsters of various sorts. Such experiments as these are producing data concerning the very fundaments of living matter and are leading biology beyond the mere description of animal structures and functions towards an insight into the elementary principles of development. Among the numerous researches, some seventy in all, which are being carried on at Woods Holl this summer, those of the most general interest are Prof. C. O. Whitman’s study of hybrids and Prof. Jacques Loeb’s study of artificial fertilization. Prof. Whitman has been breeding pigeons of a large number of species for several years, as a means of studying the phenomena of heredity shown in hybrid forms. More or less incidentally, he has discovered many notable facts about the instincts and habits of the birds and about various physiological functions connected with reproduction. Biologists everywhere are coming to realize the necessity of systematic and continuous study of families of animals through a number of generations. Prof. Whitman’s is the most extensive of such studies in this country. The detailed results of Prof. Loeb’s continuation of his experiments on the action of various salts on unfertilized eggs will naturally be awaited with great interest. We have already noticed his success in causing unfertilized eggs of the sea-urchin to develop into normal individuals as far as the pluteus stage. He has this year succeeded in producing artificial parthenogenesis not only in starfish (*Asterias*), but also in worms (*Chaetopterus*). Through a slight increase in the amount of K-ions in the sea-water, the eggs of the latter can be caused not only to throw out the polar bodies as Mead had already observed, but also to reach the *Trochophore* stage and swim about as actively as the larvæ originating from fertilized eggs.

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In the courses of instruction offered at Woods Holl there are two of more than ordinary interest. Professor Loeb's course in physiology departs from the traditional study of physiological functions in the frog and in some mammal, and offers instead experimental work on the simpler invertebrate forms. The phenomena of life are there presented in diagrammatic form, and are interpreted as far as possible in terms of physics and chemistry. The course in nature study, given this year for the first time, offers to students without technical training a chance to learn about animals and plants from specialists. It has shown clearly that the best science is popular, that really scientific work can be done without previous drill in terminology or technique. A novel feature of the course has been the systematic experimental study of the instincts and intelligent performances of animals. The method of offering to intelligent men and women, who wish to know about animal life, but have no time or need for special technical training or detailed anatomical work, a chance to get something better than mere book knowledge or haphazard personal observation, should be widely extended.

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The laboratory of the Brooklyn Institute of Arts and Sciences, situated at Cold Spring Harbor, Long Island, is nearly as old as the Woods Holl Laboratory. Prof. C. B. Davenport, its director, is probably the most active worker in this country in the quantitative study of variation, and one of the leading lines of research at Cold Spring Harbor is now and will probably be for some years the attempt to get an exact estimate of normal variation in different animals, of the production of abnormal variations and of the laws of inheritance. Professor Davenport is himself breeding mice extensively and thus securing data. Of the courses offered two deserve special mention.

One is the course for teachers of zoölogy in high schools, a chief feature of which is the study of living animals. The other is a course on ‘Variation and Inheritance,’ which gives advanced students a chance to study the most important question of biology and by the most exact methods. The Cold Spring laboratory has been growing very rapidly of late and seems likely to continue to grow. In general the evolution of the summer laboratory is of interest. An enthusiast or a modest association gathers a few sympathetic workers at some favorable locality. The informality and personal contact are inspiring and the place becomes famous for good work. Then come numbers and with numbers a rapid complication of the social life of the school. The eminent leader is replaced by a dozen different instructors; one no longer knows every one else; organization becomes complex and what was at first a sort of scientific family may turn into a formal institution. The summer laboratory should not become a big summer college at the cost of its single-mindedness.

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While special laboratories are open for work in biology, and the universities are extending their sessions through the summer, the common schools are also beginning to realize that they must adapt themselves to an urban civilization. Country schools should adjourn in the summer for obvious reasons, but in the city nothing is gained by turning the children from the schools into the streets. The vacation or play schools now in session in New York City are in every way to be commended. The only drawback is that they cannot hold half of those who wish to attend. Set free from the traditional curriculum the children learn more in the five weeks of ‘play school’ in the summer, than in twice that period of ‘work school’ in the winter. Swimming, open-air gymnastics, team games, chess, visits to parks, piers, museums and libraries, excursions in barges and into the country, sketching, whittling, cooking, sewing and the rest do not lose their educational value because the children like them. Such exercises will do a good deal toward curing the indigestion caused by being fed for five years on the three R’s, and toward correcting the anti-social atmosphere of the ordinary school-room. Among the commonplaces of modern psychology are: It is not what a person knows but what he does that counts; the way to learn is to act; progress follows from the pleasure of partial success; an individual only exists in his relations with others. Such maxims seem to be as clearly kept in view by the New York Department of Education in the summer as they are forgotten in the winter. The committee on the New York Play Schools consists of Messrs. Seth T. Stewart, John L. N. Hunt and A. P. Marble, to whom and to the teachers who have carried out their plans much honor is due. The report for 1899 is an educational document of importance. Copies can probably be obtained from the Department of Education of the City of New York.

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The Paris Exposition and its congresses may be regarded as a great summer school. The applications of science exhibited for amusement, for instruction and for the advantage of commerce and manufactures are bewildering in their multiplicity. It is interesting to note that the group ‘Education’ heads the catalogue of the Exposition. In the exhibits representing higher instruction, the United States received nine grand prizes and nine gold medals, ranking second to France. On the motion of a French juror, three Americans were mentioned as worthy of special distinction: Prof. H. A. Rowland, Johns Hopkins University; Prof. Nicholas Murray Butler, of Columbia University; Director Melvil Dewey, University of the State of New York. More than one hundred and fifty international congresses, dealing with various subjects of scientific, industrial and social importance, are held this summer in Paris, and form no small part of the interest of the Exposition, supplementing as they do the exhibits, furnishing the theory, as the exhibits set forth the accomplishments, of art and industry. The magnitude of these congresses may be seen from the fact that the thirteenth International Medical Congress had a registration of over six thousand members, of whom over four hundred were from America.

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Friends of scientific investigation and the teaching of science will rejoice at the recent decision in the courts concerning the Fayerweather will case. For the eighth time the grant of $3,000,000 to the colleges has been confirmed. The case will probably be appealed to the Supreme Court of the United States, but the probability is large that Mr. Fayerweather’s wishes will in the end be carried out. At the present time, money left to colleges is likely to be used to a very large extent to promote the progress of science. Required courses in linguistics are decreasing, and the extension of college teaching and university research is largely along scientific lines. New departments, such as those of physiography, physical chemistry, anthropology and experimental psychology are being established, while economics and sociology are becoming less speculative and more like the natural sciences in their methods. The college student of to-day gets proportionately more training in the professedly natural sciences than ever before, and gets scientific training in connection with courses which were once mere exercises in learning the opinions of more or less important people.

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We called attention last month to the completion of the plans for an international catalogue of scientific literature, and stated that Great Britain and Germany had each subscribed for forty-five of the three hundred sets that must be sold in order to defray the cost. It is obvious that the United States, with such a large number of libraries and educational institutions, should subscribe for its share of the sets, namely, not less than forty-five. The Smithsonian Institution has provisionally undertaken to represent the interests of the catalogue in the United States, and will receive promises of subscriptions. The catalogue will be issued in seventeen volumes, comprising the following subjects: Mathematics, mechanics, physics, chemistry, astronomy, meteorology (including terrestrial magnetism), mineralogy (including petrology and crystallography), geology, geography (mathematical and physical), palæontology, general biology, botany, zoölogy, human anatomy, physical anthropology, physiology (including experimental psychology, pharmacology and experimental pathology) and bacteriology. At least one volume will be given to each subject, and it is proposed that not all the volumes shall be issued at once, but in four groups, as soon as possible after the first of January, April, July and October, respectively. The subscription price for a complete set of the whole catalogue, in seventeen volumes, is £17, say $85. The volumes will vary in price and can be obtained separately, but it is necessary to secure the guarantee of the sale of forty-five sets in America during the month of September, and all libraries used for scientific research, and those individuals who can afford the cost, should send subscriptions to Dr. Richard Rathbun, Assistant Secretary of the Smithsonian Institution, Washington, D. C.

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In the July number of the Monthly Dr. H. C. Bolton gave an account of the radio-active substances which have been found in pitchblende, the chief ore of uranium. The subject continues to excite the interest of both chemists and physicists, though just at present the largest amount of work is being done by the chemists, to whom the question is of extraordinary interest as to whether these substances are or are not real chemical elements. Béla von Lengyel, of Budapest, as Dr. Bolton explained, has attacked the problem from the synthetic side, and by fusing inactive barium nitrate with uranium nitrate, he has obtained a barium sulphate which has more or less radio-activity. From this he concludes it is probable that the radio-activity is due rather to a peculiar state of the barium than to a new chemical element. On the other hand, Becquerel has in a somewhat analogous way mixed inactive barium chlorid with uranium chlorid, and from the solution has obtained likewise a radio-active barium. But he finds that the increased activity in the barium salt is attended by a corresponding decrease in the radio-activity of the uranium. Hence it cannot be settled from these experiments whether the uranium salts possess a radio-activity of their own, which can by certain methods be communicated to barium salts, or whether the radio-activity is due to an impurity in the uranium which has thus far eluded isolation.

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The director of the Blue Hill Meteorological Observatory, Mr. A. Lawrence Rotch, writes to ‘Science’ that the highest previous kite-flight was exceeded on July 19, when, by means of six kites attached at intervals to four and three-quarters miles of steel wire, the meteorograph was lifted 15,170 feet above Blue Hill, or 15,800 feet above the neighboring ocean. At the time that the temperature was 78° near the ground, it was about 30° at the highest point reached, the air being very dry and the wind blowing from the northwest with a velocity of twenty-six miles an hour. The altitude reached in this flight probably exceeds the greatest height at which meteorological observations have been made with a balloon in America.

The highest observations that have been published were made by the late Professor Hazen, of the Weather Bureau, in an ascent from St. Louis, June 17, 1887, to a height of 15,400 feet.

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The U.S. consul at St. Gall, Mr. Du Bois, sends to the Department of State the following account of the trial of the Zeppelin air-ship: At the invitation of Count Zeppelin, I was present at the trial ascent of his air-ship on the afternoon of July 2, at Manzell, on Lake Constance. At seven o'clock the great ship, 407 feet long and 39 feet in diameter, containing seventeen separate balloon compartments filled with hydrogen gas, was drawn out of the balloon house securely moored to the float. At the moment of the ascent the wind was blowing at a rate of about twenty-six feet per second, giving the operators a good opportunity of testing the ability of the air-wheels to propel the great ship against the wind. The cigar-shaped structure ascended slowly and gracefully to about thirty feet above the raft. The balances were adjusted so as to give the ship an ascending direction. The propellers were set in motion, and the air-ship, which has cost considerably over $200,000, started easily on its interesting trial trip. At first the ship moved east against the wind for about two miles, gracefully turned at an elevation of about 400 feet, and, making a rapid sail to the westward for about five miles, reached an altitude of 1,300 feet. It was then turned and headed once more east, and, traveling about a mile against the wind blowing at the rate of twenty-six feet per second, suddenly stopped; floating slowly backwards three miles to the west, it sank into the lake, the gondolas resting safely upon the water. The time of the trip was about fifty minutes; distance traveled, about ten miles; fastest time made, five miles in seventeen and one-half minutes. The cause of the sudden stoppage in the flight of the ship was proved to be a slight mishap to the steering apparatus, but the colossus floated gently with the wind until it settled upon the surface of the lake without taking any water. The raft was then brought up and the ship was easily placed upon it and brought back to the balloon house. The weight is 200 centners (22,000 pounds).

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A joint meeting of the Royal Society and the Royal Astronomical Society has been held in London to hear preliminary reports from several British expeditions that went out to observe the recent eclipse of the sun. Mr. Christie, the astronomer royal, first presented an account of the observations made by himself and Mr. Dyson at Ovar, in Portugal. There totality lasted 84½ seconds, and though the sky was rather hazy he secured some good photographs. The corona seemed distinctly inferior in brightness, structure and rays to that seen two years ago in India. Sir Norman Lockyer next described the observations made by the Solar Physics Observatory Expedition and the officers and men of H. M. S. Theseus at Santa Pola. Professor Turner spoke of the observations he had made with Mr. H. F. Newall in the grounds of the observatory near Algiers. From observations on the brightness of the corona he concluded that it was many times brighter than the moon—perhaps ten times as bright. Prof. Ralph Copeland described the observations he made on behalf of the joint committee at Santa Pola, endorsing Sir N. Lockyer's remarks as to the advantage of having the aid of a man-of-war. Mr. Evershed presented a preliminary report on his expedition to the south limit of totality. His reason for choosing a site at the limit of totality was that the flash spectrum was there visible very much longer. Unfortunately, he accepted the guidance of the Nautical Almanac Office, and found himself outside the line of totality—about two hundred meters according to his informants, who said a small speck of sunlight was visible all the time. He was successful in obtaining some fine photographs of the flash spectrum.

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During the last session of Congress a law was enacted, commonly known as the Lacey Act, which places the preservation, distribution, introduction and restoration of game and other birds under the Department of Agriculture; regulates the importation of foreign birds and animals, prohibiting absolutely the introduction of certain injurious species and prohibits interstate traffic in birds or game killed in violation of State laws. Persons contemplating the importation of live animals or birds from abroad must obtain a special permit from the Secretary of Agriculture, and importers are advised to make application for permits in advance, in order to avoid annoyance and delay when shipments reach the custom-house. The law applies to single mammals, birds or reptiles, kept in cages as pets, as well as to large consignments intended for propagation in captivity or otherwise. Permits are not required for domesticated birds, such as chickens, ducks, geese, guinea fowl, pea fowl, pigeons or canaries; for parrots or for natural history specimens for museums or scientific collections. Permits must be obtained for all wild species of pigeons and ducks. In the case of ruminants (including deer, elk, moose, antelopes and also camels and llamas), permits will be issued, as heretofore, in the form prescribed for importation of domesticated animals. The introduction of the English or European house sparrow, the starling, the fruit bat or flying fox and the mongoose, is absolutely prohibited, and permits for their importation will not be issued under any circumstances.

Transcribers’ Notes

Punctuation, hyphenation, and spelling were made consistent when a predominant preference was found in this book; otherwise they were not changed.

Simple typographical errors were corrected; occasional unbalanced quotation marks retained.

Ambiguous hyphens at the ends of lines were retained.

Page 463: “fixd rule” was printed that way.

Page 464: The Greek transliteration “hoi polloi” was added by Transcriber and enclosed in {curly braces}.

Page 466: The opening quotation mark for “half-fanatical” has no matching closing mark; Transcriber added one after “personality worship”.

Page 495: “carbon dioxid” was printed that way.

Page 515: “easily reduce” may be a misprint for “easily deduce”; “by the sign of” may be an archaic spelling of “sine”.

Page 536: “pompes funébres” is a misprint for “funèbres”.

Page 553: “Belelgueze” was printed that way, probably refers to “Betelgeuse”.

Page 559: “chlorid” was printed that way.

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